

**PROJECT NAME:**

THE PROPOSED CONSTRUCTION OF WALMER HOUSING DEVELOPMENT ON ERF 11305, PORT ELIZABETH WITHIN THE NELSON MANDELA BAY MUNICIPALITY, EASTERN CAPE PROVINCE.

DEDEAT REFERENCE NO: ECm1/C/LN2/M/07-2026

**REPORT TITLE:** DRAFT ENVIRONMENTAL IMPACT REPORT

DATE: 05 AUGUST 2026

**ENVIRONMENTAL ASSESSMENT PRACTITIONER:**



**CONSULTING ENGINEER:**



**APPLICANT:**



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## DOCUMENT REVISION STATUS

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| DATE     | July 2026                                                                                                                                                                                                                                                 | July 2026                                                                                                      | August 2026                                                                                                     |

## DECLARATION OF INDEPENDENCE

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I, **Sive Mlamla**, in my capacity as an EAP, hereby declare that I –

- Act as an independent consultant
- Do not have any financial interest in the undertaking of the activity, other than remuneration for the work performed
- Have and will not have vested interest in the activity
- Have no, and will not engage in, conflicting interests in the undertaking of the activity
- Undertake to disclose any material information that has or may have the potential to influence the decision of the competent authority or the objectivity of any report, plan, or document
- Will provide the competent authority with access to all information at my disposal regarding the report, whether such information is favourable to the Client or not
- Based on information provided to me by the Client and in addition to information obtained during the course of this study, have presented the results and conclusion within the associated document to the best of my professional ability
- Reserve the right to modify aspects pertaining to the present investigation should additional information become available through on-going research and/or further work in this field
- Undertake to have my work peer reviewed on a regular basis by a competent specialist.



Sive Mlamla (Pr.Sci.Nat, Reg. EAP (EAPASA))

Environmental Assessment Practitioner (EAP)

05/08/2026

## **UNDERTAKING (AFFIRMATION BY THE EAP)**

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*Abantu Environmental Services (Pty) Ltd hereby confirms that, to the best of our knowledge, the information provided in this report was correct at the time of compilation. Information included in this report was based on the information which was provided to us by the Engineers on behalf of the Applicant and the commissioned Specialist reports. It is further confirmed that all comments received from Stakeholders and Interested and Affected Parties (I&APs) will be included in the final reports submitted to the Competent Authorities.*

## **ASSUMPTIONS AND GAPS IN KNOWLEDGE**

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The following assumptions and potential gaps in knowledge apply:

- All information provided by the Applicant to the EAP was correct and valid at the time it was provided.
- The EAP does not accept any responsibility in the event that additional information comes to light at a later stage of the process.
- All data from unpublished research is valid and accurate at the time this report was written.
- The scope of this investigation is limited to assessing the potential environmental impacts associated with the proposed development only.

It should be noted that findings, recommendations, and conclusions provided in this report are based on the author's best scientific and professional knowledge and experience. No part of this report may be amended or extended without prior written consent of the authors. Any recommendations, statements or conclusions drawn from or based on this report must clearly cite or refer to this report. Whenever such recommendations, statements, or conclusions form part of the main report to current investigation, this report must be included in its entirety.

## **EXECUTIVE SUMMARY**

### **INTRODUCTION AND BACKGROUND**

The Nelson Mandela Bay Municipality (NMBM) is proposing the development of the Walmer Housing project and associated facilities and infrastructure on erf 11305, in Walmer, Port Elizabeth. The property is located west of Walmer along Victoria Drive and bordered by the suburb of Walmer Heights to the northwest, the Walmer Country Club to the north, and the former Arlington race course property, to the east. The proposed development was authorised on 31 July 2019. However, the Environmental Authorisation (EA) lapsed since the development has never commenced. On the basis that the authorisation was issued seven (7) years ago and EA lapsed, hence, a new application for EA is being lodged. The EA application was submitted and registered on 26 February 2026. An acceptance letter of the Final Scoping report was received on 28 May 2026. This Draft Environmental Impact report (DEIR) is circulated for the regulatory 30-days comment period from 05 August – 07 September 2026. The Environmental Impact Report includes information as required per Appendix 3 of the 2014 NEMA EIA Regulations including all inputs from the preceding Scoping phase.;

### **PROJECT DESCRIPTION**

The proposed the proposed development will provide a mix of housing typologies such as Free Basic House/RDP, GAP/FLISP, Social Housing and Open/GAP market housing. Other land uses will include a business zone, institutional zone, authority zone, open space zone, special use- conservation usage (Forest patches) and transportation zone as well as associated infrastructure.

The total area of the site is 43.73 ha. The Gross Residential Area (single dwellings) is 11.11 ha with Gross Residential Density (single dwellings) being 14.90 units/ha and a Nett Residential Density (single dwellings) being 58.69 units/ha. The average erf size for single dwellings is 170.41m<sup>2</sup>. The estimated number of residential opportunities amounts to 952 units, constituted by single dwellings (652 units) and high density residential (300 units).

### **MOTIVATION**

The proposed housing development is aimed at providing affordable housing to various income groups and will contribute to alleviating the current housing need in the municipality. The development is intended to accommodate vulnerable, poor and missing middle, those households who earn too much to qualify for government-subsidised housing but not enough to afford market-rate housing.

## **LEGISLATIVE REQUIREMENTS**

Listed Activities (LN2 [GNR 325 Activity 15] and LN3 [GNR 324 Activity 4, 12 &14]) are triggered by the proposed development and as such, an Environmental Authorisation (EA) is required prior to commencement. The EA application is subject to a Full Scoping and Environmental Impact Assessment (EIA) Process and will be adjudicated by the identified competent authority, Department of Economic Development, Environmental Affairs and Tourism (DEDEAT), under Sarah Baartman Region.

## **ALTERNATIVES**

### **1. Site Alternatives**

No site/property alternatives exist as the proposed site is municipally owned, and no land acquisition will be required. Furthermore, the site falls within an area that was allocated for residential development. Additionally, an Environmental Authorisation (EA) for a housing development on this site had been previously granted.

### **2. Design and layout Alternatives**

- Alternative 1- This option comes with a higher housing yield covering 46% of the site, however it does not entirely avoid the forest patches and the development overlaps on these forest patches.

**The layout was revised to avoid the forest, as such this layout option will not be subject to further assessment in this Scoping and EIA process.**

- Alternative 2 (preferred)- This option comes with a lower housing yield compared to Alternative 1 but is environmentally favourable because it avoids encroaching on the forest patches.

### **3. Technical Alternatives-** No alternatives have been considered. However, the Applicant should consider the use of solar energy such as solar panels as well as rainwater harvesting through the installation of barrels/tanks. Integration of climate resilient housing development through the integration of renewable energy such as solar and alternative water sources from the rain were also strongly recommended by I&APs during the public consultation process of the Scoping phase.

### **4. Operational Alternatives-** No alternatives have been considered.

5. **No-go option-** Keeping the current status quo of no activities occurring on site, which would not result in a loss of a critically endangered vegetation and the forest clumps. However, potential positive socio-economic impacts will be lost. Furthermore, current pressures such as illegal wood harvesting, cited in the consultations with I&APs would potentially degrade, if not diminish the forest ecosystems occurring within the site.

## **SPECIALIST STUDIES**

The following specialist impact assessments were undertaken to inform the Scoping & EIA Process:

- Terrestrial Biodiversity Impact Assessment- Inclusive of Plants and Animal species
- Aquatic Biodiversity Impact Assessment
- Agricultural Compliance Statement
- Paleontological Impact Assessment
- Archaeological and Cultural Heritage Impact Assessment
- Traffic Impact Assessment

None of the specialist impact assessment studies identified any fatal flaws in the development. Overall, the impacts expected to occur as a result of this development can be either avoided entirely or reduced to an acceptable level of impact if all of their recommended mitigation measures are implemented.

## **PUBLIC PARTICIPATION PROCESS**

This Draft Environmental Impact Report (DEIR) was circulated for public participation process (PPP) mainly as follows:

Public comment (30-days): 05 August 2026- 07 September 2026.

Public meetings for presenting the DEIR will be conducted as follows:

- **Physical:** Walmer High School (10am)- 01 September 2026
- **Virtual:** Microsoft Teams (18h00)- 02 September 2026

Site notices were placed on 05 August 2026.

Email notice, with a copy of the DEIR to I&APs was circulated on 05 August 2026.

All comments from stakeholders and Interested & Affected Parties (I&APs) will be incorporated into the Final Environmental Impact Report (FEIR). The FEIR will then be submitted for decision-making. This process will

ensure that environmental, social, and technical considerations are fully integrated into the final project design and authorisation.

### **Environmental Impact Statement and Reasoned Opinion of the EAP**

The assessment indicates that the proposed housing development will result in both positive and negative direct, indirect, and cumulative impacts on the receiving environment and surrounding communities. The proposed development is justified as it responds to the need for formal housing within the Nelson Mandela Bay Municipality and supports national, provincial, and municipal objectives for the delivery of integrated and sustainable human settlements. The development will contribute to improved access to housing, local economic development, and employment opportunities during the construction and operation phase.

An assessment of alternatives, including the No-Go alternative, was undertaken as part of the EIA process. The preferred development layout has been refined to avoid and minimise impacts on environmentally sensitive areas, particularly the forest patches identified on the site, while making use of municipally owned land that has been earmarked for medium- to high-density residential development. The No-Go alternative would avoid the environmental impacts associated with development but would not address the identified housing need or realize the associated socio-economic benefits. It is therefore not considered the preferred alternative.

Although negative environmental impacts have been identified, these are anticipated to be localised and can be effectively mitigated through the implementation of the Environmental Management Programme (EMPr) and the recommendations of the specialist studies. No fatal flaws have been identified that would preclude the proposed development from proceeding. Based on the findings of the Environmental Impact Assessment, the consideration of alternatives, and the anticipated socio-economic benefits, it is the opinion of the EAP that the proposed housing development is environmentally acceptable and should be authorised, subject to the implementation of the EMPr.



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## ACRONYMS AND ABBREVIATIONS

|         |                                                                       |
|---------|-----------------------------------------------------------------------|
| CBA     | Critical Biodiversity Area                                            |
| CA      | Competent Authority                                                   |
| DEDEAT  | Department of Economic Development, Environmental Affairs and Tourism |
| DEIR    | Draft Environmental Impact Report                                     |
| DFFE    | Department of Forestry, Fisheries and the Environment                 |
| DWS     | Department of Water and Sanitation                                    |
| dSR     | Draft Scoping Report                                                  |
| EA      | Environmental Authorisation                                           |
| EAP     | Environmental Assessment Practitioner                                 |
| ECPHRA  | Eastern Cape Provincial Heritage Authority                            |
| EIA     | Environmental Impact Assessment                                       |
| EAIR    | Environmental Impact Assessment Report                                |
| EMPr    | Environmental Management Programme                                    |
| FEIR    | Final Environmental Impact Report                                     |
| FLISP   | First Linked Individual Subsidy Programme                             |
| FSR     | Final Scoping Report                                                  |
| I&APs   | Interested and Affected Parties                                       |
| IEM     | Integrated Environmental Management                                   |
| IDP     | Integrated Development Plan                                           |
| NEMA    | National Environmental Management Act                                 |
| NFA     | National Forest Act                                                   |
| NMBM    | Nelson Mandela Bay Municipality                                       |
| NM MOSS | NMBM Metropolitan Open Space                                          |
| OHSA    | Occupational Health and Safety Act                                    |
| PoS     | Plan of Study                                                         |
| POS     | Public Open Space                                                     |
| PPP     | Public Participation Process                                          |
| RDP     | Reconstruction and Development Programme                              |
| S&EIR   | Scoping and Environmental Impact Assessment Report                    |
| SACAA   | South African Civil Aviation Authority                                |
| SAHRA   | South African Heritage Resources Authority                            |
| SAHRIS  | South African Heritage Resources Information System                   |
| SDF     | Spatial Development Framework                                         |
| ToR     | Terms of Reference                                                    |

# 1. INTRODUCTION

## 1.1.PROJECT DESCRIPTION

The proposed project entails the construction of the Walmer housing development (~43 ha), including associated facilities and infrastructure on Erf 11305, Walmer. The erf is located west of the existing Walmer township along Victoria Drive. The housing development will provide a mix of housing typologies and associated facilities, which include the following:

### **Housing typologies and associated facilities**

- Free Basic House/ Reconstruction and Development Programme (RDP)
- GAP/ First Linked Individual Subsidy Programme (FLISP)
- Social Housing
- Open/GAP market housing

The design, costs and earning brackets for the proposed housing typologies will be developed by the Housing Delivery Sub-Directorate during the planning and design phase of the top structures.

### **Other land uses (see Table 1)**

- Business Zone I- to provide for business's such as a supermarket, bottle store, service trades etc with parking provision at 3.33 bays per 100m<sup>2</sup>.
- Community Zone II- school site
- Authority Zone- authority uses such as police station, fire services, telecommunications facilities etc.
- Open Space Zone I- public open space
- Open Space Zone III - conservation usage
- Transportation Zone I

The total area of the site is 43.73 ha. The Gross Residential Area (single dwellings) is 11.11 ha with Gross Residential Density (single dwellings) being 14.90 units/ha and a Nett Residential Density (single dwellings) being 58.69 units/ha. The average erf size for single dwellings is 170.41m<sup>2</sup>. The estimated number of residential opportunities amounts to 952 units, constituted by single dwellings (652 units) and high density residential (300 units).

Zoning detail (Table 1). The design and typologies the current application remain the same to those previously authorised.

**Table 1.** Proposed development zoning information

| Zoning                                                                                                                    | No. Erven  | Area in m <sup>2</sup>         | % Allo         |
|---------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------|----------------|
|  General Res Zone 4                      | 1          | 24236.23 m <sup>2</sup>        | 5.54%          |
|  Single Res Zone 1 (LCH)                 | 509        | 81629.13 m <sup>2</sup>        | 18.66%         |
|  Single Res Zone 1 (Gap)                 | 143        | 29478.47 m <sup>2</sup>        | 6.74%          |
|  Business Zone 1                         | 1          | 11763.94 m <sup>2</sup>        | 2.69%          |
|  Community Zone 2                        | 1          | 15598.56 m <sup>2</sup>        | 3.57%          |
|  Authority Zone                          | 2          | 3659.82 m <sup>2</sup>         | 0.84%          |
|  Open Space Zone 1                       | 11         | 128761.40 m <sup>2</sup>       | 29.44%         |
|  Open Space Zone 3 (Conservation Usage) | 13         | 76015.22 m <sup>2</sup>        | 17.38%         |
|  Transport Zone 1                      | -          | 66130.23 m <sup>2</sup>        | 15.15%         |
| <b>Total</b>                                                                                                              | <b>681</b> | <b>437273.00 m<sup>2</sup></b> | <b>100.00%</b> |

### **Infrastructure**

No internal infrastructure is currently in place on site, however electricity, sewer and water will be connected onto existing bulk infrastructure currently servicing the surrounding areas. Confirmations from the Applicant has been received that the existing services (water, sewerage and electricity) does have sufficient capacity to accommodate the proposed development. A detailed bulk services report is nearing completion to ascertain the letter.

- **Access:**

Two access roads to the site will be provided. The access roads will be 6 m wide with a 16 m road reserve. Access to the southern portion of the site will be obtained from Victoria Drive. The second access road is proposed for the northern section of the site, from an existing road linking up to Beethoven Road.

**There will be no access from the southern portion of the site linking to the northern portion of the site except for pedestrian access alongside the business development zone portion of the site.**

This will allow pedestrians, in particular school children, to access the northern portion of the site.

Internal roads will be provided which are appropriately sized to accommodate municipal service vehicles (e.g. waste collection) and will include surface and sub-surface stormwater drainage capacity. All roads will be surfaced, and the majority will have a 12 m road reserve, except the main roads which will have a 16 m road reserve. Internal water reticulation pipelines will be situated in the road reserves.

- **Water:**

Bulk water supply will be from the Emerald Hill Reservoir via a new 450 mm diameter gravity main pipeline (4,620 metre length) running within the servitude of the existing airport reservoir feeder pipeline which runs along Victoria Drive.

- **Sewerage:**

A full waterborne sewage system will be constructed with separate connections to each erf. The sewer pipelines will be accommodated in the roads reserves and sewage discharge from the site will be from two discharge points — an existing 225 mm diameter pipeline situated north-east of the site in the golf course, and a new 250 mm diameter pipeline running south-east of the site along Victoria Drive. This pipeline has received an Environmental Authorisation as part of a separate assessment process.

- **Stormwater:**

The 1:5-year recurrence interval minor stormwater drainage will be controlled by the use of stormwater pipelines, while the 1:50 year recurrence interval major drainage will make use of open channels, detention ponds and the road system. Stormwater will be controlled during construction via temporary ponds and berms.

Three on-site stormwater detention ponds will be provided, ranging in capacity from 1331 to 7598 m<sup>2</sup>, which will be fenced off to prevent access by the public. The ponds and surrounding public open spaces will remain vegetated and will be monitored for obstructions.

The ponds will have 525 mm diameter concrete pipeline outlets. These will convey water via pipelines to the edge of the site and from there in a concrete block lined channel which can be vegetated, into the new detention pond situated in the southern portion of the Walmer Golf Course. This will attenuate flow of stormwater from Erf 11305.

- **Electricity:**

The NMBM will supply the required electrical reticulation for the development from existing powerlines in the area.

- **Solid Waste:**

The NMBM will provide a weekly refuse removal service. The waste will be disposed of at the existing licensed waste disposal site at either Arlington.

# DRAFT ENVIRONMENTAL IMPACT REPORT: WALMER HOUSING DEVELOPMENT ON ERF 11305



Figure 1. Proposed project layout





Figure 2. Locality map

## 1.2.PROPERTY INFORMATION

The property details of the development are provided in Table 2 below:

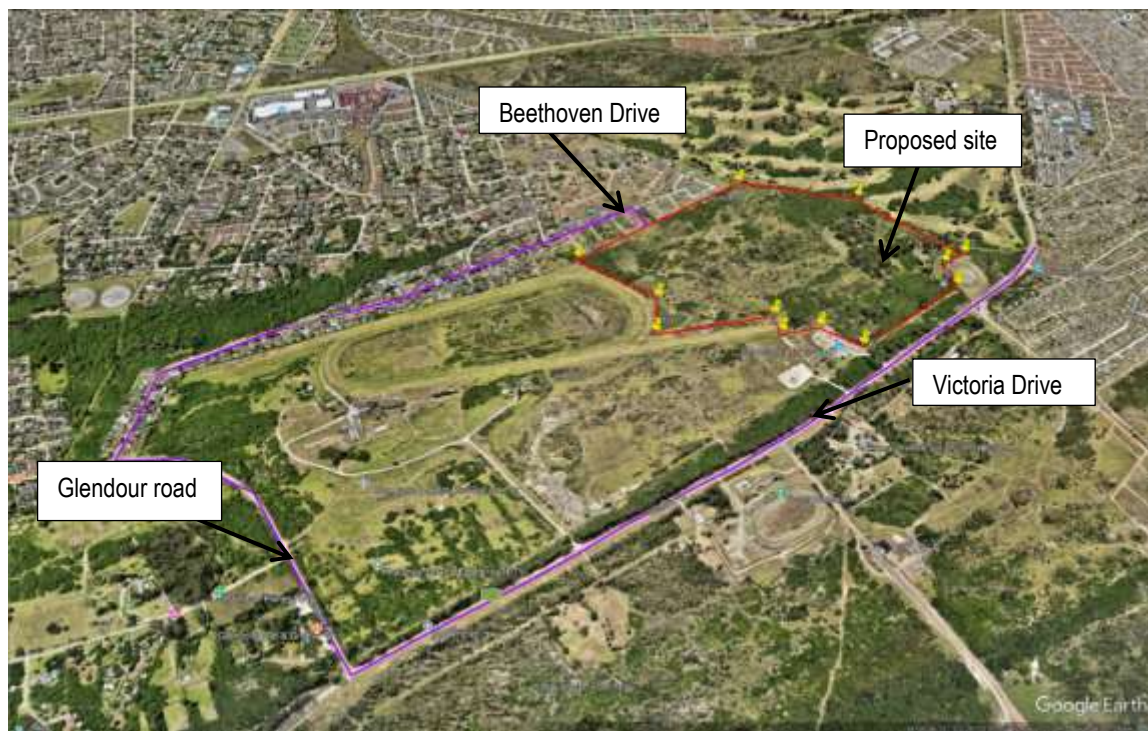
Table 2: Summary of property and technical information

| PROJECT ASPECT             | DESCRIPTION                     |
|----------------------------|---------------------------------|
| District                   | Nelson Mandela Bay Municipality |
| Local Municipality         | Nelson Mandela Bay Municipality |
| Registration Division      | Port Elizabeth                  |
| Wards                      | 1                               |
| Development footprint size | 43.73 ha                        |

| Coordinates                    | Latitude (S)        | Longitude (E) |
|--------------------------------|---------------------|---------------|
|                                | 33° 59'42.69"       | 25° 34'9.21"  |
|                                | 33° 59'45.02"       | 25° 34'22.98" |
|                                | 33° 59'52.52"       | 25° 34'33.44" |
|                                | 33° 59'53.75"       | 25° 34'30.98" |
|                                | 33° 59'56.72"       | 25° 34'31.46" |
|                                | 34° 0'3.50"         | 25° 34'19.84" |
|                                | 34° 0'1.39"         | 25° 34'15.70" |
|                                | 34° 0'2.10"         | 25° 34'11.54" |
|                                | 34° 0'0.03"         | 25° 34'11.03" |
|                                | 34° 0'2.20"         | 25° 33'58.14" |
|                                | 33° 59'58.03"       | 25° 33'58.67" |
|                                | 33° 59'53.31"       | 25° 31'49.3"  |
| Erf number                     | Erf 11305           |               |
| Surveyor General 21 Digit Code | C059003800130500000 |               |
| Nearest City                   | Port Elizabeth      |               |

### 1.3.SITE ACCESS

The site can be accessed from two access routes, from Victoria Drive (see **Figure 4**) which is on the southern section of the site. The second access route, from an existing road (Glendour road) linking up to Beethoven Drive (**Figure 5**) on the northern site of the site.



**Figure 3.** Access to the proposed site



**Figure 4.** Access to site from Victoria Drive





**Figure 5.** Access to site from Beethoven drive

#### 1.4.THE APPLICANT/PROPONENT

Details of the applicant are presented in **Table 3** below.

**Table 3:** Applicant details

|                                              |                                                                                                                                                              |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Applicant</b>                             | Nelson Mandela Bay Municipality                                                                                                                              |
| <b>Contact Person on behalf of Applicant</b> | Mr Tabiso Mfeya                                                                                                                                              |
| <b>Physical Address</b>                      | 17 <sup>th</sup> Floor, Lillian Diedericks Building, Govan Mbeki Ave, 6001                                                                                   |
| <b>Postal Address</b>                        | 17 <sup>th</sup> Floor, Lillian Diedericks Building, Govan Mbeki Ave, 6001                                                                                   |
| <b>Telephone</b>                             | 041 506 2381                                                                                                                                                 |
| <b>Fax</b>                                   |                                                                                                                                                              |
| <b>Email Address</b>                         | <a href="mailto:tmfeya@mandelametro.gov.za">tmfeya@mandelametro.gov.za</a><br><a href="mailto:sstemele@mandelametro.gov.za">sstemele@mandelametro.gov.za</a> |

### 1.5.EAP & CONTACT DETAILS

As per the requirements of the NEMA Regulations and in terms of Regulation 14 of Section 24H Registration Authority Regulations, 2016, as amended, the details and expertise levels of the EAP who prepared the report are provided in **Table 4** below.

**Table 4:** Environmental Assessment Practitioner details

| Contact Details      |                                                |
|----------------------|------------------------------------------------|
| Consultant           | Abantu Environmental Services (Pty) Ltd        |
| EAP                  | Sive Mlamla                                    |
| SACNASP Registration | 118495 – Environmental Sciences                |
| EAPASA Registration  | 2022/5204                                      |
| Cell                 | 078 207 8278                                   |
| Postal Address       | 41 Avon Place<br>Overbaakens<br>Gqeberha, 6070 |
| Fax                  | 086 685 9536                                   |
| Email                | info@abantuenvironmental.co.za                 |
| Website              | Website: www.abantuenvironmental.co.za         |

Expertise of the Lead EAP are entailed in **Table 5**.

**Table 5:** Lead EAP Profile

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Environmental Assessment Practitioner:</b> | Mr Sive Mlamla                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Expertise:</b>                             | <p>Sive Mlamla is an environmental scientist with 11 years of experience in scientific research, lecturing and environmental consulting. His key expertise includes environmental impact assessments, environmental compliance auditing, waste planning, proposal &amp; business development, project management, strategic planning, research design &amp; execution, remote sensing &amp; GIS applications for land use/cover change mapping and fluvial geomorphology.</p> <p><b>Qualifications</b></p> <ul style="list-style-type: none"> <li>• M.Sc. Geography (Catchment Hydrology)</li> <li>• B.Sc. Honours Environmental Geography</li> <li>• B.Sc. Environmental Sciences (Botany and Geography)</li> </ul> <p><b>Professional registrations:</b></p> <ul style="list-style-type: none"> <li>• Registered Professional Natural Scientist (Pr.Sci.Nat) - Reg. No. 118495 – SACNASP</li> <li>• Registered Environmental Assessment Practitioner (Reg. EAP (EAPASA)) - Reg. No. 2022/5204</li> </ul> <p><b>Training:</b></p> <ul style="list-style-type: none"> <li>• Practical Implementation of Environmental Auditing and Monitoring (2025)</li> <li>• WET-Health Tool &amp; WET-EcoServices for Level 1B and Level 2 assessment (2025)</li> </ul> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• WET-Health Tool for Level 1A assessment (2025)</li> <li>• Environmental Impact Assessments (2025)</li> <li>• Introduction to Synthetic Aperture Radar and Applications (2024)</li> <li>• Earth Observation Data Analysis with Google Earth Engine (2023)</li> <li>• People Management Skills (2017)</li> <li>• Biodiversity spatial data management, acquisition, and analysis (2018)</li> <li>• Water Use Licence Applications via e-WULAAS (2020)</li> <li>• Remote Sensing (land use/cover classification) and SWAT hydrologic modelling (2017)</li> </ul> <p><b>Registrations and memberships:</b></p> <ul style="list-style-type: none"> <li>• Member of the South African Wetlands Society Reg. No. S9BPP477</li> <li>• Member of the Southern African Association of Geomorphologists</li> <li>• Institute of Waste Management of Southern Africa (IWMSA) Reg No. 40120001</li> </ul> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

As per the requirements of the NEMA Regulations and Regulation 14(6) in the S24H Registration Authority Regulations, 2016, as amended, details of the Candidate EAP who assisted in the compilation of the report under the supervision of the EAP are provided in **Table 6** below.

**Table 6:** Candidate Environmental Assessment Practitioner details

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Candidate Environmental Assessment Practitioner:</b> | Ms Zimkita Dalasile                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Expertise:</b>                                       | <p>Junior Environmental Scientist with 4 years' experience in environmental consulting and scientific research. My expertise includes environmental management, water resource management, water quality management, aquatic ecosystems, environmental impact assessments, scientific report writing, data analysis, environmental monitoring and environmental compliance audits.</p> <p><b>Qualifications</b></p> <ul style="list-style-type: none"> <li>• MSc. Environmental Management with Ecological Water Requirements</li> <li>• Postgraduate Diploma Integrated Water Resource Management</li> <li>• BSc. Environmental and Water Science</li> </ul> <p><b>Professional registration:</b></p> <ul style="list-style-type: none"> <li>• Candidate Environmental Assessment Practitioner (Cand. EAP (EAPASA)) - Reg. No. 2025/20366</li> <li>• Candidate Natural Scientist (Cand.Sci.Nat.) - Reg. No. 132501 – SACNASP</li> </ul> <p><b>Training:</b></p> <ul style="list-style-type: none"> <li>• Introduction Environmental Impact Assessments (2024)</li> </ul> |

## 1.6.ENVIRONMENTAL TEAM AND SPECIALISTS

Details of the project team, including the appointed independent specialists are presented in **Table 7** below.

**Table 7:** Project team

| ROLE                                                                                                          | PERSONNEL              | QUALIFICATION                                                   | PROFESSIONAL REGISTRATION                                                                                           | EXPERIENCE YEARS |
|---------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Environmental Assessment Practitioners</b>                                                                 |                        |                                                                 |                                                                                                                     |                  |
| <b>Environmental Assessment Practitioner (EAP)</b>                                                            | Mr. Sive Mlamla        | MSc Geography                                                   | <ul style="list-style-type: none"> <li>Pr.Sci.Nat Reg No. 118495</li> <li>Reg EAP Reg No. 2022/5204</li> </ul>      | 10               |
| <b>Assistant EAP</b>                                                                                          | Ms Zimkita Dalasile    | MSc Environmental Management with Ecological Water Requirements | <ul style="list-style-type: none"> <li>Cand.Sci.Nat. Reg No. 132501</li> <li>Cand EAP Reg No. 2025/20366</li> </ul> | 4                |
| <b>Environmental Specialists</b>                                                                              |                        |                                                                 |                                                                                                                     |                  |
| <b>Biodiversity Specialist</b> <ul style="list-style-type: none"> <li>Terrestrial</li> <li>Aquatic</li> </ul> | Mr Roy de Kock         | MSc Botany                                                      | <ul style="list-style-type: none"> <li>Pr.Sci.Nat Reg No. 400216/16</li> </ul>                                      | 19               |
| <b>Agricultural Specialist</b>                                                                                | Mr Roy de Kock         | MSc Botany                                                      | <ul style="list-style-type: none"> <li>Pr.Sci.Nat Reg No. 400216/16</li> </ul>                                      | 19               |
| <b>Palaeontology Specialist</b>                                                                               | Mr Silindokuhle Mavuso | MSc Palaeontology                                               | <ul style="list-style-type: none"> <li>Palaeontological Society of South Africa (PSSA)</li> </ul>                   | 12               |
| <b>Heritage Specialist</b>                                                                                    | Ms Celeste Booth       | BSc Hons Archaeology                                            | <ul style="list-style-type: none"> <li>-</li> </ul>                                                                 | 15               |

The Curriculum Vitae (indicating the experience with environmental impact assessment and relevant application processes) of the consultants that is involved in the EIA process and the compilation of this EIR is presented in **APPENDIX C**.

### **1.7.PURPOSE OF THE REPORT**

The National Environment Management Act, 1998 (Act No. 107 of 1998) (NEMA) introduced the environmental impact management regime, in particular the Environmental Impact Assessment (EIA) process. The EIA process is a tool which requires the integration of social, economic and environmental factors in the planning, implementation and evaluation of decisions to ensure that development serves the present and future generations. The EIA is South Africa's key regulatory instrument used to mitigate and/or manage the impacts of new developments and activities that are considered to potentially impact on the right to an environment that is not harmful to health and well-being. It is considered as one of the processes that actively promotes or ensures sustainable development. The EIA process is outlined in the Environmental Impact Assessment Regulations and Listing Notices aimed at implementing Chapter 5 of the NEMA.

The Regulations provide for listing of activities which may not commence without an Environmental Authorisation (EA) and identify the process and reports to be submitted to the Competent Authority (CA) for decision making purposes. The process provides the Proponent with an opportunity to assess the potential environmental impacts of a proposed development as well as provide for identification of mitigation measures to be in place to ensure that environmental impacts are avoided, minimised or mitigated. Key to this process is the public participation element, which is also legislated. It forms the integral part of the EIA process and comments and inputs from the interested and/or affected are taken into consideration by the competent authority when making decisions on applications (DEA, 2018).

The environmental assessment process is used to understand the potential environmental impacts of a development, and to inform environmental decision-making before the development (and more particularly, the listed activities that require an EA under NEMA) is authorised. The information recorded during the EIA process provides the basis for a decision to grant (with or without conditions) or refuse authorization in respect of a given application, and with regard to the authorisation of an application, informs the selection of the most appropriate alternative (DEA, 2019).

In terms of the NEMA EIA Regulations, 2017 as amended, the purpose of the EIA Report is as follows:

- (1) The environmental impact assessment process must be undertaken in line with the approved plan of study for environmental impact assessment.
- (2) The environmental impacts, mitigation and closure outcomes as well as the residual risks of the proposed activity must be set out in the environmental impact assessment report.



## 1.8.OBJECTIVE OF THE ENVIRONMENTAL IMPACT ASSESSMENT PROCESS

The objective of the environmental impact process is to, through a consultative process—

- a) determine the policy and legislative context within which the activity is located and document how the proposed activity complies with and responds to the policy and legislative context;
- b) describe the need and desirability of the proposed activity, including the need and desirability of the activity in the context of the development footprint on the approved site as contemplated in the accepted scoping report;
- c) identify the location of the development footprint within the approved site as contemplated in the accepted scoping report based on an impact and risk assessment process inclusive of cumulative impacts and a ranking process of all the identified development footprint alternatives focusing on the geographical, physical, biological, social, economic, heritage and cultural aspects of the environment;
- d) determine the—
  - i) nature, significance, consequence, extent, duration and probability of the impacts occurring to inform identified preferred alternatives; and
  - ii) degree to which these impacts—
    - (aa) can be reversed;
    - (bb) may cause irreplaceable loss of resources, and
    - (cc) can be avoided, managed or mitigated;
- e) identify the most ideal location for the activity within the development footprint of the approved site as contemplated in the accepted scoping report based on the lowest level of environmental sensitivity identified during the assessment;
- f) identify, assess, and rank the impacts the activity will impose on the development footprint on the approved site as contemplated in the accepted scoping report through the life of the activity;
- g) identify suitable measures to avoid, manage or mitigate identified impacts; and
- h) identify residual risks that need to be managed and monitored.

## 1.9.REPORT STRUCTURE

The EIA Report includes information as required per Appendix 3 of the 2014 NEMA EIA Regulations. The structure of the EIA Report is presented in **Table 8**.

**Table 8:** Environmental Impact Report structure

| APPENDIX 3 | CONTENT AS REQUIRED BY NEMA                                                                                                                                                                                                                                                                                                                                                                                                           | SECTION/CHAPTER                  |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| (1)        | An environmental impact assessment report must contain the information that is necessary for the competent authority to consider and come to a decision on the application and must include—                                                                                                                                                                                                                                          |                                  |
| (a)        | details of—<br>(iii) the EAP who prepared the report; and<br>(iv) the expertise of the EAP, including a curriculum vitae;                                                                                                                                                                                                                                                                                                             | Section 1.5                      |
| (b)        | the location of the development footprint of the activity on the approved site as contemplated in the accepted scoping report, including:<br>(i) the 21-digit Surveyor General code of each cadastral land parcel;<br>(ii) where available, the physical address and farm name; and<br>(iii) where the required information in items (i) and (ii) is not available, the coordinates of the boundary of the property or properties;    | Section 1.2 and 1.3              |
| (c)        | a plan which locates the proposed activity or activities applied for as well as the associated structures and infrastructure at an appropriate scale, or, if it is—<br>(i) a linear activity, a description and coordinates of the corridor in which the proposed activity or activities is to be undertaken;<br>(ii) on land where the property has not been defined, the coordinates within which the activity is to be undertaken; | Section 1.1<br><b>APPENDIX E</b> |
| (d)        | a description of the scope of the proposed activity, including—<br>(i) all listed and specified activities triggered and being applied for; and<br>(ii) a description of the associated structures and infrastructure related to the development;                                                                                                                                                                                     | Section 2.3                      |
| (e)        | a description of the policy and legislative context within which the development is located and an explanation of how the proposed development complies with and responds to the legislation and policy context;                                                                                                                                                                                                                      | Section 2                        |
| (f)        | a motivation for the need and desirability for the proposed development, including the need and desirability of the activity in the context of the preferred development footprint within the approved site as contemplated in the accepted scoping report;                                                                                                                                                                           | Section 3                        |
| (g)        | a motivation for the preferred development footprint within the approved site as contemplated in the accepted scoping report;                                                                                                                                                                                                                                                                                                         | Section 3<br>Section 4           |
| (h)        | a full description of the process followed to reach the proposed development footprint within the approved site as contemplated in the accepted scoping report, including:<br>(i) details of the development footprint alternatives considered;                                                                                                                                                                                       | Section 4                        |

| APPENDIX 3 | CONTENT AS REQUIRED BY NEMA                                                                                                                                                                                                                                                                                         | SECTION/CHAPTER        |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|            | (ii) details of the public participation process undertaken in terms of regulation 41 of the Regulations, including copies of the supporting documents and inputs;                                                                                                                                                  | Section 6 and 6.2      |
|            | (iii) a summary of the issues raised by interested and affected parties, and an indication of the manner in which the issues were incorporated, or the reasons for not including them;                                                                                                                              | Section 6              |
|            | (iv) the environmental attributes associated with the development footprint alternatives focusing on the geographical, physical, biological, social, economic, heritage and cultural aspects;                                                                                                                       | Section 5              |
|            | (v) the impacts and risks identified including the nature, significance, consequence, extent, duration and probability of the impacts, including the degree to which these impacts—<br>(aa) can be reversed;<br>(bb) may cause irreplaceable loss of resources; and<br>(cc) can be avoided, managed or mitigated;   | Section 8 and 8.2      |
|            | (vi) the methodology used in determining and ranking the nature, significance, consequences, extent, duration and probability of potential environmental impacts and risks;                                                                                                                                         | Section 8.1            |
|            | (vii) positive and negative impacts that the proposed activity and alternatives will have on the environment and on the community that may be affected focusing on the geographical, physical, biological, social, economic, heritage and cultural aspects;                                                         | Section 9.2            |
|            | (viii) the possible mitigation measures that could be applied and level of residual risk;                                                                                                                                                                                                                           | Section 8.3            |
|            | (ix) if no alternative development footprints for the activity were investigated, the motivation for not considering such; and                                                                                                                                                                                      | N/A                    |
|            | (x) a concluding statement indicating the location of the preferred alternative development footprint within the approved site as contemplated in the accepted scoping report;                                                                                                                                      | Section 9              |
| (i)        | a full description of the process undertaken to identify, assess and rank the impacts the activity and associated structures and infrastructure will impose on the preferred development footprint on the approved site as contemplated in the accepted scoping report through the life of the activity, including— | Section 4<br>Section 9 |
|            | (i) a description of all environmental issues and risks that were identified during the environmental impact assessment process; and                                                                                                                                                                                | Section 8              |
|            | (ii) an assessment of the significance of each issue and risk and an indication of the extent to which the issue and risk could be avoided or addressed by the adoption of mitigation measures;                                                                                                                     | Section 8.3            |
| (j)        | an assessment of each identified potentially significant impact and risk, including—<br>(i) cumulative impacts;<br>(ii) the nature, significance and consequences of the impact and risk;<br>(iii) the extent and duration of the impact and risk;<br>(iv) the probability of the impact and risk occurring;        | Section 8.2            |

# DRAFT ENVIRONMENTAL IMPACT REPORT: WALMER HOUSING DEVELOPMENT ON ERF 11305

| APPENDIX 3 | CONTENT AS REQUIRED BY NEMA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SECTION/CHAPTER  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|            | (v) the degree to which the impact and risk can be reversed;<br>(vi) the degree to which the impact and risk may cause irreplaceable loss of resources;<br>and<br>(vii) the degree to which the impact and risk can be mitigated;                                                                                                                                                                                                                                                                                                                                                                        |                  |
| (k)        | where applicable, a summary of the findings and recommendations of any specialist report complying with Appendix 6 to these Regulations and an indication as to how these findings and recommendations have been included in the final assessment report;                                                                                                                                                                                                                                                                                                                                                | Section 7        |
| (l)        | an environmental impact statement which contains—<br>(i) a summary of the key findings of the environmental impact assessment;<br>(ii) a map at an appropriate scale which superimposes the proposed activity and its associated structures and infrastructure on the environmental sensitivities of the preferred development footprint on the approved site as contemplated in the accepted scoping report indicating any areas that should be avoided, including buffers; and<br>(iii) a summary of the positive and negative impacts and risks of the proposed activity and identified alternatives; | Section 9        |
| (m)        | based on the assessment, and where applicable, recommendations from specialist reports, the recording of proposed impact management outcomes for the development for inclusion in the EMP as well as for inclusion as conditions of authorisation;                                                                                                                                                                                                                                                                                                                                                       | Section 7 and 12 |
| (n)        | the final proposed alternatives which respond to the impact management measures, avoidance, and mitigation measures identified through the assessment;                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Section 11       |
| (o)        | any aspects which were conditional to the findings of the assessment either by the EAP or specialist which are to be included as conditions of authorisation;                                                                                                                                                                                                                                                                                                                                                                                                                                            | Section 12       |
| (p)        | a description of any assumptions, uncertainties and gaps in knowledge which relate to the assessment and mitigation measures proposed;                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Section 10       |
| (q)        | a reasoned opinion as to whether the proposed activity should or should not be authorised, and if the opinion is that it should be authorised, any conditions that should be made in respect of that authorisation;                                                                                                                                                                                                                                                                                                                                                                                      | Section 11       |
| (r)        | where the proposed activity does not include operational aspects, the period for which the environmental authorisation is required and the date on which the activity will be concluded and the post construction monitoring requirements finalised;                                                                                                                                                                                                                                                                                                                                                     | N/A              |
| (s)        | an undertaking under oath or affirmation by the EAP in relation to—<br>(i) the correctness of the information provided in the reports;<br>(ii) the inclusion of comments and inputs from stakeholders and I&APs;<br>(iii) the inclusion of inputs and recommendations from the specialist reports where relevant; and<br>(iv) any information provided by the EAP to interested and affected parties and any responses by the EAP to comments or inputs made by interested or affected parties;                                                                                                          | Section 13       |

| APPENDIX 3 | CONTENT AS REQUIRED BY NEMA                                                                                                                                                                                                                                                       | SECTION/CHAPTER |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| (t)        | where applicable, details of any financial provision for the rehabilitation, closure, and ongoing post decommissioning management of negative environmental impacts;                                                                                                              | N/A             |
| (u)        | an indication of any deviation from the approved scoping report, including the plan of study, including—<br>(i) any deviation from the methodology used in determining the significance of potential environmental impacts and risks; and<br>(ii) a motivation for the deviation; | N/A             |
| (v)        | any specific information that may be required by the competent authority; and                                                                                                                                                                                                     | N/A             |
| (w)        | any other matters required in terms of section 24(4)(a) and (b) of the Act.                                                                                                                                                                                                       | N/A             |
| (2)        | Where a government notice gazette by the Minister provides for any protocol or minimum information requirement to be applied to an environmental impact assessment report the requirements as indicated in such notice will apply.                                                |                 |

## **2. LEGAL AND POLICY FRAMEWORK**

### **2.1 APPLICABLE ENVIRONMENTAL LEGISLATION**

This section of the report presents an overview of the governing legislation identified which may relate to the proposed project. A summary of the applicable legislation is provided in **Table 9** below. The legal compliance obligation for this project stems from the need for an EA to be granted by the competent authority, DEDEAT, in accordance with the requirements of the NEMA. In addition, there are numerous other pieces of legislation governed by many acts, regulations, standards, guidelines on a national, provincial, and local level, which should be considered in order to assess the potential applicability of these for the proposed project. More detail on the legislative framework is presented below.

Table 9. Applicable legislation

| Legislation/<br>guidelines                                                         | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Legal requirement for this project                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NATIONAL</b>                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>The Constitution of South Africa, 1996 (Act No.108 of 1996)</b>                 | <p>The Constitution is the highest and the supreme law in South Africa. The Bill of Rights in chapter 2 section 24 of the Constitution of South Africa Act (Act 108 of 1996) makes provisions for environmental issues and declares that: "Everyone has the right –</p> <ul style="list-style-type: none"> <li>a) to an environment that is not harmful to their health or well-being; and</li> <li>b) to have the environment protected, for the benefit of present and future</li> <li>c) generations, through reasonable legislative and other measures that: <ul style="list-style-type: none"> <li>i. prevent pollution and ecological degradation;</li> <li>ii. promote conservation; and</li> </ul> </li> </ul> <p>secure ecologically sustainable development and use of natural resources while promoting justifiable economic and social development".</p>                                                                                                                                                                                                           | <p>The applicant has an obligation to ensure that the project is undertaken in a manner that respects and protects the constitutional rights of all interested and affected parties. The applicant must ensure that the project environment is not harmful and that measures are implemented to prevent pollution so that future generations can enjoy the social and ecological benefits.</p>                                                                                           |
| <b>National Environmental Management Act, 1998 (Act No.107 of 1998) as amended</b> | <p>The National Environmental Management Act, 1998 (Act No.107 of 1998) (NEMA) is a 'principles based Act' and is an overarching statute regulating various aspects of natural resources use, integrated environmental management and pollution control. The Act provides for the right to an environment that is not harmful to the health and wellbeing of the South African people; sustainable development, environmental protection, equitable distribution of natural resources; and the formulation of environmental management frameworks. Its definition of the environment includes the land and water of the earth, microorganisms, plant and animal life or a combination of those things, and the inter relationships among them. The Act aims to provide for cooperative environmental governance by establishing principles for decision making on matters affecting the environment, institutions that will promote co-operative governance, and procedures for coordinating environmental functions exercised by organs of state. Section 24 Provides for</p> | <p>The applicant must ensure that construction and operation of activities must be conducted according to the generally accepted principles of sustainable development, integrating social, economic, and environmental factors.</p> <p>An application for Environmental Authorisation is being submitted on behalf of the client in line with the requirements of NEMA since the proposed project will trigger listed activities which require authorization prior to commencement.</p> |

| Legislation/<br>guidelines                                                     | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Legal requirement for this project                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                | the prohibition, restriction and control of activities which are likely to have a detrimental effect on the environment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | As part of the EIA process, mitigation measures will be proposed to ensure that the significance of the predicted impacts is reduced thus protecting the environment from degradation.                                                                                                                                                                                                                                      |
| <b>Environmental Impact Assessment Regulation, 2014 as Amended</b>             | The Environmental Impact Assessment (EIA) Regulations promulgated under NEMA in 2014 provide a list of activities which are subject to an Environmental Authorisation (EA) process prior to construction or implementation. In accordance with the 2014 EIA Regulations, (as amended) an EIA process is required owing to the applicability of the activities listed in <b>Error! Reference source not found..</b> According to the NEMA Regulations these activities may not commence without environmental authorization from the competent authority which requires the investigation, assessment and statement of potential impact of activities and must follow the procedure as described in the EIA Regulations. A Basic Assessment (BA) process must be applied to an application if the authorisation applied for is in respect of an activity or activities listed in Listing Notices 1 and/or 3 and a Scoping and Environmental Impact Assessment (S&EIA) process must be applied to an application if the authorisation applied for is in respect of an activity or activities listed in Listing Notice 2. As the Proposed Project includes activities listed in Listing Notice 2 it is necessary that a S&EIA process is undertaken in order for the Eastern Cape Department of Economic Development, Environmental Affairs and Tourism (DEDEAT) to consider the application in terms of NEMA. | An application for Environmental Authorisation is being submitted on behalf of the client in line with the requirements of NEMA EIA Regulations since the proposed project will trigger listed activities which require authorization prior to commencement                                                                                                                                                                 |
| <b>National Environmental Management Biodiversity Act (Act No. 10 of 2004)</b> | <p>The National Environmental Management: Biodiversity Act (NEM:BA) makes provisions for achieving the objectives of the United Nations Convention on Biological Diversity, to which South Africa is a signatory. The Bill promotes management, conservation and sustainable use of indigenous biological resources, and provides for:</p> <ul style="list-style-type: none"> <li>the management and conservation of biological diversity;</li> <li>the use of indigenous biological resources in a sustainable manner; and</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | The proposed housing development area does not fall within an area identified as a Critical Biodiversity Area (CBA) according to the Eastern Cape Biodiversity Conservation Plan (ECBCP) as well as the NMBMM Bioregional Plan. The proposed development will be within threatened ecosystems as defined by the NEM:BA, namely the critically endangered Algoa Sandstone Fynbos and the vulnerable Sardinia Forest Thicket. |



| Legislation/<br>guidelines                                                          | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Legal requirement for this project                                                                                                                                                                  |
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|                                                                                     | <ul style="list-style-type: none"> <li>the fair and equitable sharing of benefits arising from the commercialization through bio-prospecting of traditional uses and knowledge of generic resources.</li> </ul> <p>The Bill gives effect to international agreements relating to biodiversity which are binding on the Republic and provides for co-operative governance in biodiversity management and conservation and provides for a National Biodiversity Institute to assist in achieving the above objectives. The Act gives wide powers to the National Biodiversity Institute to inter alia protect flora and fauna in appropriate enclosures, the collection of information, undertaking and promotion of research on indigenous biodiversity and the sustainable use of indigenous biological resources, the prevention, control or eradication of listed invasive species, biodiversity planning and other functions. This act lists all critically endangered, vulnerable and protected species. The potential occurrence of any such species will be investigated in the BA process.</p> |                                                                                                                                                                                                     |
| <b>National Water Act, 1998<br/>(Act No.36 of 1998)</b>                             | <p>In terms of chapter 3 section 12-20, water resources are to be protected, used, developed, conserved, managed and controlled. This Act recognizes that water is a scarce resource; it is a natural resource that belongs to all of South Africa's people. The National Department of Water and Sanitation is responsible for the nation's water resource and also the Minister of Department of Water and Sanitation ensures that the water resource is "protected, used, developed, conserved, managed and controlled" through the implementation of this Act (National Water Act 36 of 1998).</p> <p>This Act makes provisions for the protection of surface water and groundwater resources and their sustainable management for the prevention and remediation of the effects of pollution, and for the control of emergency occurrences. Section 21 of the National Water Act (NWA) lists water uses for which a Water Use Licence will be required.</p>                                                                                                                                      | <p>The proposed project is not located within 100m of a river or within 500m of a wetland therefore, Section 21 c and i of the National Water Act will not be triggered.</p>                        |
| <b>National Environmental Management: Air Quality Act, 2004 (Act No.39 of 2004)</b> | <p>The objective of the Act is to protect the environment by providing reasonable measures for the protection and enhancement of air quality and to prevent air pollution. The Act makes provision for measures to control dust, and offensive odours. Section 32 of The National Environmental Management: Air Quality</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>The EMPr which will be compiled during the EIA phase must include measures for control of dust during the construction phase, If there are any exceedances observed in terms of the National</p> |

| Legislation/<br>guidelines                                                         | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Legal requirement for this project                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                    | Act, 2004 (Act 39 of 2004) deals with dust control measures regarding dust control. The Minister or MEC may prescribe measures for the control of dust in specified places or areas, either in general or by specified machinery or in specified instances, the steps to be taken to prevent nuisance or other measures aimed at controlling dust. The National Dust Control Regulations (2013) provides for the management and monitoring of dust.                                                                                                                                                                         | Dust Regulations then a dust monitoring programme must be submitted to the Competent Authority.                                                                                                                                                                                                                                                                                                                                             |
| <b>Conservation of Agricultural Resources Act, 1983 (No. 43 of 1983)</b>           | This Act provides for the control over the utilisation of the natural agricultural resources of the country in order to promote the conservation of the soil, the water sources and the vegetation and the combating of weeds and invader plants. Section 5 of the Act prohibits the spread of weeds through the prohibition of their sale. GN R1084 (published under CARA) provides categories for the classification of the various weeds and invader plants, and restrictions where these species may occur. Regulation 15E of GN R1084 provides methods to be implemented for the control of weeds and invader species. | CARA finds application throughout the project lifecycle of the Proposed Project. As a result, soil conservation and erosion prevention management and mitigation measures need to be implemented. Furthermore, an Alien Invasives Species Control and Management Plan must be developed and implemented for the duration of the project life cycle of the Proposed Project                                                                  |
| <b>Subdivision of Agricultural Land Act, 1970 (Act No. 70 of 1970), as amended</b> | The Subdivision of Agricultural Land Act, 1970 (Act No. 70 of 1970), as amended provides for the subdivision of all agricultural land within the Republic thereby prohibiting certain activities from being undertaken without consent from relevant authority, the Minister of the Department of Agriculture, Land Reform and Rural Development.                                                                                                                                                                                                                                                                           | This Act could be relevant to the Proposed Project if any portion of land is zoned for agriculture and will need to be leased for a period exceeding 10 years is regulated by the Act                                                                                                                                                                                                                                                       |
| <b>National Forests Act, 1998 (No. 84 of 1998)</b>                                 | The National Forest Act (NFA) empowers the Minister of DFFE to declare and list a tree, group of trees, woodland, or a species of trees as protected. A list of protected tree species is included in GN R908, published in November 2014. Section 7 of the Act prohibits the cutting and disturbance of NFA-listed trees.                                                                                                                                                                                                                                                                                                  | A permit may be required for the removal of NFA-listed tree species in terms of Section 4 of the Act, if any will be impacted. Prior to the submission of the permit application to the competent authority, a survey of the project site is required in order to ascertain the presence and distribution of NFA-listed tree species.<br>There are forest patches within the development area, these need to be avoided as far as possible. |
| <b>Occupational Health and Safety Act, 1993 (Act No.85 of 1993)</b>                | The Occupational Health and Safety Act makes provisions in regulations Section 8 for the general duties of employers to their employees. The act                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | The applicant must ensure that a safe working environment is provided for its employees during construction and operational phases of the project. This includes obtaining the relevant work permits,                                                                                                                                                                                                                                       |

| Legislation/<br>guidelines                                                    | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Legal requirement for this project                                                                                                                                                                                                                                                                             |
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|                                                                               | <p>provides for the health and safety of people at work utilising machinery and the protection of others against health and safety risks associated with activities on site/work. General Administrative Regulations (2003) describe the administration of the various OHS Regulations, including the designation of health and safety committees, the reporting and recording of incidents and occupational diseases. This Act is applicable to all contractors during the planning, construction and operational phases of the project.</p> <p>To provide for the health and safety of persons at work and for the health and safety of persons in connection with the use of plant and machinery; the protection of persons other than persons at work against hazards to health and safety arising out of or in connection with the activities of persons at work.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>providing PPE and ensuring all required facilities are available for a working environment that is conducive. All stalls must have adequate training for their various duties and the applicant must ensure that compliance with the OHSA and Construction Regulations is monitored on a regular basis.</p> |
| <b>Hazardous Substance Act (No 15 of 1973)</b>                                | <p>This Act regulates the control of substances that may cause injury, or ill health, or death due to their toxic, corrosive, irritant, strongly sensitizing or inflammable nature of the generation of pressure thereby in certain instances and for the control of certain electronic products. To provide for the rating of such substances or products about the degree of danger; to provide for the prohibition and control of the importation, manufacture, sale, use, operation, modification, disposal or dumping of such substances and products. • Group I and II: Any substance or mixture of a substance that might by reason of its toxic, corrosive, etc., nature or because it generates pressure through decomposition, heat, or other means, cause extreme risk of injury etc., can be declared as Group I or Group II substance • Group IV: any electronic product; and • Group V: any radioactive material. The use, conveyance, or storage of any hazardous substance (such as distillate fuel) is prohibited without an appropriate license being in force. It is necessary to identify and list all the Group I, II, III, and IV hazardous substances that may be on the site and in what operational context they are used, stored, or handled. If applicable, a license is required to be obtained from the</p> | <p>Relevant permits must be obtained for the storage of hazardous substances if any will be stored on site during construction. The contractor must ensure that hazardous substances are stored in a safe manner and MSDS are retained on file for all hazardous substances on site.</p>                       |
| <b>National Environmental Management: Waste Act, 2008 (Act No.59 of 2008)</b> | <p>During construction waste will be produced, in either liquid, solid and/or hazardous state, and this waste will be required to be adequately and appropriately disposed of. There are several Regulations or Acts that are applicable to the proposed development in terms of waste management. To</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>No authorization is required in terms of NEMWA, however, the applicant must make sure that waste is managed appropriately on site. This includes separation of waste, routine clean-up of the site and spillages as well as disposal at appropriately licensed waste landfills. Where possible, waste</p>   |

| Legislation/<br>guidelines                                         | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Legal requirement for this project                                                                                                                                                                                                                                                                                                                                                      |
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|                                                                    | <p>reform the law regulating waste management in order to protect health and the environment by providing reasonable measures for the prevention of pollution and ecological degradation and for securing ecologically sustainable development; to provide for institutional arrangements and planning matters; to provide for national norms and standards for regulating the management of waste by all spheres of government; to provide for specific waste management measures; to provide for the licensing and control of waste management activities; to provide for the remediation of contaminated land; to provide for the national waste information system; to provide for compliance and enforcement; and to provide for matters connected therewith.</p>                                                                                                                                                                                                                                                                                                                         | <p>should be recycled to minimizes volumes of waste disposed to landfills</p>                                                                                                                                                                                                                                                                                                           |
| <p><b>National Heritage Resources Act (Act No. 25 of 1999)</b></p> | <p>The protection of archaeological and paleontological resources is the responsibility of a provincial heritage resources authority and all archaeological objects, paleontological material and meteorites are the property of the State. "Any person who discovers archaeological or paleontological objects or material or a meteorite in the course of development must immediately report the find to the responsible heritage resources authority, or to the nearest local authority offices or museum, which must immediately notify such heritage resources authority". According to Section 34 of NHRA, No person may alter or demolish any structure or part of a structure which is older than 60 years without a permit issued by the relevant provincial heritage resources authority.</p> <p>Section 38 Listed Activities:</p> <p>(a) the construction of a road, wall, power line, pipeline, canal or other similar form of linear development or barrier exceeding 300m in length;</p> <p>(b) the construction of a bridge or similar structure exceeding 50 m in length;</p> | <p>The site is located in close proximity to a grade IIIb heritage site. A Heritage specialist has been appointed to assess the presence of other heritage resources within the site as well as the significance thereof. The ECPHRA has been identified as a stakeholder in this project and will be provided an opportunity to comment of the findings of the specialist studies.</p> |

| Legislation/<br>guidelines                                                             | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Legal requirement for this project                                                                                                                                                                                             |
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|                                                                                        | <p>(c) any development or other activity which will change the character of a site—</p> <p>(i) exceeding 5 000 m<sup>2</sup> in extent; or</p> <p>(ii) involving three or more existing erven or subdivisions thereof; or</p> <p>(iii) involving three or more erven or divisions thereof which have been consolidated within the past five years; or</p> <p>(iv) the costs of which will exceed a sum set in terms of regulations by SAHRA or a provincial heritage resources authority;</p> <p>(d) the re-zoning of a site exceeding 10 000 m<sup>2</sup> in extent; or</p> <p>(e) any other category of development provided for in regulations by SAHRA or a provincial heritage resources authority,</p> |                                                                                                                                                                                                                                |
| <b>The Spatial Planning and Land Use Management Act, 2013 (No. 6 of 2013) (SPLUMA)</b> | The Spatial Planning and Land Use Management Act, 2013 (Act No. 6 of 2013) aims to confirm and regulate the role of municipalities in land use planning and management.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Objectives of the Act relevant to the Proposed Project ensure that the system of spatial planning and land use management promotes social and economic inclusion and to provide for the sustainable and efficient use of land. |
| <b>Housing Act, 1997 (Act No. 107 of 1997)</b>                                         | <p>The Act recognises the Constitutional right to access to adequate housing and clarifies the State's response to this right by setting out the legal plan for the sustainable development of housing. It defines the national, provincial, and local government functions concerning housing development and the financing of housing programmes. According to part 4 Section 9 of the act, every municipality must as part of the municipality's process of integrated development planning take all reasonable and necessary steps within the framework of national and provincial housing legislation and policy to</p> <p>(a) ensure that-</p>                                                          | The proposed housing development is in accordance with this Act. The NMBM shall provide residents of Walmer with adequate housing and services such as electricity, water, sanitation etc.                                     |

| Legislation/<br>guidelines                                                                | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Legal requirement for this project                                                                                                                 |
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|                                                                                           | <p>(i) the inhabitants of its area of jurisdiction have access to adequate housing on a progressive basis;</p> <p>(ii) conditions not conducive to the health and safety of the inhabitants of its area of jurisdiction are prevented or removed;</p> <p>(iii) services in respect of water, sanitation, electricity, roads, stormwater drainage and transport are provided in a manner which is economically efficient;</p> <p>(b) set housing delivery goals in respect of its area of jurisdiction;</p> <p>(c) identify and designate land for housing development;</p> <p>(d) create and maintain a public environment conducive to housing development which is financially and socially viable;</p> <p>(e) promote the resolution of conflicts arising in the housing development process;</p> <p>(f) initiate plan, co-ordinate, facilitate, promote and enable appropriate housing development in its area of jurisdiction;</p> <p>(g) provide bulk engineering services, and revenue generating services in so far as such services are not provided by specialist utility suppliers; and</p> <p>(h) plan and manage land use and development</p> |                                                                                                                                                    |
| <b>Integrated<br/>Environmental<br/>Management<br/>Information Guidelines<br/>Series:</b> | <p>This series of guidelines was published by the Department of Environmental Affairs (DEA) and refers to various environmental aspects. Applicable guidelines in the series for the proposed project include:</p> <ul style="list-style-type: none"> <li>● Guideline 5: Companion to NEMA EIA Regulations, 2010;</li> <li>● Guideline 7: Public participation; and</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>These guidelines have been consulted in the compilation of this report as well as the public participation process that will be undertaken.</p> |

| Legislation/<br>guidelines                         | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Legal requirement for this project                                                                                             |
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|                                                    | <ul style="list-style-type: none"> <li>• Guideline 9: Need and desirability. Additional guidelines published in terms of the NEMA EIA Regulations, 2014 (as amended), in particular:</li> <li>• Guideline 3: General Guide to EIA Regulations, 2006;</li> <li>• Guideline 4: Public Participation in support of the EIA Regulations, 2006; and</li> </ul> <p>Guideline 5: Assessment of alternatives and impacts in support of the EIA Regulations, 2006.</p>                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                |
| <b>Municipal Systems Act (Act 32 of 2000)</b>      | <p>The Municipal Systems Act provides for the core principles, mechanisms and processes that are necessary to enable municipalities to provide for community participation and for the integration of all activities for the overall social and economic upliftment of communities in harmony with their local natural environment. It also states that a fundamental aspect of the new local government system is the active engagement of communities in the affairs of municipalities of which they are an integral part.</p> <p>The Act requires the implementation and monitoring of Integrated Development Plans, the setting of targets and key performance indicators, including environmental targets, as well as the preparation of by-laws and policies that deal with environmental issues.</p> | The NMBM has been included as an I&AP for this project and the municipal IDP has been consulted in compilation of this report. |
| <b>PROVINCIAL LEGISLATION AND POLICY FRAMEWORK</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                |

| Legislation/<br>guidelines                                             | Summary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Legal requirement for this project                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Eastern Cape Environmental Management Bill (Notice 205 of 2019)</b> | <p>The goals of the Eastern Cape Environmental Management Bill are as follow:</p> <ul style="list-style-type: none"> <li>• To supplement national legislation in the Province where necessary to protect the environment by providing reasonable measures for – <ul style="list-style-type: none"> <li>○ The management, protection and conservation of certain areas of ecological or environmental importance;</li> <li>○ The promotion of the sustainable utilisation of the areas of ecological or environmental importance;</li> <li>○ The management, protection and conservation of biological diversity and of the components of such biodiversity; and</li> <li>○ The use of indigenous biological resources in a sustainable manner;</li> </ul> </li> <li>• To provide for sound environmental management, enhancing and encouraging sustainable use of resources;</li> <li>• To encourage conservation, a risk averse approach;</li> <li>• To set provincial requirements, norms and standards for provincial environmental management according to best practices; and</li> <li>• To provide for compliance with the measures set out in paragraph (a); and</li> <li>• Generally, to give effect to section 24 of the Constitution within the Province.</li> </ul> | The applicant must ensure that the project design, construction, and operation apply a precautionary, risk-averse approach, prioritise biodiversity protection, and demonstrate sustainable use of natural resources. The project must comply with any provincial environmental norms and standards and secure necessary provincial inputs/approvals in addition to national authorisations. |
| <b>Eastern Cape Biodiversity Conservation Plan (BCP) (2019)</b>        | <p>The primary intention of NEMBA Chapter 3 is to facilitate conservation and management of biodiversity in “biodiversity priority areas” or priority areas for conservation, outside of the Protected Area network, at a landscape level. The BCP may be used as a precursor for further refined mapping in Bioregional or Biodiversity Sector Plans undertaken at more localised scales (e.g. District or Local Municipalities). Therefore, the purpose of the BCP is to provide a map of these important biodiversity areas and develop associated land use management guidelines to inform:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | The applicant must screen the development against CBAs/ESAs and avoid development in no-go biodiversity areas. Where avoidance is not feasible, impacts must be minimized and offset according to BCP guidelines. Rehabilitation and restoration plans must be implemented to maintain ecological connectivity.                                                                              |
| <b>MUNICIPAL</b>                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Nelson Mandela Bay Bioregional Plan (2014)</b>                      | <p>The purpose of the final Bioregional Plan is to provide a map of biodiversity priorities and accompanying guidelines to inform land-use planning, environmental assessment and authorisations, and natural resource management by a range of sectors whose policies and decisions impact on biodiversity. The Bioregional Plan is a spatial plan that shows terrestrial and aquatic features that are critical for conserving biodiversity and maintaining ecosystem functioning. These areas are referred to as Critical Biodiversity Areas (CBAs).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The NMBM was used to screen the proposed development against CBAs and ESAs to identify biodiversity constraints.                                                                                                                                                                                                                                                                             |



| Legislation/<br>guidelines                                                       | Summary                                                                                                                                                                                                                                                                                                                                                   | Legal requirement for this project                                                                                                                                                      |
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| <b>Nelson Mandela Bay Municipality Municipal Spatial Development Plan (2025)</b> | The strategic framework for the NMBM SDF provides an overarching guide for developing a long-term vision and actions to manage spatial development, balancing economic, social, and environmental goals through a coordinated process. It serves as a strategic plan that establishes the basis for more detailed spatial plans and their implementation. | The municipal SDF was consulted to ensure that the proposed development aligns with the spatial development objectives, land use guidelines and development vision of the municipality. |

## 2.2 SOUTH AFRICAN GUIDELINES

The guidelines listed in the table below have been or will be considered during the EIA process. See **Table 10** below.

**Table 10:** Guidelines Considered in the EIA Process

| Guideline                                                                                                                                                                                                                                          | Governing Body | Relevance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Procedures for the Assessment and Minimum Criteria for reporting on identified environmental themes in terms of sections 24(5)(a) and (h) and 44 of the National Environmental Management Act, 1998, when applying for Environmental Authorisation | DFFE           | Provides procedures for ground-truthing environmental themes identified by the web-based Screening Tool. The baselines studies used in the Scoping Report have been undertaken in accordance with the requirements of this Notice, where relevant.                                                                                                                                                                                                                                           |
| Public Participation in terms of NEMA, EIA Regulations (2017)                                                                                                                                                                                      | DFFE           | The purpose of this guideline is to ensure that an adequate public participation process is undertaken during the S&EIA process.                                                                                                                                                                                                                                                                                                                                                             |
| Guideline on need and desirability in terms of the EIA Regulations (2014)                                                                                                                                                                          | DFFE           | These guidelines inform the consideration of the need and desirability aspects of the Proposed Project.                                                                                                                                                                                                                                                                                                                                                                                      |
| National Biodiversity Offset Guidelines (2023)                                                                                                                                                                                                     | DFFE           | These guidelines have been published to inform the determination of the measurable outcome of compliance with a formal requirement contained in an environmental authorisation to implement an intervention that has the purpose of counterbalancing the residual negative impacts of an activity, or activities, on biodiversity, through increased protection and appropriate management, after every effort has been made to avoid and minimise impacts, and rehabilitate affected areas; |

## 2.3 LISTED ACTIVITIES TRIGGERED

Table 11 below provides listed activities from the EIA Regulations, 2014 as amended, that are triggered by the proposed housing development.

**Table 11:** Listed Activities Applied For

| Listed activity as described in GN R.324, 325, and 327                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Description of project activity that triggers listed activity – if activities in GN R. 324 are triggered, indicate the triggering criteria as described in the second column of GN R. 324                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Listed activity as described in GN R. 324                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                              |
| <p><b>GNR 324 (Listing Notice 3): Activity 4</b><br/>The development of a road wider than 4 metres with a reserve less than 13,5 metres.</p> <p><b>a. Eastern Cape</b></p> <p><b>ii. Inside urban areas:</b></p> <p>(aa) Areas zoned for use as public open space;</p> <p>(bb) Areas designated for conservation use in Spatial Development Frameworks adopted by the competent authority or zoned for a conservation purpose; or</p> <p>(cc) Seawards of the development setback line or within urban protected areas.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Development of surfaced access roads that are 6 metres wide with a majority having 12 metre road reserves, except the main roads which will have a 16-metre road reserve on land zoned as Open Space.</p> |
| <p><b>GNR 324 (Listing Notice 3) Activity 12 (a)(i)(v)</b></p> <p>The clearance of an area of 300 square metres or more of indigenous vegetation except where such clearance of indigenous vegetation is required for maintenance purposes undertaken in accordance with a maintenance management plan.</p> <p><b>a. Eastern Cape</b></p> <p>i) Within any critically endangered or endangered ecosystem listed in terms of section 52 of the NEMBA or prior to the publication of such a list, within an area that has been identified as critically endangered in the National Spatial Biodiversity Assessment 2004;</p> <p>ii. Within critical biodiversity areas identified in bioregional plans;</p> <p>iii. Within the littoral active zone or 100 metres inland from the high water mark of the sea, whichever distance is the greater, excluding where such removal will occur behind the development setback line on erven in urban areas;</p> <p>iv. Outside urban areas, within 100 metres inland from an estuarine functional zone; or</p> | <p>The proposed project will require the clearance of 43.73 hectares of indigenous vegetation for the development of residential houses.</p>                                                                 |

| Listed activity as described in GN R.324, 325, and 327                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Description of project activity that triggers listed activity – if activities in GN R. 324 are triggered, indicate the triggering criteria as described in the second column of GN R. 324 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v. On land, where, at the time of the coming into effect of this Notice or thereafter such land was zoned open space, conservation or had an equivalent zoning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                           |
| <p><b>GNR 324 (Listing Notice 3) Activity 15 (a)(ii)(aa):</b></p> <p>The transformation of land bigger than 1000 square metres in size, to residential, retail, commercial, industrial or institutional use, where, such land was zoned open space, conservation or had an equivalent zoning, on or after 02 August 2010.</p> <p>a. Eastern Cape</p> <p>i. Outside urban areas, or</p> <p>ii. Inside urban areas:</p> <p>(aa) Areas zoned for conservation use or equivalent zoning, on or after 02 August 2010;</p> <p>(bb) A protected area identified in terms of NEMPAA, excluding conservancies; or</p> <p>(cc) Sensitive areas as identified in an environmental management framework as contemplated in chapter 5 of the Act as adopted by the competent authority.</p> | <p>The proposed development will entail transformation of land bigger than 1000 square metres in size (43.73ha) for residential houses.</p>                                               |
| <b>Listed activity as described in GN R. 325</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                           |
| <p><b>GN.R 325 (Listing Notice 2): Activity 15</b></p> <p>The clearance of an area of 20 hectares or more of indigenous vegetation, excluding where such clearance of indigenous vegetation is required for—</p> <p>(i) the undertaking of a linear activity; or</p> <p>maintenance purposes undertaken in accordance with a maintenance management plan.</p>                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>The activity is applicable for the clearance of vegetation for a housing (residential) development (43.73 ha).</p>                                                                     |

### **3. NEED AND DESIRABILITY**

One of the objectives of the EIA process is to motivate for “the need and desirability for the proposed development, including the need and desirability of the activity in the context of the preferred development footprint”. Consideration should be given to the need and desirability of development in determining whether this is the right time and place for the proposed land use or activity to be established. Hence, it is therefore, equated with rational land use and should be able to answer the question of what the most sustainable use of land is.

#### **3.1 MOTIVATION**

The development of integrated and sustainable human settlements remains a national priority in South Africa. This recognizes that access to adequate housing is not only a fundamental human right, as stipulated in the Constitution, but importantly it serves as a key instrument to address features of apartheid regime which sought to promote spatial segregation and inequality. Section 26 of the Constitution; “(1) Everyone has the right to have access to adequate housing, (2) The state must take reasonable legislative and other measures, within its available resources, to achieve the progressive realisation of this right”. Despite significant progress since 1994 with approximately 5.2 million housing opportunities delivered, the country continues to face a persistent housing backlog (Department of Human Settlements, 2024). This is manifested in the proliferation of informal settlements, substandard living conditions, and limited access to basic services and socio-economic infrastructure, particularly in urban centres (Department of Human Settlements, 2024).

The proposed housing development is in line with the 2024 White Paper for Human Settlements, which aims to create an enabling environment for the development of integrated and sustainable human settlements. It also advocates for the creation of sustainable human settlements with a view to providing housing, safety, and comfort for all with an emphasis on vulnerable, poor and missing middle, those households who earn too much to qualify for government-subsidised housing but not enough to afford market-rate housing in our society.

The Eastern Cape faces a challenge of spatial planning and a growing housing crisis. The province continues to urbanise with informal settlement growth around urban and peri-urban areas. According to the Eastern Cape Vision 2030 Provincial Development Plan, significant population growth is expected in the metropolitan areas of Nelson Mandela Bay and Buffalo City. One of the province's key strategic goals is to ensure that every person has a decent home by 2030, which requires the upgrading of informal settlements, the construction of new residential developments, rental housing, and rural housing. Additionally, the Eastern Cape Department of Human Settlements has a mandate to create integrated, sustainable human settlements

and improve the quality of household life. It supports key provincial outcomes such as informal settlement upgrading, security of land tenure, and the provision of affordable residential and rental housing (Eastern Cape Planning Commission, 2014).

The Nelson Mandela Bay Municipality continues to face challenges in meeting its housing demand, with a relatively slow pace of delivery. One of its strategic objectives is to accelerate access to dignified housing, sanitation, and improved basic services for indigent households, thereby ensuring safe and decent living conditions for all residents. To date, NMBM has delivered approximately 685 state-subsidised housing units, which is not sufficient to meet the growing need. As part of the Priority Human Settlement and Housing Development Areas (PHSHDA's) promulgated by Minister of Human Settlements, Water and Sanitation, Walmer area was identified as a priority area that can advance spatial transformation and housing delivery (Nelson Mandela Bay IDP, 2022/23-2026/27).

The proposed development site is located in close proximity to the Walmer Gqebera township, enabling the provision of housing opportunities to communities already residing in the area. This promotes spatial integration and reduces the need for relocation to peripheral areas. The land is already owned by the municipality and has been earmarked for medium- to high-density residential development. Its location adjacent to existing residential areas allows for integration with existing municipal services and infrastructure, thereby reducing development costs and minimising environmental disturbance associated with the construction of bulk infrastructure.

Although the site contains environmentally sensitive forest patches, the development has been carefully designed to avoid and minimise impacts on these areas. Mitigation measures include the establishment of buffer zones around forest patches, and the relocation of stormwater detention ponds away from the forest patches. Furthermore, the site has already been subject to various informal and disturbed land uses, including grazing, informal recreational activities, and illegal dumping. There are no surface water bodies, such as rivers or wetlands, present on the site.

The proposed development is expected to result in positive socio-economic impacts, including employment opportunities during the construction phase, stimulation of the local economy, and improved infrastructure provision. It will also contribute to reducing the risk of illegal land occupation by providing formal housing opportunities.

The proposed housing aligns with national, provincial, and municipal policy objectives. It also addresses a critical housing need within a priority development area.

### 3.2 NEED AND DESIRABILITY

The following **Table 12** outlines the need and desirability of the proposed BWSS, addressing ecological sustainability, socio-economic considerations, and environmental justice in relation to the project's location, scale, and anticipated impacts.

**Table 12:** Need and Desirability assessment for the development

| <b>ASPECTS RELATED TO SECURING ECOLOGICAL SUSTAINABLE DEVELOPMENT AND USE OF NATURAL RESOURCES</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1.1 How will this development (and its separate elements/aspects) impact the ecological integrity of the area, disturb or enhance ecosystems, and/or result in the loss or protection of biological diversity?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>The proposed development falls within two regional vegetation types; the Algoa Sandstone Fynbos (Critically Endangered) and the Sardinia Forest Thicket (Vulnerable). The proposed development will require clearing and transformation of these natural habitats for the construction of a housing development, which will result in the loss of terrestrial biodiversity and faunal habitat. Although the site does not fall within any CBAs, it is located within an aquatic ESA 1 which is crucial in supporting the functioning of CBAs. However, impact on this ecosystem is not envisaged as there are no wetlands or watercourses within the proposed development area.</p> <p>Forest patches were identified within the proposed development site. The forest patches will be protected through the implementation of a 20m buffer within which no development will occur. In addition, the layout will be updated to ensure that stormwater ponds are located outside of these forest patches to further minimise potential impacts.</p> |
| <b>1.2 What measures were explored to firstly avoid these negative impacts, and where these negative impacts could not be avoided altogether, what measures were explored to minimise and remedy (including offsetting) the impacts?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>The proposed development area contains forest patches, to minimise impact on these the layout was amended to reduce encroachment on the forest patches. In addition, the layout will be updated to ensure that stormwater ponds are located outside forest patches to further minimise potential impacts.</p> <ul style="list-style-type: none"> <li>• Layout 1 does not entirely avoid the forest patches and results in a housing yield of 1720 units, though this layout has a higher residential capacity it shall result in the loss of the forest patches.</li> <li>• Layout 2 has a 20m buffer around the forest patches to protect and minimise potential environmental impacts. However, it will result in a significant reduction in housing yield.</li> </ul>                                                                                                                                                                                                                                                                           |
| <b>ASPECTS RELATED TO INTRA- AND INTER-GENERATIONAL EQUITY IN THE CONTEXT OF SUSTAINABILITY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>2.1 Does the proposed location, type, and scale of development promote a reduced dependency on resources? For example, can the development be located more appropriately to reduce the dependency on resources needed for service infrastructure?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>The proposed development is located on an area that is already allocated for housing units, thereby aligning with the NMBM spatial development framework. The proposed Walmer housing development will be connected into existing municipal service infrastructure including, water, sewage, electricity which currently serve the Walmer area. As no new infrastructure will be required, the development will significantly reduce dependency on resources and minimise environmental disturbance.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>2.2 How will the ecological impacts resulting from this development impact people's environmental right?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The proposed housing development is not anticipated to have significant negative impacts on people's environmental rights. The anticipated negative impacts during the construction phase such as noise, air quality, visual impact etc will be minimal as possible. The proposed development is anticipated to have positive impacts as formal housing and services will be provided for residents.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>2.3 Describe the linkages and dependencies between human well-being, livelihoods, and ecosystem services applicable to the area in question and how the development's ecological impacts will result in socio-economic impacts (e.g. on livelihoods, loss of heritage site, opportunity costs, etc.)</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>The development will contribute to improved livelihoods and human wellbeing for the prospective beneficiaries. Although ecosystems services of the site where the development is proposed shall be negatively impacted, the preferred layout option avoids loss forests and retains considerable biodiversity patches to continue to render ecosystems to the inhabitants. Loss of heritage may be eminent, however, mitigations shall assist to manage the impacts. Furthermore, reduced property values concerns are adequately addressed in the preferred layout to buffer such impacts on the upmarket neighbouring Walmer Heights properties.</p>                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p><b>2.4 Based on all the above, how will this development positively or negatively impact ecological integrity objectives/targets/considerations of the area?</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>The proposed Walmer housing development will negatively impact the ecological integrity of the area due to the irreversible loss of natural habitat, including Critically Endangered Algoa Sandstone Fynbos and Vulnerable Sardinia Forest Thicket. The transformation of this area into residential infrastructure will fragment habitats and reduce biodiversity.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p><b>2.5 Describe the positive and negative cumulative ecological/biophysical impacts bearing in mind the size, scale, scope, and nature of the project its location, and existing and other planned developments in the area.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>A summary of both positive and negative impacts of the proposed housing development has been provided in this report, please refer to Section 5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>ASPECTS RELATED TO PROMOTING JUSTIFIABLE ECONOMIC AND SOCIAL DEVELOPMENT</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>3.1 What is the socio-economic context of the area, based on, amongst other considerations, the following considerations: The IDP and any other strategic plans, frameworks of policies applicable to the area; spatial priorities and desired spatial patterns (e.g. need for integrated or segregated communities, need to upgrade informal settlements, need for densification, etc.), spatial characteristics (e.g. existing land uses, planned land uses, cultural landscapes, etc.), and Municipal Economic Development Strategy?</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>The proposed development aligns with the strategic objectives as outlined in the Nelson Mandela Bay Municipality Integrated Development Plan (2022/23-2026/27) which include;</p> <ul style="list-style-type: none"> <li>• 'Provide dignified housing and sanitation and accelerate access to improved services to indigent households to create safe and decent living conditions for all residents'.</li> <li>• 'Provide a built environment that promotes integration, inclusivity and accessibility'.</li> <li>• 'Ensure implementation of a spatial transformation agenda which addresses the spatial disparities of the past'</li> </ul> <p>The surrounding area includes a mix of formal and informal settlements, and the development supports spatial priorities, such as the integration of previously segregated communities, the upgrading of informal areas, and densification within serviced urban zones. Furthermore, the development aligns with existing land uses in the area, as the proposed site was allocated for residential units.</p> |
| <p><b>3.2 Considering the socio-economic context, what will the socio-economic impacts be of the development (and its separate elements/aspects), and specifically also on the socio-economic objectives of the area?</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

The proposed development aligns and supports the socio-economic needs and objectives of the NMBM. One of the municipality's key objectives is to accelerate access to dignified housing, sanitation, and improved basic services for indigent households, thereby ensuring safe and decent living conditions for all residents. This development contributes directly to that objective by providing formal housing and associated infrastructure such as water, electricity and sanitation. In addition, the development will have positive impacts by creating employment opportunities during construction and operational phase.

**3.3 How will this development disturb or enhance landscapes and/or sites that constitute the nation's cultural heritage? What measures were explored to firstly avoid these impacts, and where impacts could not be avoided altogether, what measures were explored to minimise and remedy (including offsetting) the impacts? What measures were explored to enhance positive impacts?**

The proposed housing development is not expected to impact on any heritage resources. However, a Specialist has been appointed to conduct the Archaeological and Cultural Heritage Impact Assessment, and the findings will be incorporated in the Scoping and EIA reports.

**3.4 How will this development address the specific physical, psychological, developmental, cultural, and social needs and interests of the relevant communities? Will the development result in equitable (intra- and inter-generational) impact distribution, in the short- and long-term? Will the impact be socially and economically sustainable in the short- and long-term?**

The proposed development addresses the physical and developmental needs by providing formal housing with access to basic services. It includes a school site to support educational development and long-term learning opportunities. Job creation during the construction phase will further contribute to the developmental and social needs of the surrounding community.

The proposed development promotes equity and inclusion by offering a mix of housing typologies (low, middle, and high-income housing opportunities) thereby preventing socio-economic segregation and ensuring that vulnerable and low-income groups are not excluded.

The proposed development will have a socially and economically sustainable impact in both the short and long term. In short term, the development will provide local employment during construction phase and stimulate the local economy. In the long term, the development's mixed use which includes a business zone with facilities such as a supermarket, bottle store, service trades etc, will support economic activity, and contribute to the municipality's economy.

**3.5 How were a risk-averse and cautious approach applied in terms of socio-economic impacts? What are the limits of current knowledge (note: the gaps, uncertainties, and assumptions must be clearly stated)? What is the level of risk (note: related to inequality, social fabric, livelihoods, vulnerable communities, critical resources, economic vulnerability, and sustainability) associated with the limits of current knowledge? Based on the limits of knowledge and the level of risk, how and to what extent was a risk-averse and cautious approach applied to the development (and its alternatives)?**

The EMPr shall describe all reasonable and feasible mitigation measures and addresses long-term environmental management.

**3.6 How will the socio-economic impacts resulting from this development impact people's environmental right?**

The proposed housing development will have both positive and negative impacts. The proposed development may negatively impact adjacent residents during construction due to noise and dust but these will be temporary and can easily be mitigated. The positive impacts that will come with the proposed development include improvement in the NMBM's economy through job opportunities as well as objectives of the municipality of provided formal housing and upgrading of informal settlements.



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3.7 What measures were taken to pursue the selection of the “best practicable environmental option” in terms of socio-economic considerations? What measures were taken to pursue environmental justice so that adverse environmental impacts shall not be distributed in such a manner as to unfairly discriminate against any person, particularly vulnerable and disadvantaged persons (who are the beneficiaries and is the development located appropriately)?</b>                                                                                                                                                               |
| The proposed Walmer housing development will involve a mix of housing typologies (low, middle, and high-income housing opportunities) which promotes social integration and avoids spatial marginalisation of vulnerable communities. In this way the proposed development ensures that disadvantaged and low income residents are not excluded from housing opportunities. Additionally, the proposed site is within an area which is already allocated for residential use and shall make use of existing municipal services infrastructure reducing the need of new infrastructure thereby reducing associated environmental impacts. |
| <b>3.8 What measures were taken to ensure that the responsibility for the environmental health and safety consequences of the development has been addressed throughout the development’s life cycle?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| An EMPr has been compiled for the proposed development which provides mitigation measures for environmental health and safety, this is to ensure that impacts encountered during the construction phase are addressed adequately.                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>3.9 What measures were taken to ensure that the interests, needs, and values of all interested and affected parties were considered and that adequate recognition was given to all forms of knowledge, including traditional and ordinary knowledge?</b>                                                                                                                                                                                                                                                                                                                                                                              |
| The public participation process shall take note of interests, needs and values of all I&APs based on all forms of knowledge, this will be done by ensuring that every comment received is addressed and considered.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>3.10 Considering the interests, needs, and values of all the interested and affected parties, describe how the development will allow for opportunities for all the segments of the community (e.g. a mixture of low-, middle-, and high-income housing opportunities) that is consistent with the priority needs of the local area (or that is proportional to the needs of an area)?</b>                                                                                                                                                                                                                                            |
| The proposed Walmer housing development will involve a mix of housing typologies (low, middle, and high-income housing opportunities) and is aligned with the priority need of the Walmer area. According to the Nelson Mandela Bay Municipality IDP, the Walmer area was identified as a priority area that can advance spatial transformation and housing delivery.                                                                                                                                                                                                                                                                    |

**Table 13.** Need and Desirability assessment from a socio-economic perspective.

| <b>NEED</b>                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Is the land use (associated with activity being applied for) considered within the timeframe intended by the existing approved Spatial Development Framework agreed to by the relevant environmental authority?</b>                                                                                                                                          |
| Yes, the land use associated with the proposed housing development is considered to be within the timeframe and intent of the municipal SDF. The site has been allocated for a residential use and aligns with spatial planning priorities. The supports the municipality’s goals to address housing demand and spatial transformation, as outlined in the IDP. |
| <b>Should development, or if applicable, expansion of the town/area concerned in terms of this land use (associated with the activity being applied for) occur here at this point in time?</b>                                                                                                                                                                  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Yes, development in the form of the proposed land use should occur at this location and point in time. The area has been identified for residential use and was previously granted an environmental authorisation, which has since lapsed.                                                                                                                                                                                                                                                                                                                                               |
| <b>Does the community/area need the activity and the associated land use concerned? (is it a societal priority)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| The proposed development addresses a societal need within both the municipal area and the Walmer area. Access to affordable housing is a national and local development priority particularly in urban centres facing high population growth and informal settlement expansion such as NMBM. The development's integrated approach of mix housing typologies particularly for low and middle income households, ensures that vulnerable and disadvantaged groups have access to adequate and secure housing opportunities, thereby contributing to spatial equity and social upliftment. |
| <b>Is this project part of a national programme to address an issue of national concern or importance?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Yes, the proposed housing development aligns with the national programme such as the National Development Plan 2030, which prioritise the delivery of integrated, affordable and sustainable housing. Therefore, the proposed development contributes in addressing the national concern of housing backlogs and inadequate housing.                                                                                                                                                                                                                                                     |
| <b>DESIRABILITY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Would the approval of this application compromise the integrity of the existing approved municipal IDP and SDF as agreed to by the relevant authorities?</b>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| No, the proposed development will not compromise the integrity of the existing municipal IDP and SDF. It aligns with the strategic objectives and spatial priority as outlined in the IDP.                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Do location factors favour this land use (associated with the activity applied for) at this place? (relates to the contextualization of the proposed land use on this site within its broader context)</b>                                                                                                                                                                                                                                                                                                                                                                            |
| Yes, the proposed housing development area is located within an area allocated for residential use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Will the proposed activity or the land use associated with the activity applied for, result in unacceptable opportunity costs?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| No, the proposed activity and associated land use will not result in unacceptable opportunity costs. The site identified for the development has already been allocated for residential use. The development shall make use of existing municipal existing infrastructure, meaning no need for new infrastructure on greenfield land which would be more costly and could potentially impact on natural or sensitive areas.                                                                                                                                                              |
| <b>Will the proposed land use result in unacceptable cumulative impacts?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| The proposed land use is not expected to result in unacceptable cumulative impacts. The development will be connected to existing municipal infrastructure, which limits the need for new infrastructure and associated environmental disruption. Potential environmental impacts such as loss of terrestrial biodiversity will be mitigated through appropriate mitigation measures.                                                                                                                                                                                                    |

## 4. PROJECT ALTERNATIVES

The identification of alternatives is a key aspect of the success of the environmental scoping phase. All reasonable and feasible alternatives must be identified and screened to determine the most suitable alternatives to consider and assess in the EIA phase. There are, however, some significant constraints that have to be considered when identifying alternatives for a project with this scope. Such constraints include social, financial and environmental issues, which will be discussed as part of the evaluation of the alternatives for this project.

“Alternatives”, in relation to a proposed activity, is defined as different means of meeting the general purpose and requirements of the activity, which may include alternatives to; -

- a) the property on which or location where it is proposed to undertake the activity;
- b) the type of activity to be undertaken;
- c) the design or layout of the activity;
- d) the technology to be used in the activity;
- e) the operational aspects of the activity; and

### 4.1 PROCESS FOLLOWED TO REACH THE PREFERRED ALTERNATIVES

The process followed to reach the preferred alternative may be described as follows:

- Developer provides description of project and proposed site area;
- The preliminary sensitivity of the site area is determined using the National Web Based Screening Tool;
- Verification of site sensitivity is done through visual site assessment;
- Options of how the desired project can be implemented with least disturbance of sensitive areas are considered (i.e. alternatives);
- The feasibility of the options/alternatives presented is assessed, only feasible and reasonable alternatives are considered for impact assessment;
- The potential Impacts that may be caused by the feasible alternatives are identified, inputs from specialist reports and I&APs are taken into consideration in this process;
- The technical justification for the alternatives are also considered;
- An assessment of the significance of the impacts is done using the methodology described in Section 8.1;
- The alternative with the lowest (negative) overall combined impact significance score is selected as the preferred alternative.

## 4.2 PROPERTY OR LOCATION ALTERNATIVES

The proposed Walmer housing development entails construction of a mix of housing typologies and associated infrastructure on Erf 11305 in Ward 1, Walmer within the Nelson Mandela Bay Municipality. The site is in close proximity to the Walmer settlement and is intended to form a logical extension for beneficiaries who are currently living in the area and need to be accommodated in new destination areas. The proposed site is municipally owned, and no land acquisition will be required. Furthermore, the site falls within an area that was allocated for residential development, therefore service infrastructure will be connected on to existing municipal infrastructure. An Environmental Authorisation (EA) for a housing development on this site had been previously granted, however, this authorisation has lapsed. Thus, no other location alternatives were considered for the proposed development.

## 4.3 ACTIVITY ALTERNATIVES

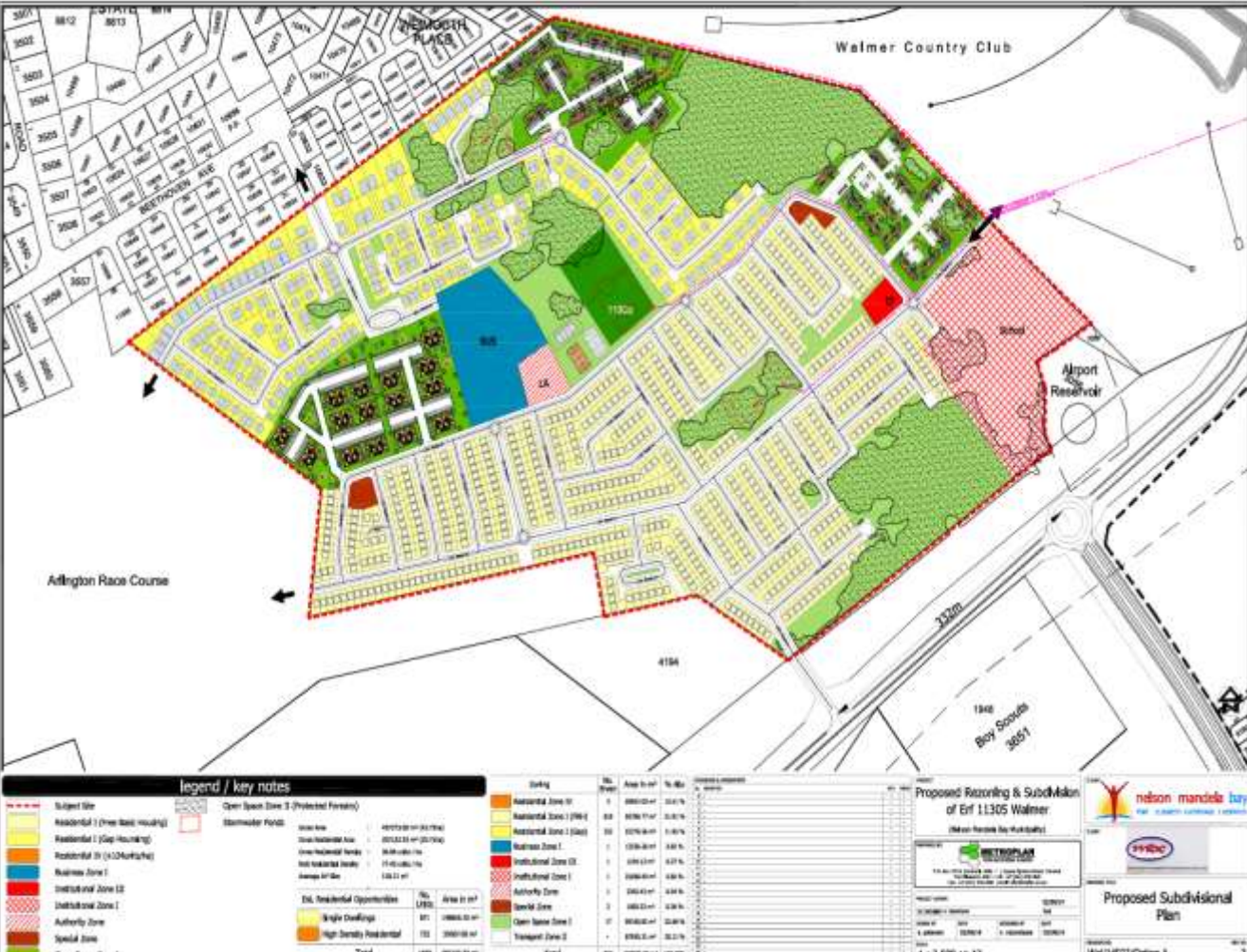
The activity involves construction of a housing development and associated infrastructure. The use of the land is for residential development as per the municipal spatial development framework. Thus, no other activity alternatives are assessed.

## 4.4 DESIGN AND LAYOUT ALTERNATIVES

As per the previous application for the proposed development two layout alternatives were considered. These alternatives were based on ensuring that the development does not overlap with protected forest patches.

- Option A (see **Figure 6**)- this layout option includes development of a total housing yield of approximately 1720 units which covers 46% of the site. Layout one does not entirely avoid the forest patches and the development overlaps on these forest patches. This layout offered some positive impact such as a higher housing yield but also presented a potential impact on the forest.

**The layout was revised to avoid the forest, as such this layout option will not be subject to further assessment in the EIA process.**



**Figure 6.** Proposed layout option A



## Preferred Alternative

Option 2 (see **Figure 7**)- is layout option includes development of a total housing yield of approximately 1236 units which covers 38% of the site. Although this results in a lower housing yield compared to Option A, the layout avoids encroaching on forest patches. In addition, the layout will be updated to ensure that stormwater ponds are located outside forest patches to further minimise potential impacts. A 20m buffer is maintained around these sensitive ecological areas, minimising environmental disturbances and protect the existing forest. This layout option was authorised in the previous application.



Figure 7. Proposed layout option B (Preferred)

#### **4.5 TECHNOLOGY ALTERNATIVES**

No technology alternatives have been considered in this assessment. The infrastructure required such as water, sewer and electricity will be connected onto existing bulk infrastructure currently servicing the surrounding areas. However, the applicant is encouraged to consider the incorporation of solar energy systems for residential units, as well as rainwater harvesting systems.

- **Energy efficiency**

The installation of solar panels is recommended to supplement and reduce dependence on the existing electricity network. The integration of solar panels will lead to a decrease in greenhouse gas emissions, thereby lowering the overall carbon footprint of the proposed development. Furthermore, solar panels have the potential to mitigate the effects of the urban heat island phenomenon and contribute to the moderation of the local climate.

- **Water efficiency**

Rainwater harvesting systems such as rainwater barrels or tanks can be incorporated in the development. These systems provide an essential alternative water supply in periods of drought or service interruptions. Harvested rainwater can be utilized for irrigation and domestic purposes. Additionally, rainwater barrels help intercept rainfall and stormwater, thereby reducing runoff and mitigating soil erosion.

#### **4.6 OPERATIONAL ALTERNATIVES**

No operational alternatives have been considered in this assessment.

#### **4.7 OPTION OF NOT IMPLEMENTING- “NO-GO ALTERNATIVE”**

The no-go alternative means doing nothing or keeping the current status quo of no activities occurring on-site. The proposed site is situated within a residential area, by not developing the site would result in the loss of potential direct and indirect socio-economic benefits as such job creation, housing shortages and upgrading of informal settlements would not materialise. From an ecological perspective, the site has a critically endangered vegetation type and forest, not developing the site will protect the natural environment. However, the site is not in its pristine state and has been disturbed by illegal dumping and livestock grazing. Developing the area will also assist in the protection of the forest on site.

## 4.8 SUMMARY OF ALTERNATIVES

Table 14 below provides a summary of these alternatives, their key characteristics, and the preferred.

Table 14: Summary of alternatives

| CATEGORY                             | ALTERNATIVE                                                    | DESCRIPTION                                                                                                                                                                                                                                                                                                                               | PREFERRED ALTERNATIVE |
|--------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| NO-GO Alternative                    | Status Quo                                                     | Keeping the current status quo of no activities occurring on site, which would not result in a loss of a critically endangered vegetation and a forest. However, potential positive socio-economic impacts will be lost.                                                                                                                  | No                    |
| Site Alternatives                    | Alternative 1                                                  | No site/property alternatives exist as the proposed site is municipally owned, and no land acquisition will be required. Furthermore, the site falls within an area that was allocated for residential development. Additionally, an Environmental Authorisation (EA) for a housing development on this site had been previously granted. | Yes                   |
| Design and layout Alternatives       | Alternative 1                                                  | This option comes with a higher housing yield covering 46% of the site, however it does not entirely avoid the forest patches and the development overlaps on these forest patches.                                                                                                                                                       | No                    |
|                                      | Alternative 2 (Preferred)                                      | This option comes with a lower housing yield compared to Alternative 1. Although Alternative 1 would have been preferred this option avoids encroaching on the forest patches. In addition, the layout will be updated to ensure that stormwater ponds are located outside forest patches.                                                | Yes                   |
| Technical & Operational Alternatives | No feasible and reasonable alternatives evaluated and assessed | No alternatives have been considered.                                                                                                                                                                                                                                                                                                     | N/A                   |

## 4.9 PREFERRED ALTERNATIVES

The preferred alternatives for the proposed development include the following:

- Site Alternatives: Alternative 1
- Design and layout Alternative: Alternative 2

The reason for the abovementioned preferred alternative options is attributed to the outcome of advantages and disadvantages as described above. The preferred alternatives are likely to have less environmental impacts.



## 5. DESCRIPTION OF THE AFFECTED ENVIRONMENT

This section of the report provides a description of the environment that may be affected by the proposed project. Aspects of the biophysical, social and economic environment that could be directly or indirectly affected by, or could affect, the proposed project have been described. Baseline information sourced from various spatial datasets, the National Screening tool report, as well as specialist biodiversity and ecosystem mapping resources. These include the National Ecosystem freshwater Priority Areas (NEFPA) wetlands and rivers, the National Wetland Map Version 5 (NWM5), the Eastern Cape Biodiversity Conservation Plan (ECBCP) 2019 and the National Vegetation Map (Mucina & Rutherford, 2018). In addition to environmental sensitivities, this section also describes the existing infrastructure within the project area, including roads, bulkwater pipelines, internal reticulations pipelines, and reservoirs. The biophysical environment descriptions below include the areas close to the proposed construction area.

### 5.1 CLIMATE

The weather patterns of the Nelson Mandela Bay area change throughout the day as it lies at the confluence of several climatic regimes, the most important of which are temperate and subtropical (Stone, 1988). Gqeberha is dominated by topographical or gradient winds for most of the year (Grobler, 2012). The area experiences westerly winds throughout the year, though in summer the percentage of easterly winds reached more than 40% (Schumann et al., 1999). Maximum and minimum mean temperatures are experienced in February and July, respectively (McCallum 1981). Exceptionally high temperatures (~30°C) can occur during berg wind conditions that develop frequently in autumn and winter. The mean annual rainfall for the Port Elizabeth area is approximately 600 mm (Stone 1988). The strongest winds occur during October and November, with weakest winds during May and June (Schumann et al. 1991).

### 5.2 TOPOGRAPHY AND HYDROLOGY

The area is characterized by flats to undulating terrain (**Error! Reference source not found.**). Elevation ranges from 104 m to 110 m, with an average slope of 1.9% to 2.4%.



**Figure 8.** Topography of the proposed housing development

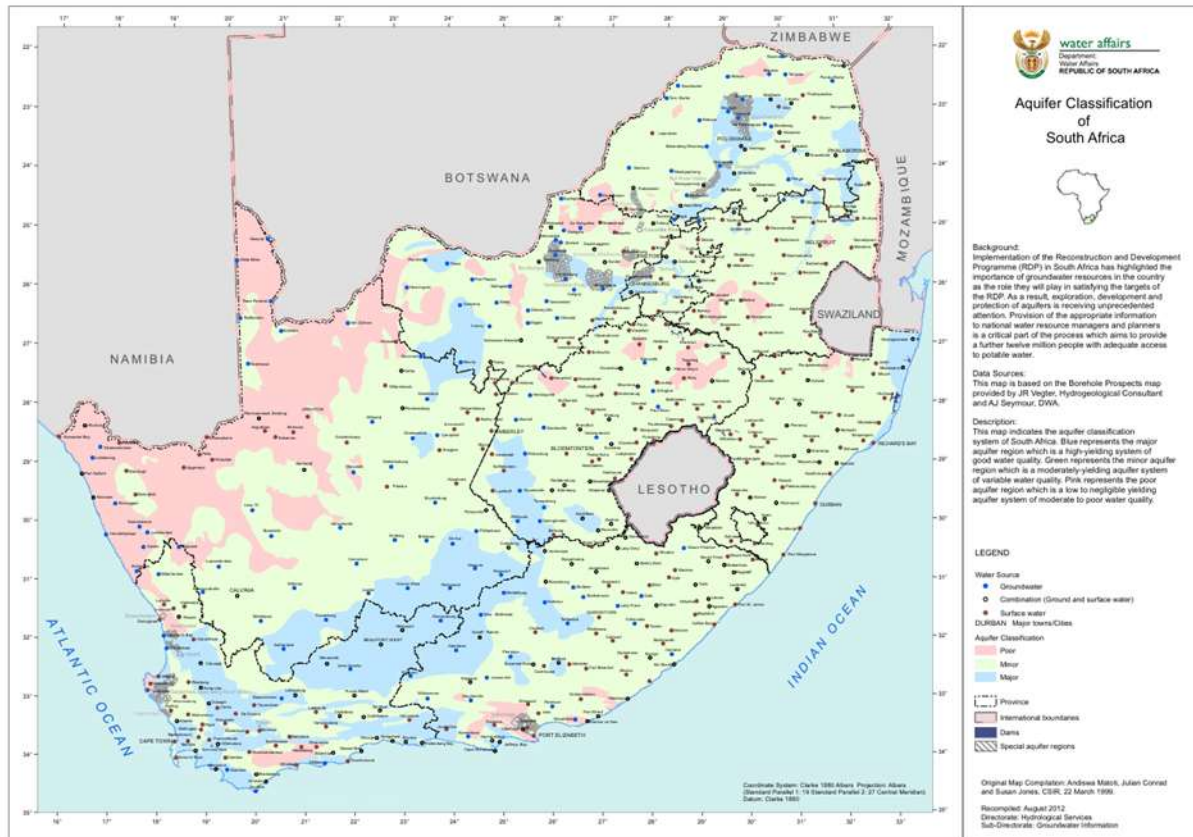
The study area is situated within quaternary catchment M20A, located within the Mzimvubu-Tsitsikamma Water Management Area. The site occurs within the catchment area of the Baakens River, however there is no connectivity between the Baakens and the proposed site (Scherman Colloty & Associates, 2014). According to the NFEPA database, a NFEPA artificial wetland is located southeast of the boundary of the site, this feature is a reservoir.



**Figure 9.** Conditions within the proposed development area

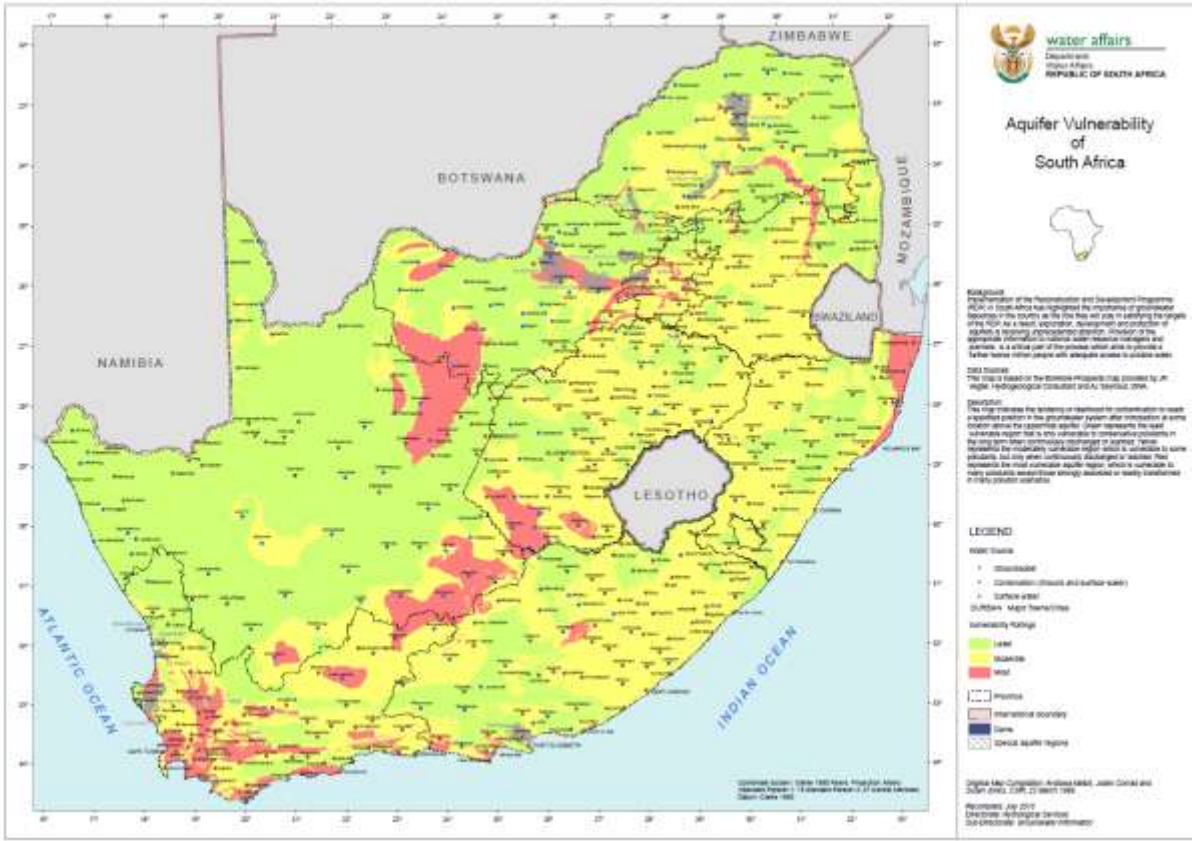
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Based on the Aquifer Classification and Aquifer Vulnerability Map, the aquifer is classified as a poor aquifer region which is a low to negligible yielding aquifer system of moderate to poor water quality, with least vulnerability meaning is only vulnerable to conservative pollutants in the long term when continuously discharged or leached.



**Figure 10.** Aquifer classification map of South Africa





**Figure 11. Aquifer vulnerability map of South Africa**

### 5.3 GEOLOGY AND SOILS

The proposed area is underlain by coastal aeolianites (ancient, wind-blown dune sands) of the Nanaga Formation (Algoa Group) of Pliocene to Early Pleistocene age. These ancient dune sands crop out extensively to the west and east of Port Elizabeth. In the study area they unconformably overlie Palaeozoic sandstones and quartzites of the Peninsula Formation (Almond, 2014). The Nanaga Formation overlies the Alexandria Formation paraconformably, semi-consolidated, medium-grained calcareous sandstone beds are characterized by large-scale cross-bedding, interpreted as aeolian deposits (Booth et al., 1999). The Nanaga Formation is very prominent as fine-grained red sands.

## 5.4 TERRESTRIAL AND AQUATIC BIODIVERSITY

### 5.4.1 TERRESTRIAL

According to the National South African Vegetation Map (2018), the study area falls within both the Eastern Fynbos-Renosterveld Bioregion, within the Fynbos Biome and the Albany Thicket Biome. The vegetation types that occur in this area include the Algoa Sandstone Fynbos and Sardinia Forest Thicket. Algoa Sandstone Fynbos is the dominant vegetation type, covering most of the site, while Sardinia Forest Thicket occurs in a smaller section along the southern boundary. The vegetation types are described below:

#### (a) Algoa Sandstone Fynbos

Algoa Sandstone Fynbos is a grassy shrubland (graminoid fynbos) on coastal flats between Van Stadens River and Summerstrand in the Gqeberha area. It occurs on acidic lithosol soils derived from Ordovician sandstones of the Table Mountain Group (Cape Supergroup) (SRK Consulting, 2024). This vegetation type occurs on flat to slightly undulating plain and grasses become dominant especially in wet habitats.

The Algoa Sandstone Fynbos is classified as **Critically Endangered (CR)** according to The Revised National List of Threatened Terrestrial Ecosystems –2022.

#### (b) Sardinia Forest Thicket

This vegetation type is dominated by thicket clumps typical of Algoa Dune Thicket. The matrix is forest characterized by Cape ash *Ekebergia capensis* and coral trees *Erythrina caffra* (SRK Consulting, 2008).

The Sardinia Forest Thicket is classified as **Vulnerable (VU)**.

The vegetation on site (**Figure 12**) can be described as sparse to moderate shrubs as well as presence of forest clumps. Based on field observations the site is no longer pristine as it has been disturbed by illegal dumping, grazing, access tracks and informal soccer fields. According to the Nelson Mandela Bay Critical Biodiversity Areas (2009), none of the proposed development footprint falls within any CBAs.



**Figure 12.** Vegetation conditions at the proposed site

A follow-up site inspection was undertaken on 18 February 2026 following a fire event that occurred within the municipality beginning 2026. **Figure 13** below shows portions of the site exhibiting visible fire damage, including burned grass cover. The burn pattern appears patchy, with some areas more affected than other areas while some sections of the site were not impacted by the fire.





**Figure 13.** Vegetation conditions post the fire events at the development site

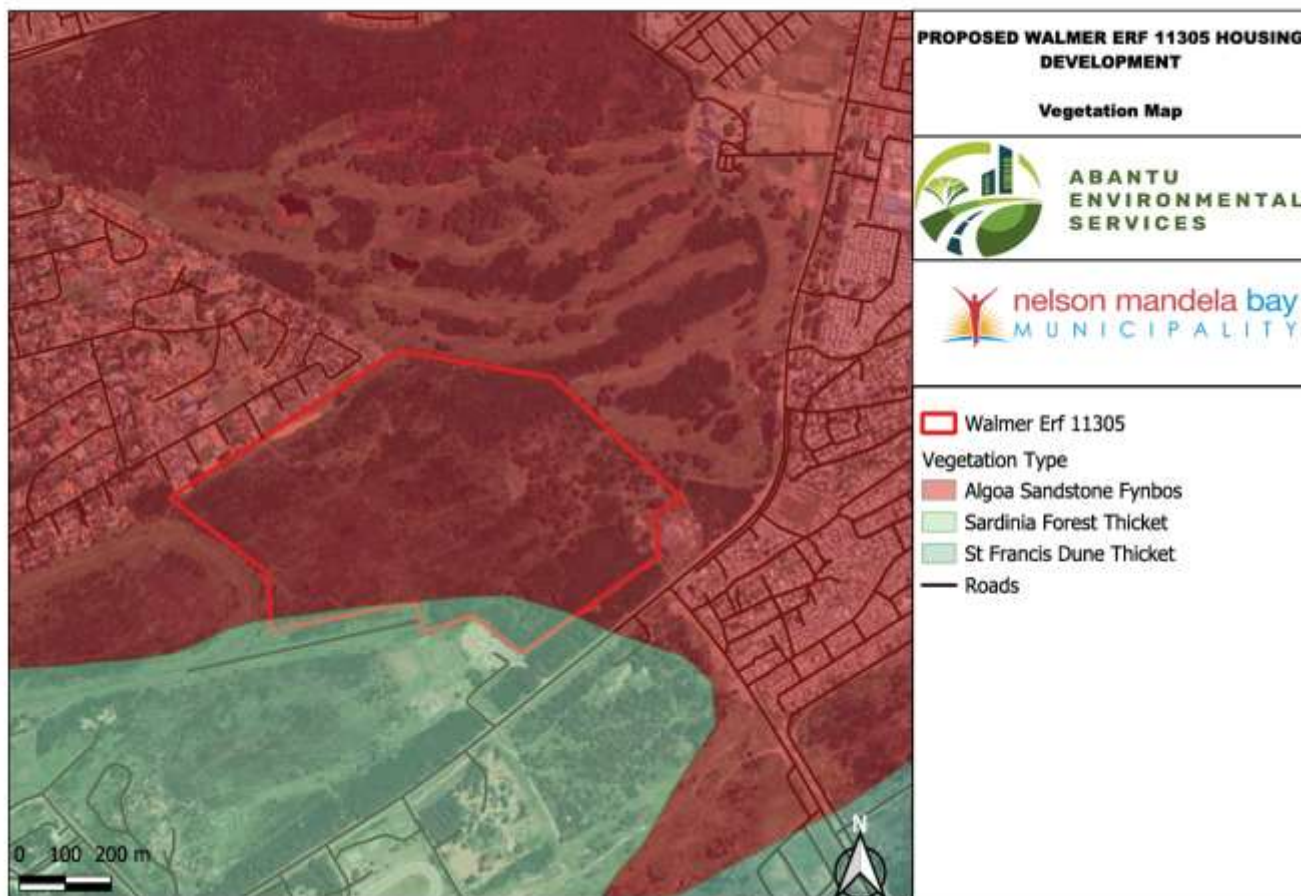


Figure 14. Terrestrial map

#### 5.4.2 AQUATIC AND STRATEGIC WATER SOURCE AREAS

ESAs are areas that must be maintained in at least fair ecological condition (seminatural/moderately modified state) in order to support the ecological functioning of a CBA or protected area, or to generate or deliver ecosystem services, or to meet remaining biodiversity targets for ecosystem types or species, when it is not possible or not necessary to meet them in natural or near-natural areas (SANBI, 2017).

The project area is located in an aquatic Ecological Support Area (ESA) 1 and other according to the 2019 Eastern Cape Biodiversity Conservation Plan. As reported by the national web based environmental screening tool, the aquatic biodiversity sensitivity on the site is very high. Although the aquatic biodiversity sensitivity is rated as high, no wetlands or watercourses were observed within the proposed development footprint.



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According to Le Maitre et al. (2018), Strategic Water Source Areas (SWSAs) are areas of land that either: (a) supply a disproportionate quantity of mean annual surface water runoff in relation to their size and are considered nationally important; or (b) have high groundwater recharge and where the groundwater forms a nationally important resource; or (c) areas that meet both criteria (a) and (b). Based on the DFFE screening tool, the project area falls within a Strategic Water Source Area for surface water (Tsitsikamma).



Figure 15. Aquatic CBA

### 5.5 FAUNA

The DFFE Screening tool listed the animal species theme for the site as high. The animal species that have potential to occur on site are listed on **Table 15** below.

**Table 15.** Animal species that have potential to occur on site.

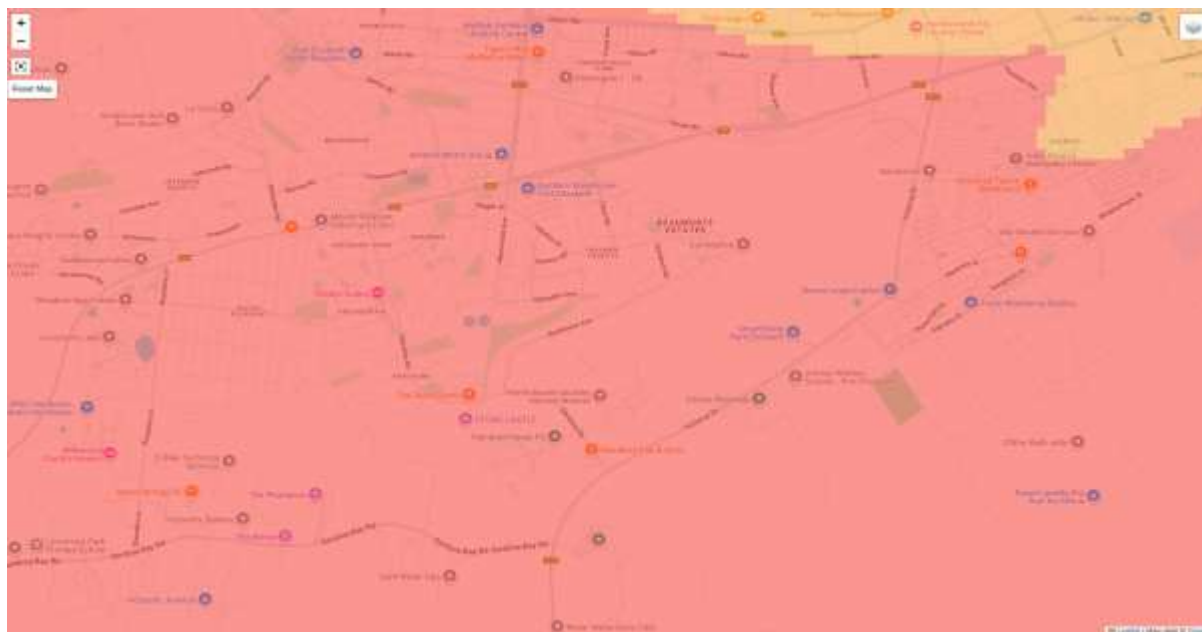
| Feature                             |
|-------------------------------------|
| <i>Aves-Circus ranivorus</i>        |
| <i>Aves-Stephanoaetus coronatus</i> |
| <i>Aves-Neotis denhami</i>          |
| <i>Aves-Bradypterus sylvaticus</i>  |
| <i>Aves-Stephanoaetus coronatus</i> |
| <i>Aves-Neotis denhami</i>          |

|                                           |
|-------------------------------------------|
| <i>Aves-Eupodotis senegalensis</i>        |
| <i>Insecta-Chrysoritis thysbe whitei</i>  |
| <i>Mammalia-Chlorotalpa duthieae</i>      |
| <i>Sensitive species 8</i>                |
| <i>Invertebrate-Aneuryphymus montanus</i> |

## 5.6 ARCHAEOLOGY AND HERITAGE RESOURCES

According to the National Web-based Screening Tool the proposed area is rated as high sensitivity and within 100m of a Grade IIIb Heritage site. Grade III Built Environment Heritage Resources are buildings and sites that have sufficient intrinsic significance to be regarded as local heritage resources and are significant enough to warrant that any alteration is regulated. According to a Phase 1 Archaeological Impact Assessment conducted by Booth (2014), the following built environment sites were observed at the proposed development area: an unused dipping tank overgrown with vegetation, ruin of a dwelling and associated infrastructure, ruin of a farmstead and contemporary substation. The ruins of the built environment were considered as having a low-medium cultural significance.

As seen on **Figure 16** below the sensitivity on palaeontological resources within the proposed site is rated as 'Very High', as such a field assessment and protocol for finds is required.



| Colour        | Sensitivity        | Required Action                                                                                                                     |
|---------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| RED           | VERY HIGH          | field assessment and protocol for finds is required                                                                                 |
| ORANGE/YELLOW | HIGH               | desktop study is required and based on the outcome of the desktop study, a field assessment is likely                               |
| GREEN         | MODERATE           | desktop study is required                                                                                                           |
| BLUE          | LOW                | no palaeontological studies are required however a protocol for finds is required                                                   |
| GREY          | INSIGNIFICANT/ZERO | no palaeontological studies are required                                                                                            |
| WHITE/CLEAR   | UNKNOWN            | these areas will require a minimum of a desktop study. As more information comes to light, SAHRA will continue to populate the map. |

Figure 16. Palaeo Map

## 5.7 LAND USE

The proposed site comprises vacant, undeveloped municipal land that has been subject to various informal and unauthorized uses. These include illegal dumping, livestock grazing, the creation of informal access tracks, and the use of portions of the site as informal soccer fields. The site is strategically located, bounded by the Walmer Golf Course to the east, Victoria Drive and the township of Walmer Gqebera to the southeast, the former Arlington Racecourse to the southwest, and the residential suburb of Walmer Heights to the north.



Figure 17. Landcover map

## **5.8 SOCIO-ECONOMIC ATTRIBUTES**

The proposed project is located in Gqeberha (formerly known as Port Elizabeth) located in Nelson Mandela Bay Municipality, in the Eastern Cape Province of South Africa and is one of eight Metropolitan Municipalities. In 2001, the Nelson Mandela Bay Metropolitan Municipality was formed as a single administrative area covering inter alia Gqeberha, Kariega (formerly Uitenhage), Despatch and a number of surrounding areas.

The city has a unique advantage in that it possesses two ports, namely Port Elizabeth Harbour and Ngqura. This presents several opportunities related to the diversification of port activities, the maritime sector, and the distinction between 'dirty' and 'clean' port uses. The NMBM Municipality has a size of 195 890.76 ha and covers the smallest area of land (1952 km<sup>2</sup>) in the Eastern Cape, which is the second largest province in South Africa (Almond et al., 2008).

### **5.8.1 POPULATION**

The current population of Nelson Mandela Bay is estimated at 1 190 496 (Community Survey, 2022), with a growth rate of 1.54% (IHS, 2017), which is lower than that of other metropolitan areas in South Africa, such as Ekurhuleni (2.1%) and Tshwane (2.6%) people. According to the census the Nelson Mandela Bay Municipality population is split 52.7% to 47.3% in favour of females, with the Black African population accounting for 62.7% of the overall population, Coloureds accounting for 19.2%, Whites are 15.7% and the rest are a mix of different ethnicities. According to the census, the elderly constitute 8.8% of the population while the young children under the 14 year make up 22.3% of the population. The literacy levels in the local municipality by 2016 were measured at 76% of the population, which is a significant increase from the 66% measured in 2006.

### **5.8.2 EDUCATION**

According to the Municipality profile of the Nelson Mandela Bay Metro (2016) there is a total number of 324 schools, which consist of 269 ordinary public schools, 11 special needs schools, 25 independent schools and 19 Early Childhood Development (ECD) Centres within the municipality. A comparison with other metros shows that 75.8% of learners aged 7 to 24 years in Nelson Mandela Bay attended an educational institution as compared to 80.2% in Buffalo City and 79.2% in Mangaung (79.2%). The lowest attendance was observed in Cape Town (69.8%) and eThekweni (71.1%). People 20 years and older with less than Grade 7 as highest level of education are 4.1%. Buffalo City is 11.4% and Cape Town is 6.1%. 0.8% of people 20 years and older in Nelson Mandela Bay have no schooling compared to a Metro average of 1.4% (STATS SA GHS, 2019).

### **5.8.3 ECONOMIC ASPECT**

Whilst the NMBM is undoubtedly an important node of activity within the economy of the Eastern Cape, it is characterized by several challenges in terms of economic development. These include:

- A high unemployment rate (36.6%), compounded by the low education levels of the labour force, including large numbers of illiterate adults with limited employment prospects.
- Ageing and inadequate investment in the maintenance and upgrading of infrastructure.
- The dependence on the automotive sector and insufficient diversification within the manufacturing and others sectors.
- A lack of up-to-date local economic statistics and monitoring and evaluation systems.

STATS SA states that: “according to ECSECC (Eastern Cape Socio Economic Consultative Council), the GDP growth rate for the Nelson Mandela Bay Municipality was 2.1% in 2010 and the GDP per capita R52 147. The largest economic sectors in the Nelson Mandela Metro are manufacturing, finance, community services and transport. Community services, trade and manufacturing sectors are the sectors that create the most employment in the Metro”. Good quality infrastructure is key to sustainable social, economic and industrial development. Poor infrastructure hampers development, growth and ability to trade in the domestic and global economy.

Economic infrastructure, which includes transport, energy, telecommunications, water and sanitation provides services which are of fundamental importance for development. In NMBM it is deficient and this is exacerbated by inadequate maintenance and thus prematurely deteriorating installations and services. Infrastructural services are often overlooked as a means to alleviate poverty and improve environmental conditions. In order for Nelson Mandela Bay to grow and develop a sustainable economy, it is important to build new economic drivers to replace or augment the ones that have served the region in the past.

#### **5.8.4 HOUSING**

According to STATS SA General Households Survey (2019), 21,6% of households in Nelson Mandela Bay listed grants as their main source of income. A total of 640,000 individuals, accounting for approximately 49.6% of the overall population, live in poverty within the Nelson Mandela Bay Metropolitan area (ECSECC, 2017). In the 2017/18 financial year, all qualifying households with an income of less than R3 200 per month (equivalent to two state pensions) received access to free basic services provided by the Municipality through its Assistance To The Poor Programme (ATTP). There is a slow pace in housing delivery within the municipality with 92% of households living in formal dwellings, 7.4% live in informal dwellings and the remaining 0.6% live in traditional and other dwellings. The current housing demand in the metro is estimated at 126 550 (Nelson Mandela Bay Municipality IDP, 2022/23-2026/27).

### **5.8.5 BASIC SERVICES**

#### **(a) Water**

According to the NMBM IDP (2022/23-2026/27), the NMBM has extent of wate provision backlog in recently established informal settlements. The municipality has about 99% of households that have access to piped water as follows:

- 82.3% have access to piped water inside dwelling.
- 11.4% have access to piped (tap) water inside the yard.
- 5.4% have access to piped (tap) water on community stand.
- 0.8% have no access to basic water.

#### **(b) Sanitation**

About 96.2% of households in the Nelson Mandela Bay Municipality has access to basic sanitation. Basic sanitation includes flush toilets connected to a public sewerage system, a septic tank or a pit toilet with a ventilation pipe. There are approximately 6010 buckets in circulation and 2 834 chemical toilets in informal settlements. The Municipality rolls out basic sanitation in informal settlements and relocates households to formal sites with waterborne sanitation (NMBM IDP 2022/23-2026/27).

#### **(c) Electricity**

According to Census 2022, 96.5% of households have access to electricity from the mains electricity supply, 3.4% use other sources of energy for lighting whilst the remaining 0.1% have no access to any source of energy.



## 6. PUBLIC PARTICIPATION PROCESS

The Public Participation Process (PPP) is a requirement of several pieces of South African Legislation and aims to ensure that all relevant I&AP's are consulted, involved and their opinions are taken into account and a record included in the reports submitted to Authorities. The process ensures that all stakeholders are provided this opportunity as part of a transparent process which allows for a robust and comprehensive environmental study.

The public participation (PP) process has been undertaken in line with the requirements of Regulations 39 to 44 of the EIA Regulations as amended. The primary aim of the public participation process is to ensure that:

- Information that reasonably has or may have the potential to influence any decision regarding an application is made available to potential stakeholders and I&APs;
- Potential or registered interested and affected parties, including the competent authority, may be provided with an opportunity to comment on reports and plans; and
- Comments received from potential stakeholders and I&APs are recorded and incorporated into the FEIR to be submitted to the Competent Authority.

The primary objectives of the public participation process are to:

- Inform and notify potentially I&APs of the proposed application (explain steps that were taken to achieve this);
- Initiate or promote meaningful and timeous participation of I&APs by providing proof that notice boards, advertisements and notices notifying potentially interested and affected parties of the proposed application have been displayed, placed, or given;
- Maintain a list of all persons, organization and organs of state that were registered as interested and affected parties in relation to the application;
- Identify issues and concerns of key stakeholders and I&APs with regards to the application for the proposed project;
- Provide a summary of the issues raised by interested and affected parties, the date of receipt of and the response of the EAP to those issues; and
- Provide responses to I &AP's queries.

The PPP for this project has been undertaken in accordance with the requirements of the NEMA, as well as the principles of Integrated Environmental Management (IEM). IEM implies an open and transparent participatory process, whereby stakeholders and other I&APs are afforded an opportunity to comment on the project.



## 6.1 SCOPING PHASE

During the Scoping phase of this project, the key objective of public participation was to provide I&APs with an opportunity to provide comment and input in the planning phase of the project.

### (a) Initial Notice

The first stage of public consultation was undertaken during the initial notification phase prior to the circulation of the draft scoping for public review. The initial notice was circulated on 07 November 2025 notifying the public about the proposed development.

### (b) Advertisement and Notices

An advertisement was placed with The Herald, site notices were erected in strategic areas around the site, notifying potential Interested and Affected Parties (I & AP's) of the availability of the Draft Scoping Report and inviting I & AP's to register on the database as Registered Interested & Affected Parties.

### (c) Circulation of the DSR/ Submission of FSR

A draft Scoping Report (dSR) was compiled and made available for public comment for a period of 30 days, from **02 March 2026- 01 April 2026** where after the Final Scoping Report (FSR) including Comments and Responses from the public was submitted to the Competent Authority, DEDEAT on 13 April 2026. On **28 May 2026**, the DEDEAT accepted the FSR and confirmed that the project may proceed to the Impact Assessment Phase.

## 6.2 EIA PHASE

After acceptance of the FSR the assessment moves into the EIA phase. This entails the compilation of the Draft Environmental Impact Report (DEIR) and EMPr, 30 days comment period by I&APs and review by competent authority.

### (a) Advertisement and Notices

An advertisement is placed with The Daily Dispatch, site notices were erected in strategic areas around the site, notifying potential Interested and Affected Parties (I & AP's) of the availability of the Draft Environmental Impact Report.

### (b) Draft EIR and EMPr Public review

The DEIR and EMPr is made available to all I&APs for a 30-day comment period commencing from **05 August 2026- 07 September 2026**. Public meetings for presenting the DEIR will be conducted as follows:

- **Physical:** Walmer High School (10am)- 01 September 2026
- **Virtual:** Microsoft Teams (18h00)- 02 September 2026

**(c) Final Environmental Impact Report (FEIR)**

All comments made on the DEIR during public review will be captured and adequately responded to in the Comments and Response Report. Once the FEIR has been finalised, it will be submitted to the DEDEAT for decision making. When a decision is received from the department, I&APs will also be notified accordingly as per requirements.

**(d) Notice of EA decision**

- I&APs will be notified of the decision within 14 days from receipt of such
- A 20 day period will be allowed for appeals

## 6.3 SUMMARY OF COMMENTS RECEIVED

A full account of comments and responses as well as copies of all correspondences between the EAP and I&APs will be appended in a Comments and Responses Report in to be appended in Error! Reference source not found. of this report. The key issues expected to emanate during the public review period of this report will be summarised as follows;

**Table 16.** Key issues

| Aspects        | How issue(s) have been addressed |
|----------------|----------------------------------|
| PPP is ongoing | PPP is ongoing                   |
|                |                                  |

## 7. SUMMARY AND RECOMMENDATIONS OF THE SPECIALIST ASSESSMENTS

This section provides the key findings and recommendations for the conducted specialist studies. Specialist studies are appended as Appendix C of this report.

### TERRESTRIAL IMPACT ASSESSMENT (de Kock, 2026)

#### Key Findings

- The proposed development area environmental sensitivity for the plant species theme is identified as high sensitivity. Based on the site sensitivity verification the site sensitivity is reduced to medium sensitivity. This is due to the following reasons:
  - None of the plant SCC listed in the Screening Tool Report were identified on site at the time of the site survey.
  - Three plant SCC listed in the Screening Tool Report may plausibly occur on site, as the site hosts patches of indigenous forest/thicket vegetation, which may host these species.
  - No other plant SCC, sensitive or threatened plant species were observed on site.
  - One protected tree was observed on site i.e. *Sideroxylon inerme subsp. inerme* (White Milkwood).
  - Six protected plants (protected under the Nature and Environmental Conservation Ordinance (NECO), No. 19 of 1974) were identified on site.
  - The activity will not have an impact on plant SCC beyond the boundary of the Primary PAOI.
- The proposed development area environmental sensitivity for the animal species theme is identified as high sensitivity. Based on the site sensitivity verification the site sensitivity was reduced to low sensitivity. This is due to the following reasons:
  - None of the sensitive animal species listed in the Screening Tool Report were observed on site at the time of the site survey.
  - Nine sensitive animal species are listed in the Screening Tool Report – five bird, two insect and two mammal species.
  - Mostly high-sensitivity bird species were identified in the Screening Tool Report. Given the high degree of past and current disturbance in the greatest portion of the site, as well as in the surrounding areas, these species are unlikely to breed or nest on site and will be able to relocate to the green open space located northeast and southwest of the site.
  - No other sensitive or threatened animal species or tracks/traces thereof were observed on site.
  - The activity will not have an impact on SCC beyond the boundary of the project site/PAOI.
- The proposed development area environmental sensitivity for the terrestrial biodiversity theme is identified as very high sensitivity. Based on the site sensitivity verification (conducted further below) the site sensitivity remains as very high sensitivity. This is due to the following reasons:

- The site is not located within nor near any surface or groundwater strategic water source areas (SWSA's).
- Vegetation on site is largely mapped and described as FFs29 Algoa Sandstone Fynbos (VegMap, 2018/2024), which is classified as a Critically Endangered ecosystem (NEMBA, 2022; NBA, 2025) – on a national level. However, the remnant patches of indigenous vegetation present in the proposed project site are not representative of FFs29 Algoa Sandstone Fynbos and most likely was not in the past, either, before any past vegetation clearing or development was undertaken, or before the site became invaded by Australian Acacia and Eucalyptus trees.
- The southernmost portions of the site, along the southern boundary, are classified as AT48 Sardinia Forest Thicket (VegMap, 2018/2024). It is, therefore, most likely that vegetation present on site, before settlement of the area, was representative of AT48 Sardinia Forest Thicket rather than FFs29 Algoa Sandstone Fynbos. AT48 Sardinia Forest Thicket is also classified as a Critically Endangered ecosystem (NBA, 2025).
- The municipal-scale vegetation classification and map for the site i.e. the NMBM Final Bioregional Plan (2015), maps the project site as hosting Sardinia Bay Forest Thicket – a Vulnerable ecosystem. The site is, therefore, classed as hosting a Vulnerable vegetation/ecosystem type on a bioregional or municipal scale.
- The greatest portion of the site has been cleared of the indigenous vegetation originally present on site (for more than the past 22 years) and transformed to grassland dominated by Rooikrans and Port Jackson Willow trees. Only an approx. 15.95 ha area of Sardinia Bay Forest Thicket vegetation remains as separate patches of partially intact vegetation on site – 7.6 ha of which will not be developed and will be zoned Open Space Zone 3 (conservation usage) (out of the total 43.73 ha project area or Primary PAOI).
- The site is not classified as a Critical Biodiversity Area (CBA), Ecological Support Area (ESA) or a Critical Ecological Process Area (CEPA) in the NMBM Bioregional Plan (2015).

### **Recommendations and conclusion**

The following general rehabilitation measures are recommended for inclusion into the EMP:

- Vegetation within the proposed direct development footprint (total 43.73 ha) is mostly secondary and degraded/transformed in nature i.e. invaded grassland and remnant, partially intact forest/thicket patches with the potential to host plant and animal SCC (though not observed at the time of the site survey).
- Though no longer representative of the original Critically Endangered or Vulnerable vegetation/ecosystem types mapped for the site i.e. Sardinia Forest Thicket (SANBI VegMap, 2018/2024) or Sardinia Bay Forest Thicket (NMBM Bioregional Plan, 2015), respectively – these areas of 'green open space' do serve as corridors for the movement/browsing of animals and the transfer of the genetic material of plants.
- All forest patches must be avoided.

## **DRAFT ENVIRONMENTAL IMPACT REPORT: WALMER HOUSING DEVELOPMENT ON ERF 11305**

- The proposed development will not result in altered ecological services and functionality, provided that all mitigation measures proposed in this report are implemented.
- The proposed Walmer Housing Development is NOT considered to be Fatally Flawed, and no components of the proposed development have been identified as flawed. No site or layout alternatives are proposed. Biodiversity offsetting is not required.
- The ecological impacts of all aspects for the proposed development were assessed and considered to be acceptable, provided that all mitigation measures provided in this report are implemented.

### **AQUATIC BIODIVERSITY IMPACT ASSESSMENT (de Kock, 2026)**

#### **Key Findings**

- Five artificial wetlands occur within the Total PAOI. A desktop analysis as well as a site inspection has confirmed the presence of these artificial wetland systems.
- All of the wetland systems are classified and identified as artificial wetlands with artificial wetland 5 classified as a reservoir as per site inspection and desktop analysis. Artificial wetlands 1 – 4 are scattered across a green landscape which appears to be a golf course. These are man-made water features.
- According to the NFEPA database, only one wetland was observed within the Total PAOI. A site inspection has confirmed that this wetland is actually a reservoir and should be classified as such.
- No rivers were observed within the Total PAOI.
- The artificial wetlands found within the PAOI are all categorized as Low, which is not ecologically important and sensitive at any scale.

#### **Recommendations and conclusion**

- The proposed development is considered as an acceptable change to the environment provided all proposed mitigations are implemented.
- All freshwater systems must be avoided. Provided that all mitigation measures proposed in this report are implemented, these risks will be reduced and therefore considered as an acceptable change to the local environment.
- The proposed development is NOT considered to be fatally flawed, and no components of the project have been identified as flawed.
- The combination of the desktop survey, analysis of historical imagery and site visit found no evidence of the presence of any natural watercourses on the property.
- All observed wetlands on site are artificial. While the ecological importance of these artificial wetlands is low, they are moderately important in terms of their hydro-functional attributes and could provide a useful function with respect to attenuating stormwater runoff.
- However, no development must be allowed within the delineated artificial dams and the demarcated 10m no development buffer. These areas should be considered as No-Go Areas for construction

activities. All construction proposed within these areas should be reconsidered to not infringe within these areas.

### **AGRICULTURAL COMPLIANCE STATEMENT (de Kock, 2026)**

#### **Key findings**

- The screening tool classifies the site as being of a very high agricultural sensitivity. Based on the outcome of the site verification the agricultural theme is reduced to low.
- The land has a high land capability classification indicating that the soil, climate, and terrain are naturally ideal for plant growth. However, this land is unsuitable for agricultural production due to:
  - The site is mostly covered by dense natural thicket and forest vegetation in the northeast while open shrubland interspersed with woody alien vegetation dominates most of the site.
  - The site is covered by dense housing on the eastern and western boundaries, a golf course in the north and the old Arlington racetrack in the south. As per the NMBM SDF, the racetrack is being reallocated to residences.
  - The area lacks access to a reliable water source for irrigation infrastructure; it cannot support intensive agriculture.
  - No agricultural infrastructure exists on site.
  - The site has never been used for agriculture. Some informal communal grazing occurred.

#### **Conclusion**

- No agricultural land (cropland, rangelands or pastures) will be directly impacted by the proposed development.
- No supporting agricultural infrastructure exists on site or nearby.
- Agricultural sensitivity is considered as low.
- Based on the above, it is the opinion of the specialist the proposed housing development on erf 11305 in Walmer, Gqeberha will NOT require a full Agricultural Assessment. No mitigations are proposed.

### **PALEONTOLOGICAL IMPACT ASSESSMENT (Mavuso, 2026)**

#### **Key Findings**

- The proposed Walmer Housing Development on ERF 11305 is located within Cenozoic coastal deposits of the Algoa Group, most likely belonging to the Salnova Formation.
- These deposits are known to contain marine fossil assemblages, particularly molluscs and other marine invertebrates preserved within calcareous sandstones and shell-rich horizons.
- As a result, the SAHRIS palaeosensitivity map classifies the geological units underlying the study area as having very high palaeontological sensitivity.
- Despite this classification, the development area itself is situated within an urbanised environment where natural geological exposures are absent and the underlying strata are largely covered by soil and anthropogenic deposits.

## **DRAFT ENVIRONMENTAL IMPACT REPORT: WALMER HOUSING DEVELOPMENT ON ERF 11305**

- Previous palaeontological work undertaken in the broader Walmer area has reported no fossil material within similar development footprints and has indicated that the likelihood of encountering significant fossil resources within such settings is low.
- Based on the results of this desktop assessment, the proposed development is therefore considered unlikely to result in significant impacts on palaeontological heritage resources.

### **Recommendations and conclusion**

- The proposed development may proceed from a palaeontological perspective.
- No further specialist palaeontological studies are required prior to development.
- A Chance Find Protocol must be implemented during excavation activities.
- Should any fossil material be encountered during construction (e.g., fossil shells, bones, petrified wood, or trace fossils), work must cease in the immediate area and the Environmental Control Officer (ECO) should notify a qualified palaeontologist and the relevant heritage authority so that the material can be appropriately recorded and managed.
- Subject to the implementation of these measures, the proposed development is considered acceptable from a palaeontological heritage perspective.

## **ARCHAEOLOGICAL AND CULTURAL HERITAGE IMPACT ASSESSMENT (Booth, 2026)**

### **Key Findings**

- The proposed area for the housing development is situated on the boundary of the generally accepted 5 km archaeologically sensitive coastal zone.
- However, no archaeological heritage remains or expected shell midden sites were observed within the proposed area for development.
- The ruins of a farmstead, dwelling, and associated infrastructure were documented on the property. The ruins of the buildings may be older than 60 years, however, modifications to the buildings may have been made over time.
- Within the past few years, subsequent to the 2014 assessment, archaeological human remains were found exposed by a member of the public reported to the South African Police Service (SAPS) in Walmer.
- The burial was found along the north western boundary of the proposed development area, adjacent to the suburb of Beaumont Estates. Upon inspection, the exposed burial was found within a deflation hollow containing marine shell and stone artefact scatters, confirming the archaeological nature of the remains and the broader heritage sensitivity of the locality.
- The finding of the exposed archaeological human remains along the north-western boundary of the proposed development area is of high heritage significance.



- The burial, situated within an exposed deflation hollow containing marine shell and stone artefact scatters, confirms the presence of an archaeological context associated with precolonial occupation and mortuary practices in the Walmer coastal-dune landscape.
- The occurrence of human remains in situ, together with associated cultural material, elevates the heritage sensitivity of the locality and triggers the protective provisions of Section 35 and Section 36 of the National Heritage Resources Act (Act 25 of 1999).
- This finding demonstrates that the broader area could possibly retain intact archaeological deposits and burial related features, and therefore requires careful management, controlled ground disturbance procedures, and post clearance monitoring in accordance with ECPHRA's directives.

### **Recommendations and conclusion**

- **Archaeological Monitoring during Vegetation Clearing and Excavations:** A qualified archaeologist must conduct archaeological monitoring during all vegetation-clearing activities and subsequent excavations associated with the proposed development. Monitoring is required due to the confirmed presence of archaeological human remains and associated cultural material along the north-western boundary of the project area, as well as the broader heritage sensitivity of the Walmer dune-landscape. Vegetation clearing and initial earthworks have the potential to expose additional burials, shell middens, stone-artefact scatters, or other subsurface archaeological deposits that were not visible during the desktop update

Monitoring must occur during:

- Mechanical or manual vegetation removal, particularly in areas of dune deflation, natural hollows, or previously undisturbed terrain.
  - Topsoil stripping and bulk earthworks, including trenching, levelling, and foundation excavations.
  - Any ground-disturbing activities within proximity to the previously exposed burial locality.
  - If archaeological material, human remains, or burial related features are identified at any stage, all work in the immediate area must cease, and the finds must be reported to ECPHRA in accordance with Section 36 of the National Heritage Resources Act (Act 25 of 1999). Work may only resume once the heritage authority has issued written instructions or a permit authorising further action.
  - This monitoring requirement must be incorporated into the contractor's method statements and the project's Environmental Management Programme (EMPr) to ensure compliance with NEMA and the NHRA.
- **Inclusion of Living Heritage in the Public Participation Process:** It is recommended that living heritage considerations be formally included as part of the public participation process in accordance with Sections 3(2)(b) and 38(3) of the National Heritage Resources Act (NHRA, Act 25 of 1999). Public

consultation must provide affected communities, traditional leaders, and knowledge holders with the opportunity to identify culturally meaningful places, ritual spaces, ancestral associations, and other forms of living heritage that was not visible during field assessment. The archaeologist must be informed of any living heritage spaces that may be identified during the public participation process. These spaces must be identified and documented together with the affected communities who have identified them. The results of the consultation must be included as part of an updated or revised report including mitigation measures for these sites and possible no-go zones.

- Include Heritage Awareness in Contractor Induction: Contractor teams must be briefed on heritage sensitivities, chance find procedures, and reporting requirements. Heritage awareness should form part of environmental induction and ongoing site compliance monitoring.
- Implementation of a Chance-Find Procedure: A formal chance-find procedure must be implemented for all contractors. If human remains, grave structures, artefacts, or culturally significant features are uncovered, work must stop immediately and ECPHRA must be notified. The procedure must be included in contractor induction training.
- Documentation and Reporting: All heritage-related actions, including public participation, consultation, monitoring, chance-find responses, and assessments—must be documented and included in the final heritage compliance report submitted to ECPHRA.
- Management of Pre Colonial Material and Human Remains: If concentrations of pre-colonial archaeological heritage material, including subsurface dense artefact accumulations, associated cultural deposits, or human remains such as graves or burials—are uncovered during construction, all work must cease immediately and the discovery must be reported without delay to the Eastern Cape Provincial Heritage Resources Agency (ECPHRA) at 043 492 1370. No further disturbance may occur until the heritage authority has provided guidance.
- Should ECPHRA require further investigation, Phase 2 mitigation may be undertaken in the form of controlled test-pitting, sampling, or systematic excavation and collection of the archaeological or heritage material. The purpose of Phase 2 mitigation is to establish the contextual status, extent, and significance of the site and, where appropriate, to remove the archaeological deposit prior to the continuation of development activities. All costs associated with Phase 2 mitigation, including specialist fees, excavation, documentation, and reporting, will be the responsibility of the developer.

## 8. IDENTIFIED ENVIRONMENTAL IMPACTS

It is expected that proposed development will have environmental and social impacts during the construction and operational phases. The impacts have been identified by the EAP, specialist studies and informed public participation process conducted in the scoping phase.

**Table 17:** Summary of identified impacts

| Impacts/ Aspects                                                                 | Description                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PLANNING AND DESIGN PHASE</b>                                                 |                                                                                                                                                                                                                                                                                                  |
| <b>Legal and social compliance</b>                                               | <ul style="list-style-type: none"> <li>Commencement of project construction and activities without lodging and obtaining relevant authorisation applications and being approved will result in authorities halting the projects, issuing fines, implying no development commencement.</li> </ul> |
| <b>Layout issues</b>                                                             | <ul style="list-style-type: none"> <li>Failure to adequately address site layout requirements may result in the authorities halting the project.</li> </ul>                                                                                                                                      |
| <b>Site establishment</b>                                                        | <ul style="list-style-type: none"> <li>The establishment of a construction site requires appropriate planning to achieve a holistic balance between efficacy and cost-efficiency of construction activities as well as minimised environmental damage.</li> </ul>                                |
| <b>Existing civil services, infrastructure and properties</b>                    | <ul style="list-style-type: none"> <li>Disruption and damage of water supply, powerlines, infrastructure and other properties in close proximity to the proposed site.</li> </ul>                                                                                                                |
| <b>Freshwater utilisation</b>                                                    | <ul style="list-style-type: none"> <li>Failure to consider technologies that reduce consumption of water can place further pressure on already stressed water resources.</li> </ul>                                                                                                              |
| <b>Bulk services (Electricity, water (domestic and stormwater) and sewerage)</b> | <ul style="list-style-type: none"> <li>The connection of bulk services to existing municipal infrastructure must be planned and designed to ensure that the municipality has sufficient capacity, preventing the overburdening of existing infrastructure.</li> </ul>                            |
| <b>Stormwater management</b>                                                     | <ul style="list-style-type: none"> <li>Inadequate stormwater design may result in uncontrolled surface runoff, causing downstream flooding, erosion and sedimentation, and contamination of water bodies.</li> </ul>                                                                             |
| <b>CONSTRUCTION PHASE</b>                                                        |                                                                                                                                                                                                                                                                                                  |
| <b>Legal and social compliance</b>                                               | <ul style="list-style-type: none"> <li>Failure to fulfil legal environmental compliance may lead to overall disapproval and failure of the project.</li> </ul>                                                                                                                                   |
| <b>Site establishment and security</b>                                           | <ul style="list-style-type: none"> <li>The establishment of a construction site requires appropriate planning to achieve a holistic balance between efficacy and cost-efficiency of construction activities as well as minimised environmental damage.</li> </ul>                                |
| <b>Water consumption</b>                                                         | <ul style="list-style-type: none"> <li>Increased freshwater use for construction and drinking purposes.</li> </ul>                                                                                                                                                                               |
| <b>Sanitation</b>                                                                | <ul style="list-style-type: none"> <li>Lack of supply of ablution facilities for construction workers may result in the pollution of the immediate and surrounding receiving environment of the facility and may pose a health hazard to users.</li> </ul>                                       |
| <b>Loss of indigenous vegetation</b>                                             | <ul style="list-style-type: none"> <li>Clearing vegetation for housing development may result in loss of vegetation and habitat fauna.</li> </ul>                                                                                                                                                |
| <b>Disturbance of fauna and loss of habitats.</b>                                | <ul style="list-style-type: none"> <li>Clearing of vegetation and construction activities may lead to the displacement of fauna and alteration of habitats, potentially resulting in loss of biodiversity and ecosystem fragmentation.</li> </ul>                                                |
| <b>Loss of SCC</b>                                                               | <ul style="list-style-type: none"> <li>Direct loss of terrestrial plant SCCs and ToPS and their habitat due to construction activities.</li> </ul>                                                                                                                                               |
| <b>Impact on forest</b>                                                          | <ul style="list-style-type: none"> <li>Clearing of vegetation on site may result in the loss of forest including its species.</li> </ul>                                                                                                                                                         |

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stormwater and erosion impacts</b>                    | <ul style="list-style-type: none"> <li>Vegetation clearing and soil disturbance during construction expose the site to erosion by water and wind. This can result in increased sediment in stormwater runoff, localised soil loss, and destabilised slopes if not properly controlled.</li> </ul>                                                                                     |
| <b>Alien invasion proliferation</b>                      | <ul style="list-style-type: none"> <li>Uncontrolled vegetation clearance during construction may lead to loss of natural vegetation and habitats. Opportunistic alien and invasive vegetation encroachment may result in post degradation of native vegetation.</li> </ul>                                                                                                            |
| <b>Waste management</b>                                  | <ul style="list-style-type: none"> <li>Lack of adequate waste management during construction could result in spread of litter which may pollute or affect the surrounding environment.</li> </ul>                                                                                                                                                                                     |
| <b>Hazardous material storage and handling</b>           | <ul style="list-style-type: none"> <li>Environmental pollution as well as health and safety impact due to poor handling of hazardous material</li> </ul>                                                                                                                                                                                                                              |
| <b>Traffic impact</b>                                    | <ul style="list-style-type: none"> <li>During construction increase in traffic on the nearby road is expected due to construction vehicles. This may lead to increased pedestrian, animal and vehicle accidents.</li> </ul>                                                                                                                                                           |
| <b>Visual impact</b>                                     | <ul style="list-style-type: none"> <li>Temporary degradation of the visual aesthetics of the area due to the presence of construction equipment and site camp. This may cause nuisance to nearby residents.</li> </ul>                                                                                                                                                                |
| <b>Noise pollution</b>                                   | <ul style="list-style-type: none"> <li>During construction noise from machinery and vehicles may potentially cause nuisance to residents living close to the proposed site e.g Walmer Heights.</li> </ul>                                                                                                                                                                             |
| <b>Air pollution</b>                                     | <ul style="list-style-type: none"> <li>During construction dust from vehicles, material stockpiles and cleared areas may potentially cause nuisance to residents living close to the proposed site e.g Walmer Heights.</li> </ul>                                                                                                                                                     |
| <b>Paleontological and heritage resource disturbance</b> | <ul style="list-style-type: none"> <li>Damage to cultural, heritage, archaeological and paleontological resources during construction activities such as excavations and plant movement.</li> </ul>                                                                                                                                                                                   |
| <b>Employment opportunities</b>                          | <ul style="list-style-type: none"> <li>During the construction phase the development will create temporary job opportunities both direct and indirect in return stimulating the local economy.</li> </ul>                                                                                                                                                                             |
| <b>OPERATIONAL PHASE</b>                                 |                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Open spaces and Forest patches management</b>         | <ul style="list-style-type: none"> <li>Inadequate management and maintenance of open spaces and forest patches during the operational phase may lead to vegetation degradation and impact on the integrity and ecological functioning of the forest.</li> </ul>                                                                                                                       |
| <b>Waste management</b>                                  | <ul style="list-style-type: none"> <li>Lack of adequate waste management during operation by residents and facilities on the area may result into littering, illegal dumping and pollution of the immediate and surrounding receiving environment.</li> </ul>                                                                                                                         |
| <b>Freshwater utilization</b>                            | <ul style="list-style-type: none"> <li>Inappropriate use of water resources could lead to a reduction of these resources.</li> </ul>                                                                                                                                                                                                                                                  |
| <b>Bulk services (Electricity and sewerage)</b>          | <ul style="list-style-type: none"> <li>Failure to maintain bulk services (i.e. Sewerage pipelines and manholes) may result in leakages which may pollute the surrounding environment.</li> <li>During operation of the proposed project, additional energy will be consumed, resulting in an increased demand on this resource. Energy efficiency resources are essential.</li> </ul> |
| <b>Stormwater management</b>                             | <ul style="list-style-type: none"> <li>The increase in impermeable surfaces associated with the operational phase of the development will reduce stormwater infiltration into the soil, resulting in higher runoff volumes and potentially exacerbating stormwater impacts.</li> </ul>                                                                                                |
| <b>Increase in traffic levels in the area</b>            | <ul style="list-style-type: none"> <li>Increase in traffic and pedestrian volumes is expected during operation phase on existing roads.</li> </ul>                                                                                                                                                                                                                                    |
| <b>Neighbouring upmarket houses value impacts</b>        | <ul style="list-style-type: none"> <li>The development of low-income housing close to Walmer Heights may negatively impact the property prices in the area.</li> </ul>                                                                                                                                                                                                                |

|                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Impacts on traffic safety</b>                                                          | <ul style="list-style-type: none"> <li>Possible collisions with current pedestrian and vehicle traffic as a result of additional vehicle movements</li> </ul>                                                                                                                                                                                                                            |
| <b>Increased water demand</b>                                                             | <ul style="list-style-type: none"> <li>Increased water demand may occur as a result of domestic use and this may place pressure on existing municipal water supply infrastructure which may result to local water shortages.</li> </ul>                                                                                                                                                  |
| <b>Increased pressure on groundwater reserves to augment deficit surface water supply</b> | <ul style="list-style-type: none"> <li>If municipal surface water supply is insufficient to meet operational demand, there may be increased reliance on groundwater abstraction to supplement water supply. This may place increased pressure on local aquifer systems, potentially resulting in declining groundwater levels and reduced borehole yields over the long term.</li> </ul> |
| <b>Employment opportunities</b>                                                           | <ul style="list-style-type: none"> <li>Permanent employment opportunities are expected to directly result from the development.</li> </ul>                                                                                                                                                                                                                                               |
| <b>Contribution to meeting housing need</b>                                               | <ul style="list-style-type: none"> <li>Provision of formal housing as well as services will significantly improve the standard of living of the beneficiaries currently living in informal settlements, and people with different accommodation needs and income levels will have access to housing as different housing typologies are proposed.</li> </ul>                             |

## 8.1 IMPACT ASSESSMENT METHODOLOGY

This section provides the methodology for assessing the significance of impacts associated with the activity. The criterion for determining impact significance has been defined in accordance with the criteria drawn from Appendix 3 of the Environmental Impact Assessment Regulations, 2014. The levels of detail described in the EIA regulations were fine-tuned by assigning specific values to each impact identified. The impact ratings will be informed by the findings of specialist assessments conducted, fieldwork, and desk-top analysis. The significance of potential impacts that may result from the proposed development will be determined in order to assist the competent authority in making a decision.

In order to establish a coherent framework within which all impacts could be objectively assessed, it is necessary to establish a rating system, to be applied consistently to all the criteria. **Table 18** provides a summary of the criteria and the rating scales, which will be used in the assessment of the impacts.

Impacts identified were assessed according to the criteria outlined below. Each impact was ranked according to the nature, extent, duration, magnitude, probability, irreplaceable loss of resources and reversibility. These criteria are based on the Department of Environmental Affairs and Tourism (DEAT) Guideline Document to the EIA Regulations (1998). A significance rating was calculated as per the methodology outlined below.

The significance rating of each identified impact / effect was further reviewed by the EAP and/or specialist by applying professional judgement. Where possible, mitigatory measures were recommended for the impacts identified.

**Table 18.** Impact rating methodology

| EXTENT OF THE IMPACT                                         |                                                                                                                                                                                                                               |   |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| This addresses the physical and spatial scale of the impact. | <b>Site</b> – The impact area extends only as far as the activity – i.e. within the boundaries of the development site.                                                                                                       | 1 |
|                                                              | <b>Local</b> - The impacted area extends slightly further than the actual physical disturbance footprint and could affect the whole, or a measurable portion of adjacent areas (within approx. 5 km of the development site). | 2 |
|                                                              | <b>Landscape</b> - The impact could affect all areas generally visible to the naked eye, as well as those areas essentially linked to the site in terms of ecosystem functioning                                              | 3 |
|                                                              | <b>Regional</b> - The impact could affect the site including the neighbouring areas, transport routes and surrounding towns etc.                                                                                              | 4 |
|                                                              | <b>Ecosystem</b> - The impact could affect areas essentially linked to the site in terms of significantly impacting ecosystem functioning.                                                                                    | 5 |

## DRAFT EIR REPORT: BLIKANA BULK WATER SUPPLY SCHEME

|                                                                                 |                                                                                                                                                                                                                                              |   |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                                                                                 | <b>National</b> - The impacted area extends beyond provincial boundaries.                                                                                                                                                                    | 6 |
|                                                                                 | <b>International</b> - The impacted area extends beyond national boundaries.                                                                                                                                                                 | 7 |
| <b>DURATION OF IMPACT</b>                                                       |                                                                                                                                                                                                                                              |   |
| This describes the predicted lifetime / temporal scale of the predicted impact. | <b>Short term</b> - Quickly reversible. Less than the project lifespan. The impact will either disappear with mitigation or will be mitigated through natural process in a span shorter than any of the project phases or within 0 -5 years. | 1 |
|                                                                                 | <b>Medium term</b> – Some mitigation will be required to reduce the duration of the impact – 6-15 years.                                                                                                                                     | 3 |
|                                                                                 | <b>Long term</b> - the impact will cease when the operation stops.                                                                                                                                                                           | 5 |
|                                                                                 | <b>Permanent</b> - no mitigation measure will reduce the impact after construction.                                                                                                                                                          | 7 |

|                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                    |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>MAGNITUDE OF THE IMPACT</b>                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                    |   |
| This provides a qualitative assessment of the severity of a predicted impact / effect.                                                                                                                                                                                                                                             | <b>None</b> - where the aspect will have no impact on the environment                                                                                                                                                                                              | 0 |
|                                                                                                                                                                                                                                                                                                                                    | <b>Minor</b> - The affected environment is altered, but natural function and process continue.                                                                                                                                                                     | 1 |
|                                                                                                                                                                                                                                                                                                                                    | <b>Low</b> - where the impact affects the environment in such a way that the natural, cultural and social functions / processes are slightly affected.                                                                                                             | 2 |
|                                                                                                                                                                                                                                                                                                                                    | <b>Moderate</b> - where the affected environment is altered but natural, cultural and social functions / processes continue, albeit in a modified way                                                                                                              | 3 |
|                                                                                                                                                                                                                                                                                                                                    | <b>High</b> - natural, cultural or social functions / processes are altered to the extent that they will temporarily cease.                                                                                                                                        | 4 |
|                                                                                                                                                                                                                                                                                                                                    | <b>Very High</b> - natural, cultural or social functions / processes are altered to the extent that they will permanently cease.                                                                                                                                   | 5 |
| <b>IRREPLACEABLE LOSS OF RESOURCES</b>                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                    |   |
| Environmental resources cannot always be replaced; once destroyed, some may be lost forever. It may be possible to replace, compensate for or reconstruct a lost resource in some cases, but substitutions are rarely ideal. The loss of a resource may become more serious later, and the assessment must take this into account. | <b>Short-term</b> – Quickly recoverable. Less than the project lifespan. The resource can be renewed / recovered with mitigation or will be mitigated through natural process in a span shorter than any of the project phases, or in a time span of 0 to 5 years. | 1 |
|                                                                                                                                                                                                                                                                                                                                    | <b>Loss of an ‘expendable’ resource</b> - one that is not deemed critical for biodiversity targets, planning goals, community welfare, agricultural production, or other criteria.                                                                                 | 2 |
|                                                                                                                                                                                                                                                                                                                                    | <b>Medium term</b> – The resource can be recovered within the lifespan of the project. The resource can be renewed / recovered with mitigation or will be mitigated through natural process in a span between 5 and 15 years.                                      | 3 |



## DRAFT EIR REPORT: BLIKANA BULK WATER SUPPLY SCHEME

|  |                                                                                                                                                                                                                                             |   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|  | <b>Loss of an 'at risk' resource</b> - one that is not deemed critical for biodiversity targets, planning goals, community welfare, agricultural production, or other criteria, but cumulative effects may render such loss as significant. | 4 |
|  | <b>Long term</b> – The loss of a non-renewable / threatened resource which cannot be renewed / recovered with, or through, natural process in a time span of over 15 years, but can be mitigated by other means.                            | 5 |
|  | <b>Permanent</b> – The loss of a non-renewable / threatened resource which cannot be renewed / recovered with, or through, natural process in a time span of over 15 years, or by artificial means.                                         | 7 |

### REVERSIBILITY / POTENTIAL FOR REHABILITATION

|                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                        |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| The distinction between reversible and irreversible impacts is a very important one and the irreversible impacts not susceptible to mitigation can constitute significant impacts in an EIA (Glasson et al, 1999). The potential for rehabilitation is the major determinant factor when considering the temporal scale of most predicted impacts. | <b>Short term</b> – The impact / effect will be returned to its benchmark state through mitigation or natural processes in a span shorter than any of the phases of the project, or in a time span of 0 to 5 years.    | 1 |
|                                                                                                                                                                                                                                                                                                                                                    | <b>Medium term</b> – The impact / effect will be returned to its benchmark state through mitigation or natural processes in a span shorter than the lifetime of the project, or in a time span between 5 and 15 years. | 3 |
|                                                                                                                                                                                                                                                                                                                                                    | <b>Long term</b> - The impact / effect will be returned to its benchmark state through extensive mitigation or natural processes in a time span between 15 and 25 years.                                               | 5 |
|                                                                                                                                                                                                                                                                                                                                                    | <b>Permanent</b> – The impact/ effect is permanent and will never be returned to its benchmark state                                                                                                                   | 7 |

### PROBABILITY OF OCCURRENCE

|                                                  |                                                  |   |
|--------------------------------------------------|--------------------------------------------------|---|
| The likelihood of the impact actually occurring. | <b>Remote possibility / unlikely</b>             | 0 |
|                                                  | <b>Possibility</b>                               | 1 |
|                                                  | <b>Low probability / anticipated</b>             | 2 |
|                                                  | <b>Medium probability / strongly anticipated</b> | 3 |
|                                                  | <b>High probability / to be expected</b>         | 4 |
|                                                  | <b>Absolute certainty / will occur</b>           | 5 |

### IMPACT SIGNIFICANCE

The overall significance of an impact / effect has been ascertained by attributing numerical ratings to each identified impact. The numerical scores obtained for each identified impact have been multiplied by the probability of the impact occurring before and after mitigation. High values suggest that a predicted impact / effect is more significant, whilst low values suggest that a predicted impact / effect is less significant.

**((Spatial Extent + Severity + Duration + Resource Lost + Reversibility) \* Probability) = Significance.**

|  |                      |
|--|----------------------|
|  | <b>Overall Score</b> |
|--|----------------------|

## DRAFT EIR REPORT: BLIKANA BULK WATER SUPPLY SCHEME

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>Insignificant</b> – the impact is meaningless has no influence on the decision to develop                                                                                                                                                                                                                                                                                                                                                                                                                        | < 15    |
| <b>Low</b> – the impact would not have a direct influence on the decision to develop in the area;                                                                                                                                                                                                                                                                                                                                                                                                                   | 16 - 35 |
| <b>Medium</b> – the impact could influence the decision to develop in the area unless it is effectively managed / mitigated; and                                                                                                                                                                                                                                                                                                                                                                                    | 36 - 65 |
| <b>High</b> - the impact must have an influence on the decision-making process for development in the area.                                                                                                                                                                                                                                                                                                                                                                                                         | > 65    |
| <b>MITIGATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| In terms of the assessment process the potential to mitigate the negative impacts is determined and rated for each identified impact and mitigation objectives that would result in a measurable reduction or enhancement of the impact are considered. The significance of environmental impacts has therefore been assessed considering any proposed mitigation measures. The significance of the impact “without mitigation” is therefore the prime determinant of the nature and degree of mitigation required. |         |

### **Status of impact**

*The environmental impacts of an activity are those resultant changes in environmental parameters, in space and time, compared with what would have happened had the activity not been undertaken. It is an appraisal of the type of effect the activity would have on the affected environmental parameter. Its description includes what is being affected, and how*

|                                                                               |                               |
|-------------------------------------------------------------------------------|-------------------------------|
| Indication whether the impact is adverse (negative) or beneficial (positive). | + ve (positive – a ‘benefit’) |
|                                                                               | – ve (negative – a ‘cost’)    |

### **Confidence of assessment**

|                                                                                                                     |        |
|---------------------------------------------------------------------------------------------------------------------|--------|
| The degree of confidence in predictions based on available information, EAP’s judgment and/or specialist knowledge. | Low    |
|                                                                                                                     | Medium |
|                                                                                                                     | High   |

## 8.2 IMPACT ASSESSMENT AND SIGNIFICANCE

A summary of all the identified impacts, their associated phase, as well as their impact calculations and significance are presented in **Table 19** below. The impact significance for the No-Go alternative is not calculated as it is negligible.

**Table 19:** Impact Assessment and Significance

| PHASE               | Impact                                                 | Mitigation | Extent | Duration | Magnitude | Irreplaceability | Reversibility | Probability | Significance | Significance Rating | Status | Confidence |
|---------------------|--------------------------------------------------------|------------|--------|----------|-----------|------------------|---------------|-------------|--------------|---------------------|--------|------------|
| Planning and Design | DIRECT IMPACT                                          |            |        |          |           |                  |               |             |              |                     |        |            |
|                     | Legal and social compliance                            | Without    | 2      | 5        | 3         | 5                | 5             | 4           | 80           | High                | - ve   | High       |
|                     |                                                        | With       | 1      | 1        | 1         | 2                | 1             | 3           | 18           | Low                 | - ve   | High       |
|                     | Layout issues                                          | Without    | 1      | 5        | 4         | 3                | 3             | 4           | 64           | Medium              | - ve   | High       |
|                     |                                                        | With       | 1      | 1        | 1         | 1                | 1             | 3           | 15           | Insignificant       | - ve   | High       |
|                     | Site establishment                                     | Without    | 1      | 3        | 3         | 3                | 3             | 4           | 52           | Medium              | - ve   | High       |
|                     |                                                        | With       | 1      | 1        | 1         | 2                | 1             | 3           | 18           | Low                 | - ve   | High       |
|                     | Existing civil services, infrastructure and properties | Without    | 1      | 3        | 3         | 3                | 3             | 4           | 52           | Medium              | - ve   | High       |
|                     |                                                        | With       | 1      | 1        | 1         | 2                | 1             | 3           | 18           | Low                 | - ve   | High       |
|                     | Freshwater utilisation                                 | Without    | 1      | 1        | 1         | 1                | 1             | 3           | 15           | Insignificant       | - ve   | High       |
|                     |                                                        | With       | 1      | 1        | 1         | 1                | 1             | 2           | 10           | Insignificant       | - ve   | High       |
|                     | Bulk services (electricity and sewerage)               | Without    | 1      | 3        | 3         | 3                | 3             | 4           | 52           | Medium              | - ve   | High       |
|                     |                                                        | With       | 1      | 1        | 2         | 1                | 1             | 3           | 18           | Low                 | - ve   | High       |
|                     | Stormwater management                                  | Without    | 1      | 1        | 3         | 3                | 3             | 4           | 44           | Medium              | - ve   | High       |
|                     |                                                        | With       | 1      | 1        | 1         | 2                | 1             | 3           | 18           | Low                 | - ve   | High       |
|                     | DIRECT IMPACT                                          |            |        |          |           |                  |               |             |              |                     |        |            |
| Construction        | Legal and social compliance                            | Without    | 2      | 5        | 3         | 5                | 5             | 4           | 80           | High                | - ve   | High       |
|                     |                                                        | With       | 1      | 1        | 1         | 2                | 3             | 3           | 24           | Low                 | - ve   | High       |
|                     |                                                        | Without    | 1      | 1        | 3         | 2                | 1             | 4           | 32           | Low                 | - ve   | High       |

# DRAFT EIR REPORT: BLIKANA BULK WATER SUPPLY SCHEME

|  |                                           |         |   |   |   |   |   |   |    |               |      |      |
|--|-------------------------------------------|---------|---|---|---|---|---|---|----|---------------|------|------|
|  | Site establishment and security           | With    | 1 | 1 | 1 | 1 | 1 | 3 | 15 | Insignificant | - ve | High |
|  | Water consumption                         | Without | 1 | 1 | 2 | 1 | 1 | 2 | 12 | Insignificant | - ve | High |
|  |                                           | With    | 1 | 1 | 1 | 1 | 1 | 1 | 5  | Insignificant | - ve | High |
|  | Sanitation                                | Without | 1 | 1 | 2 | 1 | 1 | 2 | 12 | Insignificant | - ve | High |
|  |                                           | With    | 1 | 1 | 1 | 1 | 1 | 2 | 10 | Insignificant | - ve | High |
|  | Loss of indigenous vegetation             | Without | 1 | 5 | 4 | 4 | 5 | 4 | 76 | High          | - ve | High |
|  |                                           | With    | 1 | 5 | 3 | 3 | 3 | 3 | 45 | Medium        | - ve | High |
|  | Disturbance of fauna and loss of habitats | Without | 1 | 3 | 3 | 3 | 3 | 4 | 52 | Medium        | - ve | High |
|  |                                           | With    | 1 | 1 | 2 | 2 | 1 | 3 | 21 | Low           | - ve | High |
|  | Loss of SCC                               | Without | 1 | 3 | 3 | 3 | 3 | 4 | 52 | Medium        | - ve | High |
|  |                                           | With    | 1 | 1 | 2 | 2 | 1 | 3 | 21 | Low           | - ve | High |
|  | Loss of animal SCC                        | Without | 2 | 5 | 2 | 2 | 2 | 2 | 26 | Low           | - ve | High |
|  |                                           | With    | 1 | 1 | 1 | 1 | 1 | 1 | 5  | Insignificant | - ve | High |
|  | Loss of biodiversity value                | Without | 2 | 5 | 4 | 4 | 5 | 4 | 80 | High          | - ve | High |
|  |                                           | With    | 1 | 3 | 3 | 3 | 3 | 4 | 52 | Medium        | - ve | High |
|  | Habitat fragmentation                     | Without | 2 | 5 | 3 | 5 | 3 | 4 | 72 | High          | - ve | High |
|  |                                           | With    | 1 | 3 | 2 | 4 | 1 | 3 | 33 | Low           | - ve | High |
|  | Impact on forest                          | Without | 1 | 5 | 4 | 4 | 5 | 4 | 76 | High          | - ve | High |
|  |                                           | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | - ve | High |
|  | Stormwater and erosion impacts            | Without | 2 | 1 | 3 | 2 | 1 | 4 | 36 | Medium        | - ve | High |
|  |                                           | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | - ve | High |
|  | Alien invasion proliferation              | Without | 1 | 3 | 3 | 2 | 3 | 4 | 48 | Medium        | - ve | High |
|  |                                           | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | - ve | High |
|  | Waste management                          | Without | 1 | 1 | 2 | 2 | 1 | 4 | 28 | Low           | - ve | High |
|  |                                           | With    | 1 | 1 | 1 | 1 | 1 | 3 | 15 | Insignificant | - ve | High |
|  | Traffic impact                            | Without | 1 | 3 | 3 | 1 | 1 | 3 | 27 | Low           | - ve | High |
|  |                                           | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | - ve | High |

# DRAFT EIR REPORT: BLIKANA BULK WATER SUPPLY SCHEME

|  |                                                                                    |         |   |   |   |   |   |   |    |               |      |      |
|--|------------------------------------------------------------------------------------|---------|---|---|---|---|---|---|----|---------------|------|------|
|  | Visual impact                                                                      | Without | 1 | 1 | 3 | 1 | 1 | 4 | 28 | Low           | - ve | High |
|  |                                                                                    | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | - ve | High |
|  | Noise pollution                                                                    | Without | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | - ve | High |
|  |                                                                                    | With    | 1 | 1 | 1 | 1 | 1 | 2 | 10 | Insignificant | - ve | High |
|  | Air pollution                                                                      | Without | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | - ve | High |
|  |                                                                                    | With    | 1 | 1 | 1 | 1 | 1 | 2 | 10 | Insignificant | - ve | High |
|  | Paleontological and heritage resource disturbance                                  | Without | 1 | 3 | 3 | 2 | 1 | 4 | 40 | Medium        | - ve | High |
|  |                                                                                    | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | - ve | High |
|  | Health and safety                                                                  | Without | 1 | 1 | 3 | 1 | 1 | 4 | 28 | Low           | - ve | High |
|  |                                                                                    | With    | 1 | 3 | 1 | 1 | 1 | 4 | 28 | Low           | - ve | High |
|  | Employment opportunities                                                           | Without | 2 | 1 | 3 | 1 | 1 | 3 | 24 | Low           | + ve | High |
|  |                                                                                    | With    | 2 | 1 | 2 | 1 | 1 | 3 | 21 | Low           | + ve | High |
|  | CUMULATIVE IMPACTS                                                                 |         |   |   |   |   |   |   |    |               |      |      |
|  | Cumulative loss of indigenous vegetation                                           | Without | 1 | 7 | 3 | 3 | 3 | 4 | 68 | High          | -ve  | High |
|  |                                                                                    | With    | 1 | 3 | 2 | 2 | 1 | 3 | 27 | Low           | -ve  | High |
|  | Increase in traffic levels                                                         | Without | 1 | 3 | 4 | 3 | 3 | 3 | 42 | Medium        | -ve  | High |
|  |                                                                                    | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low           | -ve  | High |
|  | Increased water demand                                                             | Without | 2 | 3 | 4 | 3 | 3 | 4 | 60 | High          | -ve  | High |
|  |                                                                                    | With    | 2 | 1 | 2 | 2 | 1 | 3 | 24 | Low           | -ve  | High |
|  | Increased pressure on groundwater reserves to augment deficit surface water supply | Without | 2 | 1 | 3 | 2 | 1 | 4 | 36 | Medium        | -ve  | High |
|  |                                                                                    | With    | 2 | 1 | 1 | 1 | 1 | 2 | 12 | Insignificant | -ve  | High |
|  |                                                                                    |         |   |   |   |   |   |   |    |               |      |      |
|  | Open spaces and Forest                                                             | Without | 1 | 3 | 3 | 3 | 3 | 4 | 52 | Medium        | -ve  | High |
|  |                                                                                    | With    | 1 | 1 | 1 | 1 | 1 | 3 | 15 | Insignificant | -ve  | High |

# DRAFT EIR REPORT: BLIKANA BULK WATER SUPPLY SCHEME

|                 |                                            |         |   |   |   |   |   |   |    |        |      |      |
|-----------------|--------------------------------------------|---------|---|---|---|---|---|---|----|--------|------|------|
| Operation Phase | patches management                         |         |   |   |   |   |   |   |    |        |      |      |
|                 | Waste management                           | Without | 1 | 3 | 3 | 2 | 1 | 4 | 40 | Medium | - ve | High |
|                 |                                            | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low    | - ve | High |
|                 | Freshwater utilisation                     | Without | 2 | 3 | 3 | 2 | 1 | 4 | 44 | Medium | - ve | High |
|                 |                                            | With    | 2 | 1 | 2 | 1 | 1 | 3 | 21 | Low    | - ve | High |
|                 | Bulk services (electricity and sewerage)   | Without | 1 | 5 | 2 | 3 | 3 | 3 | 42 | Medium | - ve | High |
|                 |                                            | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low    | - ve | High |
|                 | Stormwater management                      | Without | 2 | 3 | 3 | 3 | 3 | 3 | 42 | Medium | - ve | High |
|                 |                                            | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low    | - ve | High |
|                 | Increase in traffic levels in the area     | Without | 2 | 3 | 4 | 1 | 1 | 4 | 44 | Medium | - ve | High |
|                 |                                            | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low    | - ve | High |
|                 | Neighbouring upmarket houses value impacts | Without | 2 | 5 | 4 | 3 | 3 | 4 | 68 | High   | - ve | High |
|                 |                                            | With    | 2 | 1 | 2 | 2 | 1 | 3 | 24 | Low    | - ve | High |
|                 | Impacts on traffic safety                  | Without | 1 | 5 | 3 | 3 | 3 | 4 | 60 | High   | - ve | High |
|                 |                                            | With    | 1 | 1 | 2 | 1 | 1 | 3 | 18 | Low    | - ve | High |
|                 | Employment opportunities                   | Without | 2 | 5 | 2 | 1 | 1 | 4 | 44 | Medium | + ve | High |
|                 |                                            |         |   |   |   |   |   |   |    |        |      |      |
|                 | Contribution to meeting housing need       | Without | 1 | 5 | 2 | 1 | 1 | 4 | 40 | Medium | + ve | High |
|                 |                                            |         |   |   |   |   |   |   |    |        |      |      |

### 8.3 PROPOSED MITIGATION MEASURES

Appendix 3 of the EIA Regulations, 2014 requires that possible mitigation measures that could be applied to avoid or mitigate negative impacts and optimize positive impacts must be identified in the EIA Report. Many of the impacts can be readily mitigated and it is not foreseen that they are likely to pose a significant risk. Where necessary, the EMPr will identify and recommend specific mitigation measures applicable to the project. **Table 20** identifies mitigation measures that have been proposed by specialists in the respective reports as well as those tabled by the EAP as identified during the Environmental Impact Assessment. These mitigation measures will also be included in the EMPr for implementation during the construction and operational phases.

**Table 20:** Proposed mitigation measures

| Impacts/ Aspects                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PLANNING AND DESIGN PHASE</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Legal and social compliance</b> | <ul style="list-style-type: none"> <li>• Ensure environmental authorisations namely EA is obtained for the development.</li> <li>• All relevant plant and animal removal/relocation permits must be obtained from the competent authorities to remove any protected plant and animal species.</li> <li>• Environmental Control Officer (ECO) to be appointed to monitor compliance with permits conditions.</li> <li>• All health and safety permits and certification to be in place.</li> <li>• Appointment of a Community Liaison Officer (CLO).</li> <li>• Appointment of a Social Facilitator (SF) to manage social-economic aspects of the project relating to community and local business interest, particularly small, medium and micro-enterprise (SMMEs).</li> <li>• A health and safety plan in terms of the Occupational Health and Safety Act (Act No 85 of 1993) must be drawn up by the HSE officer prior to construction to ensure workers safety.</li> <li>• Environmental files to be compiled prior to commencement of construction activities.</li> <li>• Methods statement to be compiled for all works with environmental aspects and impacts, for approval of the ECO.</li> <li>• Employment and sub-contracting of locals.</li> </ul> |
| <b>Layout issues</b>               | <p>The development layout must conform to the following;</p> <ul style="list-style-type: none"> <li>• Stormwater ponds should be located outside forest patches.</li> <li>• The business site to be designed such that there is no vehicular access from both the northern and southern portions of the development but that no through flow of vehicular traffic is possible. Separate parking areas to be provided for around the business sites.</li> <li>• A pedestrian walkway along the edge of the business site to be provided, with suitable concrete bollards or similar barrier to prevent vehicular access via the pedestrian walkway</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



## DRAFT ENVIRONMENTAL IMPACT REPORT: WALMER HOUSING DEVELOPMENT ON ERF 11305

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                               | <ul style="list-style-type: none"> <li>• Zoning of the forest patches must be Open Space III and managed as for conservation usage. The buffers around the patches as well as other open spaces to be zoned as Public Open Space (POS) I</li> <li>• Stormwater ponds to be restricted to the edges of any forest patches and in accordance with any conditions related to any authorisation obtained in terms of the NFA from DFFE</li> </ul>                                                                                                                                                                                                                                                                                                   |
| <b>Site establishment</b>                                     | <ul style="list-style-type: none"> <li>• Develop a site establishment method statement</li> <li>• The Contractor must prepare a Construction Site Plan prior to establishing on-site. This plan must indicate: <ul style="list-style-type: none"> <li>• The boundaries of the site that encompass all construction-related activities.</li> <li>• Vehicle and pedestrian access points and routes (safe diversions of routes when necessary).</li> <li>• Laydown area/s, offices, stockpile areas, storage areas, etc.</li> </ul> </li> <li>• Construction site to be selected by the contractor in conjunction with the developer, consulting engineer and the ECO.</li> </ul>                                                                 |
| <b>Existing civil services, infrastructure and properties</b> | <ul style="list-style-type: none"> <li>• Ensure that all existing services and infrastructure are known, mapped and marked to prevent potential damage.</li> <li>• Development of a plan to manage existing services to avoid damages</li> <li>• Wayleaves to be obtained from the relevant custodians for affected servitudes as applicable.</li> <li>• Property owners such as the local residents should be engaged throughout the project lifespan and engaged as necessary to promote conflict free development implementation.</li> <li>• Where relevant, the directly affected property owners should be informed in writing and engaged to reach a consensus and those proceedings and relevant agreements to be documented.</li> </ul> |
| <b>Water consumption</b>                                      | <ul style="list-style-type: none"> <li>• Water conservation measures must be adopted into the building design.</li> <li>• Rainwater tanks should be considered.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Bulk services (Electricity and sewerage)</b>               | <ul style="list-style-type: none"> <li>• A Bulk Services Report should be conducted.</li> <li>• The design of all residential dwellings, as well as community facilities, school buildings and business buildings should consider the use of alternative renewable energy sources such as CFL or LED lighting, solar geysers, solar street lighting etc.</li> <li>• All buildings should have a rainwater harvesting tanks.</li> <li>• Ensure that the municipality will have sufficient capacity for connection to existing infrastructure.</li> </ul>                                                                                                                                                                                         |
| <b>Stormwater management</b>                                  | <ul style="list-style-type: none"> <li>• Infrastructure should be planned and designed in such a way as to take increased stormwater runoff in consideration.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>CONSTRUCTION PHASE</b>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Legal and social compliance</b>                            | <ul style="list-style-type: none"> <li>• ECO to be appointed to audit environmental compliance with conditions in the EMP and other permits and authorisations.</li> <li>• CLO and Social Facilitator to be retained throughout the construction tenure to represent the surrounding communities and local business, particularly SMMEs.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Site establishment and security</b>                        | <ul style="list-style-type: none"> <li>• Fulltime security officers to be appointed to safeguard the site, its personnel, plant and equipment</li> <li>• Necessary health and safety as well as road construction signage to be installed.</li> <li>• Construction areas are to be clearly demarcated, and all construction activities are to be restricted to within these demarcated areas.</li> </ul>                                                                                                                                                                                                                                                                                                                                        |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Water consumption</b>                          | <ul style="list-style-type: none"> <li>• A sustainable and lawful water supply should be utilised.</li> <li>• Consistent water supply for construction staff and activities.</li> <li>• If water is stored on-site, drinking water and multi-purposed water storage facilities shall be clearly distinguished and demarcated.</li> <li>• Implement rainwater harvesting techniques, where possible.</li> <li>• Re-use wash water for selected activities such as concrete batching.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Sanitation</b>                                 | <ul style="list-style-type: none"> <li>• Ensure availability of ablutions for males and females. One ablution facility for every 10 workers during construction phase.</li> <li>• Ensure regular servicing of the ablution toilets by a service provider.</li> <li>• Ablution facilities to be located outside the regulated water area by all means practically possible.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Loss of indigenous vegetation</b>              | <ul style="list-style-type: none"> <li>• Minimise vegetation clearance and confine to the proposed infrastructural footprint.</li> <li>• No blanket clearing of vegetation is to occur on the development site. Vegetation removal must be done in phases, to reduce the area of any any exposed surfaces and potential for dust pollution to occur at any one time.</li> <li>• When excavating, ensure that topsoil is separated from the subsoil to preserve the indigenous vegetation seedbanks of the area.</li> <li>• Use existing access roads as opposed to initiating new route, the latter should be authorised first prior to initiating.</li> <li>• Construction should not take place outside the authorised site footprint</li> <li>• Minimise vegetation clearance as practically possible to do so.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Disturbance of fauna and loss of habitats.</b> | <ul style="list-style-type: none"> <li>• A suitably qualified individual is to carry out a faunal “search and rescue”, which fauna are to be relocated to a suitably protected area prior commencement of vegetation clearing. Should any injured fauna be found, they are to be taken to be examined by a veterinarian and if deemed suitable for rehabilitation, such are to be taken to a suitable faunal rehabilitation centre, in consultation with the department’s biodiversity unit. Any species protected in terms of the Cape Nature and Environmental Ordinance (Ordinance 19 of 1974) which need to be removed or relocated require the necessary permits to be obtained from DEDEAT.</li> <li>• Any fauna or avi-fauna on site is to remain undisturbed as far as possible and shall not be trapped/killed or otherwise caught by any contractor or individuals carrying out any clearing or construction work.</li> <li>• No termite mounds should be intentionally destroyed. Any lizards, geckos, agamids, monitors or snakes encountered should be allowed to escape to a suitable habitat away from the disturbance.</li> <li>• The construction must be completed as quickly as possible - no fauna species may be killed.</li> <li>• Any bird nests that are found during the construction period must be reported to the Environmental Control Officer (ECO).</li> </ul> |
| <b>Loss of plant SCC</b>                          | <ul style="list-style-type: none"> <li>• No plant SCC were observed. Other plants protected by provincial legislation do occur.</li> <li>• All relevant plant permits must be obtained prior to commencement of clearing.</li> <li>• A qualified botanist must be appointed to oversee all protected plant removal.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | <ul style="list-style-type: none"> <li>Removed plants must be planted in a similar environment nearby. Alternatively, plants can be kept in an onsite nursery or donated to a local rehabilitation nursery.</li> <li>Some of the plants may be used during rehabilitation of the site after development.</li> <li>No plant harvesting by construction staff will be allowed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Loss of animal SCC</b>             | <ul style="list-style-type: none"> <li>No animal SCC were identified.</li> <li>Animals protected by provincial legislation may occur. These include local reptiles (snakes, lizards, geckos, tortoises and chameleons). All reptiles in the Eastern Cape are protected.</li> <li>Permits (if required) must be obtained to relocate any protected animals identified.</li> <li>Animals must only be relocated into the surrounding habitats.</li> <li>Animals must be relocated by a trained and qualified handler as some of them may be venomous.</li> <li>No poaching by construction staff will be allowed.</li> <li>No killing of wild animals will be allowed.</li> </ul>                                                                                          |
| <b>Loss of biodiversity value</b>     | <ul style="list-style-type: none"> <li>All areas outside the demarcated construction footprint will be considered as No-Go areas. – Forest patches must be avoided. Permits must be obtained from Forestry to impact in any of the delineated forest patches.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Habitat fragmentation</b>          | <ul style="list-style-type: none"> <li>No areas of forest/thicket vegetation occurring outside the areas designated 'Open Space Zone 3 (conservation usage)' in the site layout plan/rezoning and subdivision plan should be lost because of the clearing.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Impact on forest</b>               | <ul style="list-style-type: none"> <li>Permits must be obtained from Forestry to impact in any of the delineated forest patches.</li> <li>Forest patches should be no-go areas during construction.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Stormwater and erosion impacts</b> | <ul style="list-style-type: none"> <li>Minimise disturbance and clearing of vegetation</li> <li>The site must be continuously monitored for erosion.</li> <li>Appropriate erosion reducing methods must be implemented if required.</li> <li>Where feasible, the river and wetlands within the Secondary PAOI should be protected from polluted runoff by cut-off drains and/or similar interventions.</li> <li>Residential stormwater runoff that exceeds the 1:5 year flood volumes can be conveyed directly into drainage, if of acceptable quality. All other stormwater runoff, including runoff from the first seasonal rains, should be directed elsewhere or treated onsite, such that the water quality has been polished to an acceptable standard.</li> </ul> |
| <b>Alien invasion proliferation</b>   | <ul style="list-style-type: none"> <li>Develop and implement an Alien Vegetation Management Plan to mitigate the establishment and spread of undesirable alien plant species during construction and operations.</li> <li>Site must be continuously monitored and managed for alien vegetation growth.</li> <li>Any alien invasive species which colonises disturbed ground must be systematically removed and destroyed prior to it attaining the seed formation stage.</li> <li>Removal must occur through appropriate methods such as hand pulling, application of chemicals, cutting, etc. in accordance with the NEMBA: Alien Invasive Species Regulations.</li> </ul>                                                                                              |
| <b>Waste management</b>               | <ul style="list-style-type: none"> <li>Develop a comprehensive solid waste management plan.</li> <li>Provide clearly labelled waste bins with lids</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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|                                                | <ul style="list-style-type: none"> <li>• Littering is strictly prohibited.</li> <li>• All waste generated on site shall be collected and appropriately disposed of at a registered municipal landfill site.</li> <li>• No on-site burning, burying or dumping of any waste materials, litter or refuse shall occur.</li> <li>• The Contractor shall identify and separate materials that can be re-used or recycled to minimise waste e.g. metals, packaging and plastics, and provide separate marked bins for these items.</li> <li>• Skips are to be provided for the temporary storage of construction waste.</li> <li>• All staff shall be trained on correct waste management.</li> <li>• Records of disposal of all waste generated on site shall be maintained for auditing purposes.</li> <li>• No wastewater shall be disposed of to the surrounding soil or natural water resources.</li> <li>• All effluent water from the camp/ office sites shall be disposed of in a properly designed and constructed system, situated so as not to adversely affect water sources.</li> <li>• All cement wastewater shall be collected in a container, allowed to evaporate, and the sludge disposed of as waste. Under no circumstances shall it be allowed to enter soil, groundwater resources or stormwater.</li> </ul> |
| <b>Hazardous material storage and handling</b> | <ul style="list-style-type: none"> <li>• No cement/concrete mixing is permissible on the soil surface. Cement mixers are to be placed on large trays to prevent accidental spills from coming into contact with soil surface.</li> <li>• Generators and fuel supply needed during construction must be placed on trays which rest on clean sand. Once construction has been completed, this sand must be removed from site and disposed of at a registered waste disposal site.</li> <li>• No servicing of vehicles and other machinery to take place on site and no fuel or other hazardous material to be stored on site.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Traffic impact</b>                          | <ul style="list-style-type: none"> <li>• Ensure construction vehicles are visible and make use of Victoria Drive to gain access to site.</li> <li>• Pedestrians and motorists should be accommodated through appropriate traffic warning signs and measures.</li> <li>• Speed should be kept at a minimum next to the construction site.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Visual impact</b>                           | <ul style="list-style-type: none"> <li>• The site camp must be clearly demarcated and screened in order to block any unappealing aesthetics within the site camp such material storage areas.</li> <li>• The construction footprint must be kept as small as possible, to avoid unnecessary disruption to the surrounding environment.</li> <li>• Disturbed areas to be rehabilitated / revegetated as soon as possible during the construction phase.</li> <li>• Areas utilised for construction at the site must be kept clean and tidy at all times.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Noise pollution</b>                         | <ul style="list-style-type: none"> <li>• Ensure that construction vehicles and working machinery are serviced and are in good condition to reduce their noise levels.</li> <li>• Construction activities should be maintained during the normal working hours (08h00-17h00).</li> <li>• Where works are to be carried outside the normal working hours, the affected adjacent residents must be informed and such be undertaken within the shortest time possible.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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|                                                                          | <ul style="list-style-type: none"> <li>• Vehicles and plant to be inspected daily for defects which may cause elevated noise generation.</li> <li>• Vehicles and plant to be adequately serviced and maintained in good condition.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Air pollution</b>                                                     | <ul style="list-style-type: none"> <li>• Implement dust suppression measures as required.</li> <li>• Vehicle speed should be kept at a minimal to reduce dust pollution.</li> <li>• Construction plant, equipment, machinery and vehicles should be well maintained and serviced regularly to minimise exhausted fumes air pollution.</li> <li>• Exposed areas should be re-vegetated as soon as all installations are completed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Paleontological and Archaeological, heritage resource disturbance</b> | <p><b>Archaeological:</b></p> <ul style="list-style-type: none"> <li>• A qualified archaeologist must conduct archaeological monitoring during all vegetation clearing activities and subsequent excavations associated with the proposed development. Monitoring is required due to the confirmed presence of archaeological human remains and associated cultural material along the north western boundary of the project area, as well as the broader heritage sensitivity of the Walmer dune landscape. Vegetation clearing and initial earthworks have the potential to expose additional burials, shell middens, stone artefact scatters, or other subsurface archaeological deposits that were not visible during the desktop update.</li> <li>• Monitoring must occur during: <ul style="list-style-type: none"> <li>○ Mechanical or manual vegetation removal, particularly in areas of dune deflation, natural hollows, or previously undisturbed terrain.</li> <li>○ Topsoil stripping and bulk earthworks, including trenching, levelling, and foundation excavations.</li> <li>○ Any ground-disturbing activities within proximity to the previously exposed burial locality.</li> </ul> </li> <li>• If archaeological material, human remains, or burial related features are identified at any stage, all work in the immediate area must cease, and the finds must be reported to ECPHRA in accordance with Section 36 of the National Heritage Resources Act (Act 25 of 1999). Work may only resume once the heritage authority has issued written instructions or a permit authorising further action.</li> <li>• This monitoring requirement must be incorporated into the contractor's method statements and the project's Environmental Management Programme (EMPr) to ensure compliance with NEMA and the NHRA.</li> <li>• It is recommended that living heritage considerations be formally included as part of the public participation process in accordance with Sections 3(2)(b) and 38(3) of the National Heritage Resources Act (NHRA, Act 25 of 1999). Public consultation must provide affected communities, traditional leaders, and knowledge holders with the opportunity to identify culturally meaningful places, ritual spaces, ancestral associations, and other forms of living heritage that was not visible during field assessment. The archaeologist must be informed of any living heritage spaces that may be identified during the public participation process. These spaces must be identified and documented together with the</li> </ul> |

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|                                        | <p>affected communities who have identified them. The results of the consultation must be included as part of an updated or revised report including mitigation measures for these sites and possible no-go zones.</p> <ul style="list-style-type: none"> <li>• Contractor teams must be briefed on heritage sensitivities, chance find procedures, and reporting requirements. Heritage awareness should form part of environmental induction and ongoing site compliance monitoring.</li> <li>• All heritage related actions, including public participation, consultation, monitoring, chance find responses, and assessments—must be documented and included in the final heritage compliance report submitted to ECPHRA.</li> <li>• If concentrations of pre-colonial archaeological heritage material, including subsurface dense artefact accumulations, associated cultural deposits, or human remains such as graves or burials—are uncovered during construction, all work must cease immediately and the discovery must be reported without delay to the Eastern Cape Provincial Heritage Resources Agency (ECPHRA) at 043 492 1370. No further disturbance may occur until the heritage authority has provided guidance.</li> </ul> <p><b>Paleontological:</b></p> <ul style="list-style-type: none"> <li>• The Environmental Compliance Officer (ECO) responsible for overseeing the project should be well-informed about the potential presence of fossils within the project area. It is essential that they are aware of the likelihood of uncovering fossils during excavation into the underlying strata.</li> <li>• The construction managers and supervisory personnel involved in the project should also be made aware of the possibility of encountering significant fossils on-site.</li> <li>• During excavation, it is crucial that any geological exposures encountered are thoroughly inspected for the presence of fossil remains, particularly fossils of high scientific value e.g. vertebrate teeth and bones. If any fossils of significance are discovered, the ECO should contact the appropriate authority to further investigate the area.</li> <li>• In the event of significant fossil discoveries, a qualified palaeontologist should be promptly notified. This palaeontologist should possess the expertise and qualifications necessary for accurately assessing and documenting the findings.</li> <li>• Any fossils of scientific value found within the excavation area should be regarded as of palaeontological interest. The palaeontologist should be responsible for recording and sampling these fossils. The costs associated with this activity should be borne by the developer as part of their responsibility for preserving and managing palaeontological resources.</li> <li>• If a new paleontological resource is discovered, it is the jurisdiction of the SAHRA to recover and store such resource. Hence, further exploration must be conducted by a professional palaeontologist.</li> </ul> |
| <p><b>Employment opportunities</b></p> | <ul style="list-style-type: none"> <li>• Employment of local labour and contracting of SMMEs.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p><b>OPERATIONAL PHASE</b></p>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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| <b>Open spaces and Forest patches management</b>             | <ul style="list-style-type: none"> <li>No forest patches are to be developed or impacted on. These sites are to be appropriately zoned to Open Space III for conservation purposes and the protection thereof and must be incorporated into the NMBM's Metropolitan Open Space Management Plan/Bioregional Plan and managed as a conservation area.</li> <li>The forest patches must be fenced off with a suitable fencing solution such as "Bettafence" to prevent dumping and destruction of the forest.</li> <li>Regular inspections of the fenced areas are to be made to monitor whether dumping of rubble or refuse is occurring.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Waste management</b>                                      | <ul style="list-style-type: none"> <li>Regular (weekly) waste collection service to be provided.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Freshwater utilization</b>                                | <ul style="list-style-type: none"> <li>Water-wise practices should be considered such as use of rainwater for non-portable water requirements.</li> <li>Leak monitoring, detection repairs be implemented.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Bulk services</b>                                         | <ul style="list-style-type: none"> <li>Consider use of alternative energy sources such as solar panels placed on the infrastructure roofs to power the facility.</li> <li>The municipality must ensure sewer system is not overloaded.</li> <li>Regular maintenance of the system required.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Stormwater management</b>                                 | <ul style="list-style-type: none"> <li>Once the storm water facilities have been completed, and fenced to prevent access by the public, the maintenance and monitoring will remain the responsibility of the municipality. These are some of the suggested Best Management Practices (BMPs) for the development <ul style="list-style-type: none"> <li>Public open space – Vegetation to be maintained such that flood water levels are not increased.</li> <li>Storm water structures – Ponds frequently monitored and obstructions (debris) to inlet/outlet works, removed.</li> <li>Open Channels – Channels to be regularly monitored to identify any erosion gullies or silt deposition.</li> <li>Road Construction – No concentrated flow points are allowed onto road area.</li> <li>Pipes – All pipes are to be frequently monitored and cleared of obstructions.</li> </ul> </li> <li>Develop a comprehensive stormwater management program which addresses management of stormwater detention ponds, the discharge outlets, inclusive of erosion protection measures and mitigation of such erosion occur.</li> <li>Include periodical site inspection that inspects the effectiveness of the run-off control system and specifically records occurrence or not of any erosion on site or downstream.</li> <li>Outlet structures at road culverts to be designed in a manner that dissipates flow energy.</li> <li>Vegetation to be maintained in public open space such that flood water levels are not increased.</li> <li>Ponds frequently monitored and obstructions (debris) to inlet and outlet works, removed.</li> <li>Channels to be regularly monitored to identify any erosion gullies or silt deposition.</li> </ul> |
| <b>Increase in traffic and pedestrian levels in the area</b> | <ul style="list-style-type: none"> <li>Upgrade intersections; provide additional facilities for public transport and vulnerable road users (pedestrians and cyclists).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



## DRAFT ENVIRONMENTAL IMPACT REPORT: WALMER HOUSING DEVELOPMENT ON ERF 11305

|                                                   |                                                                                                                                                                                                                                                                |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Impact on traffic safety</b>                   | <ul style="list-style-type: none"><li>• Implement public transport and NMT facilities and services, e.g. sidewalks, street lighting, public transport feeder services.</li></ul>                                                                               |
| <b>Neighbouring upmarket houses value impacts</b> | <ul style="list-style-type: none"><li>• No deviating from definite pricing contours in the layout.</li><li>• The area directly adjacent Walmer Heights must be reserved for higher income households, with lower priced housing along Victoria Drive</li></ul> |
| <b>Increased water demand</b>                     | <ul style="list-style-type: none"><li>• Implement water-efficient building design.</li><li>• Promote water conservation awareness among residents.</li></ul>                                                                                                   |
| <b>Employment opportunities</b>                   | <ul style="list-style-type: none"><li>• N/A</li></ul>                                                                                                                                                                                                          |
| <b>Contribution to meeting housing need</b>       | <ul style="list-style-type: none"><li>• N/A</li></ul>                                                                                                                                                                                                          |

## 9. ENVIRONMENTAL IMPACT STATEMENT

The proposed housing development is intended to contribute to the provision of formal housing opportunities within the Nelson Mandela Bay Municipality, where housing demand and informal settlement pressures remain a key planning and service delivery challenge. The development is considered desirable as the site is municipally owned, located close to the existing Walmer Gqebera township, and has been earmarked for medium- to high-density residential development. Its location adjacent to existing residential areas also supports spatial integration and allows for potential connection to existing municipal services and infrastructure.

The No-Go alternative would avoid the direct environmental impacts associated with construction and land transformation. However, it would also prevent the realisation of the socio-economic benefits associated with the proposed development, including the provision of housing opportunities, employment creation during construction, improved infrastructure provision, and the formal planning of land that is currently subject to disturbed and informal land uses such as grazing, recreational use, and illegal dumping.

Based on the findings of the assessment, the proposed development is associated with both positive and negative impacts. The positive impacts relate to housing delivery, employment opportunities, and support for local socio-economic development. Potential negative impacts are expected to be mainly associated with the construction phase and may include vegetation clearance, disturbance to sensitive vegetation or habitat areas, soil disturbance, erosion, dust generation, noise, waste generation, and temporary traffic-related impacts. The site contains environmentally sensitive forest patches, which have been considered in the planning and design of the proposed development. The layout has been designed to avoid and minimise impacts on these areas as far as possible, including through the establishment of buffer areas and the relocation of stormwater detention ponds away from the forest patches.

According to the terrestrial biodiversity assessment the site contains remnant patches of indigenous forest/thicket vegetation within a landscape that has largely been transformed through historical disturbance, invasive alien plant infestation, grazing, recreational activities, and illegal dumping. The screening tool identified the sensitivities for plant and animal species as high however these were reduced to medium and low respectively, primarily because no Species of Conservation Concern (SCC), or other sensitive or threatened plant and animal species, were recorded during the survey. The terrestrial biodiversity sensitivity, however, remains very high due to the presence of remnant patches associated with Critically Endangered vegetation types, including Algoa Sandstone Fynbos and Sardinia Forest Thicket. However, the specialist

concluded that the vegetation within the proposed development footprint is largely degraded and consists predominantly of secondary grassland and dense stands of invasive alien species, including Rooikrans (*Acacia cyclops*) and Port Jackson Willow (*Acacia saligna*). Additionally the vegetation is no longer representative of the original and Critically Endangered/Vulnerable vegetation/ecosystem type(s) present before development of the project site. The ecological impacts of all aspects for the proposed development were assessed and considered to be acceptable, provided that all mitigation measures provided in this report are implemented.

Based on the assessment by the EAP and specialists it is anticipated that the identified impacts can be mitigated to acceptable levels through the implementation of the Environmental Management Programme (EMPr) and adherence to specialist recommendations.

## **9.1 SUMMARY OF IDENTIFIED ENVIRONMENTAL IMPACTS**

The tables below summarise all the potential impacts anticipated during the construction and operation phases of the proposed development.

### **CONSTRUCTION PHASE**

**Table 21.** Identified impacts on construction phase

| <b>Impact</b>                             | <b>Mitigation</b> | <b>Significance rating</b> |
|-------------------------------------------|-------------------|----------------------------|
| Legal and social compliance               | Without           | High                       |
|                                           | With              | Low                        |
| Site establishment and security           | Without           | Low                        |
|                                           | With              | Insignificant              |
| Water consumption                         | Without           | Insignificant              |
|                                           | With              | Insignificant              |
| Sanitation                                | Without           | Insignificant              |
|                                           | With              | Insignificant              |
| Loss of indigenous vegetation             | Without           | High                       |
|                                           | With              | Medium                     |
| Disturbance of fauna and loss of habitats | Without           | Medium                     |
|                                           | With              | Low                        |
| Loss of SCC                               | Without           | Medium                     |
|                                           | With              | Low                        |
| Impact on forest                          | Without           | High                       |
|                                           | With              | Low                        |
| Stormwater and erosion impacts            | Without           | Medium                     |
|                                           | With              | Low                        |
| Alien invasion proliferation              | Without           | Medium                     |
|                                           | With              | Low                        |

| Impact                                            | Mitigation | Significance rating |
|---------------------------------------------------|------------|---------------------|
| Waste management                                  | Without    | Low                 |
|                                                   | With       | Insignificant       |
| Traffic impact                                    | Without    | Low                 |
|                                                   | With       | Low                 |
| Visual impact                                     | Without    | Low                 |
|                                                   | With       | Low                 |
| Noise pollution                                   | Without    | Low                 |
|                                                   | With       | Insignificant       |
| Air pollution                                     | Without    | Low                 |
|                                                   | With       | Insignificant       |
| Paleontological and heritage resource disturbance | Without    | Medium              |
|                                                   | With       | Low                 |
| Health and safety                                 | Without    | Low                 |
|                                                   | With       | Low                 |
| Employment opportunities                          | Without    | Low                 |
|                                                   | With       | Low                 |

## **OPERATION PHASE**

**Table 22.** Identified impacts on operation phase

| Impact                                     | Mitigation | Significance rating |
|--------------------------------------------|------------|---------------------|
| Open spaces and Forest patches management  | Without    | Medium              |
|                                            | With       | Insignificant       |
| Waste management                           | Without    | Medium              |
|                                            | With       | Low                 |
| Freshwater utilisation                     | Without    | Medium              |
|                                            | With       | Low                 |
| Bulk services (electricity and sewerage)   | Without    | Medium              |
|                                            | With       | Low                 |
| Stormwater management                      | Without    | Medium              |
|                                            | With       | Low                 |
| Increase in traffic levels in the area     | Without    | Medium              |
|                                            | With       | Low                 |
| Neighbouring upmarket houses value impacts | Without    | High                |
|                                            | With       | Low                 |
| Impacts on traffic safety                  | Without    | High                |
|                                            | With       | Low                 |
| Employment opportunities                   | Without    | Medium (+)          |
|                                            | With       |                     |
| Contribution to meeting housing need       | Without    | Medium (+)          |
|                                            | With       |                     |

**Cumulative impacts****Table 23.** Identified cumulative impacts

| Impact                                                                             | Mitigation | Significance rating |
|------------------------------------------------------------------------------------|------------|---------------------|
| Cumulative loss of indigenous vegetation                                           | Without    | High                |
|                                                                                    | With       | Low                 |
| Increase in traffic levels                                                         | Without    | Medium              |
|                                                                                    | With       | Low                 |
| Increased water demand                                                             | Without    | High                |
|                                                                                    | With       | Low                 |
| Increased pressure on groundwater reserves to augment deficit surface water supply | Without    | Medium              |
|                                                                                    | With       | Insignificant       |

**9.2 POSITIVE AND NEGATIVE IMPLICATIONS OF THE PROPOSED PROJECT**

The following negative and positive impacts are anticipated from the proposed development:

**Table 24.** Positive and negative impacts

| NEGATIVE IMPACTS                                                            | POSITIVE IMPACTS                     |
|-----------------------------------------------------------------------------|--------------------------------------|
| Impacts on terrestrial ecosystems- Habitat loss due to vegetation clearance | Creation of local employment         |
| Disturbance to fauna and habitat fragmentation                              | Contribution to meeting housing need |
| Soil erosion                                                                |                                      |
| Impact on Heritage and Cultural Resources                                   |                                      |
| Risk of alien invasive species spread due to disturbance                    |                                      |
| Noise Impact                                                                |                                      |
| Waste                                                                       |                                      |
| Impact on Air Quality                                                       |                                      |
| Increase in traffic levels                                                  |                                      |

### 9.3 FINAL COMPOSITE MAP

The environmental sensitivities/constraints of the sites are illustrated in **Figure 18** below.



**Figure 18:** Final composite map

## **10. ASSUMPTIONS, UNCERTAINTIES OR GAPS IN KNOWLEDGE**

The following assumptions and limitations have been identified by the EAP and some have been extracted from the various specialist studies undertaken for this project.

### **EAP:**

- All information provided by the Applicant to the EAP was correct and valid at the time it was provided.
- The EAP does not accept any responsibility in the event that additional information comes to light at a later stage of the process.
- All data from unpublished research is valid and accurate at the time this report was written.
- The scope of this investigation is limited to assessing the potential environmental impacts associated with the proposed development only.

### **TERRESTRIAL BIODIVERSITY ASSESSMENT**

- The report is based on currently available information and, as a result, limited by the information provided by the Client.
- The report is limited by seasonality as the presented data will be based on one site survey of plant and animal species conducted within a single season (winter) of a single year (2026).

### **AQUATIC BIODIVERSITY IMPACT ASSESSMENT**

- The report is based on currently available information and, as a result, limited by the information provided by the Client.
- The report is limited by seasonality as the presented data will be based on a single site survey of the site characteristics conducted within a single season (early spring) of a single year (2024).
- A desktop delineation of the wetland and riparian area was done before the site visit. This is thought to be an acceptable method.
- The watercourses were mainly dry and aquatic assessments such as water quality measurements, macroinvertebrate (South African Scoring System Version 5 -SASS5), Integrated Habitat Assessment System (IHAS), Macro-Invertebrate Response Assessment Index (MIRAI version 2) and fish sampling (Fish Response Assessment Index - FRAI) were excluded.
- The watercourse assessment is confined to the proposed project footprint and does not include the neighbouring and adjacent properties, which were only considered as part of the desktop assessment.

- Global Positioning System (GPS) technology is inherently inaccurate and some inaccuracies due to the use of handheld GPS instrumentation (Garmin eTrex) may occur. If more accurate assessments are required, the watercourses will need to be surveyed and pegged according to surveying principles.
- The risk assessment was applied on the basis that the stipulated mitigation measures and all specialist recommendations will be implemented, and therefore the results presented demonstrate the impact significance of perceived impacts on the receiving freshwater environment post-mitigation.
- All information provided by the EAP, applicant and engineering design team to the environmental specialist was correct and valid at the time that it was provided.
- The proposed project footprint as provided by the engineering design team is correct and will not be significantly deviated from.
- Significant reliance on visual eco-morphological observations was made to derive an understanding of the state of the habitat within the subject site. This state may change under a different meteorological regime.
- Freshwater resources that fall outside of the affected catchment (but still within the 500 m DWS regulated area) and are not at risk of being impacted (such as upslope water resources) by the specific activity were not delineated nor assessed. Such features were flagged during a baseline desktop assessment prior to the site visit.
- This assessment deals primarily with inland wetlands (i.e. no existing connection to the ocean and these ecosystems are characterised by the complete absence of marine exchange and/or tidal influence) (Ollis et al., 2013).
- Selection of assessment techniques and tools were based on the assessment practitioner's knowledge and experience of these tools and their attributes and shortcomings.
- The assessment techniques and tools are currently the most appropriate available tools and techniques to undertake assessments of freshwater resources; they are rapid assessment tools that rely on qualitative information and expert judgment. While these tools have been subjected to peer review processes, the methodology for these tools are ever evolving and will likely be further refined in the future. For the purposes of this assessment, the assessments were undertaken at rapid levels with somewhat limited field verification: it therefore provides an indication of the PES of the portions of the affected systems rather than providing a definitive measure.



- PES and EIS were only determined for the affected/regulated areas even though upstream and downstream as well as catchment impacts were considered (based on available desktop information).
- The PES and EIS assessments undertaken are largely qualitative assessment tools and thus the results are open to professional opinion and interpretation.
- The EIS assessment did not specifically address the finer-scale biological aspects of the watercourses such as faunal species.
- The initial study was undertaken as a desktop assessment and as such, the information gathered must be considered with caution, as inaccuracies and data capturing errors are often present within these databases.
- The guideline document, “A Practical Field Procedure for the Identification and Delineation of Wetlands and Riparian Areas” document, as published by DWAF (2005) was followed for the delineation of the wetland and riparian areas.
- This assessment is focussed on surface water and excluded terrestrial landscapes and groundwater features.
- BlueLeaf Environmental is an independent environmental consulting firm and as such, all processes and attributes of the EIA are addressed in a fair and unbiased fashion. It is believed that through the running of a transparent and participatory process, risk associated with assumptions, uncertainties and gaps in knowledge can be, and were, minimised.

## **11. REASONED OPINION AS TO WHETHER THE PROPOSED ACTIVITY SHOULD BE AUTHORISED**

The Environmental Impact Assessment (EIA) process has been undertaken in accordance with the EIA Regulations, 2014 (as amended) published in terms of Section 24 (5) of the National Environmental Management Act (Act No 107 of 1998) (as amended). All relevant legislation has been considered to ensure compliance, and the assessment has been informed by specialist studies, stakeholder engagement, and available baseline information. These inputs have guided the identification of potential impacts, the evaluation of alternatives, and the development of appropriate mitigation and management measures.

The EIA process serves as a decision-support tool that integrates social, economic, and environmental considerations to promote sustainable development. In this regard, the assessment has considered the potential environmental risks associated with the proposed housing development against the anticipated social and economic benefits of the project.

The proposed development is considered justified from a socio-economic perspective, as it will contribute to the provision of formal housing opportunities within an area where housing demand remains a key concern. The site is municipally owned, located close to existing residential areas, and has been identified for medium- to high-density residential development. The proposed land use is therefore considered compatible with the surrounding area and municipal development objectives. The proposed development has considered the identified environmental sensitivities, including the presence of sensitive forest patches on the site. The development layout has been refined to avoid and minimise impacts. The No-Go alternative was also assessed and found to be unacceptable, while this alternative would avoid the direct impacts associated with the proposed development, it would not address the identified housing need or support the planned use of municipal land for residential development. It may also allow existing unmanaged land use pressures on the site to continue.

The EIA process has identified a range of potential environmental impacts associated with the construction and operation of the proposed development. These include impacts on terrestrial biodiversity, soil, heritage resources, and the surrounding environment. However, these impacts are generally localised and can be effectively mitigated through the implementation of appropriate mitigation measures.

Importantly, no fatal flaws have been identified by the EAP and specialists that would prevent the proposed development from proceeding. The Environmental Management Programme (EMPr) provides detailed and appropriate mitigation measures to ensure that all identified impacts are reduced to acceptable levels. Taking

into consideration the findings of the EIA, the effectiveness of proposed mitigation measures, and the significant socio-economic benefits associated with the project, it is the opinion of the EAP that the proposed Walmer housing development is environmentally acceptable and it is recommended that the development be authorised.

## 12. ASPECTS TO BE INCLUDED AS CONDITIONS OF AUTHORISATION

In order to achieve appropriate environmental management standards and ensure that the findings of the environmental studies are implemented through practical measures, the recommendations from this EIA study are included within an EMPr. The EMPr must be used to ensure compliance with environmental specifications and management measures.

The following key conditions should be included as conditions of authorization:

- Ensure environmental authorisations namely EA is obtained for the development.
- Ensure a permit is obtained from DFFE before removing any protected species within the development area.
- ECO to be appointed to audit environmental compliance with conditions in the EMPr and other permits and authorisations.
- The development layout must conform to the following;
  - Stormwater ponds should be located outside forest patches.
  - The business site to be designed such that there is no vehicular access from both the northern and southern portions of the development but that no through flow of vehicular traffic is possible. Separate parking areas to be provided for around the business sites.
  - A pedestrian walkway along the edge of the business site to be provided, with suitable concrete bollards or similar barrier to prevent vehicular access via the pedestrian walkway
  - Zoning of the forest patches must be Open Space III and managed as for conservation usage. The buffers around the patches as well as other open spaces to be zoned as Public Open Space (POS) I
  - Stormwater ponds to be restricted to the edges of any forest patches and in accordance with any conditions related to any authorisation obtained in terms of the NFA from DFFE
- Water conservation measures must be adopted into the building design.
- Rainwater tanks should be considered.
- A plant search and rescue to be conducted by a suitably qualified botanical specialist prior to commencement of any vegetation clearing for each progressive stage of clearing for construction, for any protected species and species of special concern that may be affected by the development as well as other indigenous plant specimens which can be relocated. Such specimens are to be relocated and appropriately transplanted, where applicable and practical to the open space and forest patch conservation area.

- Any protected species which need to be removed or relocated require the necessary permits to be obtained from DEDEAT for those species protected under the Cape Nature and Environmental Ordinance (Ordinance 19 of 1974). The relevant permits are to be obtained from the department of Forestry, Fisheries and the Environment for any species protected under the National Forest Act (Act 84 of 1998).
- Forest patches should be no-go areas during construction.
- Minimise disturbance and clearing of vegetation
- The site must be continuously monitored for erosion.
- No forest patches are to be developed or impacted on. These sites are to be appropriately zoned to Open Space III for conservation purposes and the protection thereof and must be incorporated into the NMBM's Metropolitan Open Space Management Plan/Bioregional Plan and managed as a conservation area.
- The forest patches must be fenced off with a suitable fencing solution such as "Bettafence" to prevent dumping and destruction of the forest.
- Regular inspections of the fenced areas are to be made to monitor whether dumping of rubble or refuse is occurring.
- Regular (weekly) waste collection service to be provided.
- The municipality must ensure sewer system is not overloaded.
- Regular maintenance of the system required.
- Once the storm water facilities have been completed, and fenced to prevent access by the public, the maintenance and monitoring will remain the responsibility of the municipality. These are some of the suggested Best Management Practices (BMPs) for the development
  - Public open space – Vegetation to be maintained such that flood water levels are not increased.
  - Storm water structures – Ponds frequently monitored and obstructions (debris) to inlet/outlet works, removed.
  - Open Channels – Channels to be regularly monitored to identify any erosion gullies or silt deposition.
  - Road Construction – No concentrated flow points are allowed onto road area.
  - Pipes – All pipes are to be frequently monitored and cleared of obstructions.

- Develop a comprehensive stormwater management program which addresses management of stormwater detention ponds, the discharge outlets, inclusive of erosion protection measures and mitigation of such erosion occur.
- Include periodical site inspection that inspects the effectiveness of the run-off control system and specifically records occurrence or not of any erosion on site or downstream.
- Outlet structures at road culverts to be designed in a manner that dissipates flow energy.
- Vegetation to be maintained in public open space such that flood water levels are not increased.
- Ponds frequently monitored and obstructions (debris) to inlet and outlet works, removed.
- Channels to be regularly monitored to identify any erosion gullies or silt deposition.

### 13. UNDERTAKING

The EAP herewith confirms:

- (a) The correctness of the information provided in the reports;
- (b) The inclusion of comments and inputs from stakeholders and I&AP's;
- (c) The inclusion of inputs and recommendations from the specialist reports where relevant;  
and
- (d) That the information provided by the EAP to interested and affected parties and any responses by the EAP to comments or inputs made by interested and affected parties are correctly reflected herein.



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Signature of the environmental assessment practitioner:

**ABANTU ENVIRONMENTAL SERVICES (PTY) LTD**

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Name of company:

Date: 05 August 2026

## 14. REFERENCES

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## 15. APPENDICES

**APPENDIX A: CV AND DECLARATION OF INDEPENDENCE BY THE EAP**

## APPENDIX B: ENVIRONMENTAL MANAGEMENT PROGRAMME

## APPENDIX C: SPECIALIST STUDIES

## APPENDIX D: CONFIRMATION OF BULK SANITATION

## APPENDIX E: LAYOUT PLAN